21.

(Mensuration)

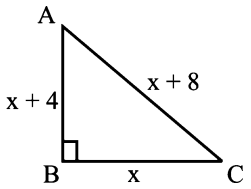
Type - 1

1. In a right angle triangle, the perpendicular is 4 cm larger than the base and the hypotenuse is 4 cm larger than the perpendicular. Calculate the length of the hypotenuse.

(a) 12 cm (b) 10 cm
(c) 20 cm (d) 8 cm

RRB RPF Constable – 18/01/2019 (Shift-III)

Ans : (c)



Pythagoras theorem

Hypotenuse^2 = perpendicular^2 + base^2

$(x + 8)^2 = (x + 4)^2 + x^2$

$x^2 + 64 + 16x = x^2 + 16 + 8x + x^2$

$x^2 - 8x - 48 = 0$

$x^2 - 12x + 4x - 48 = 0$

$x(x - 12) + 4(x - 12) = 0$

$(x - 12)(x + 4) = 0$

It is $x = 12,\ x = -4$

So $x = 12, x = -4$ not possible

$\therefore$ Hypotenuse $= 12 + 8 = 20$ cm

2. The perimeter of an isosceles triangle is 32 cm. Its base is $\frac{6}{5}$ times of equal sides. Find the area of triangle.

(a) 39 cm^2 (b) 64 cm^2
(c) 48 cm^2 (d) 57 cm^2

RRB JE - 22/05/2019 (Shift-III)

Ans : (c) Let the side of isosceles triangle is X.

According to question-

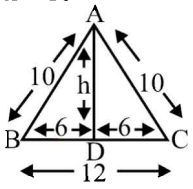
$$x + x + \frac{6x}{5} = 32$$

$$\Rightarrow \quad 2x + \frac{6x}{5} = 32$$

$$\Rightarrow \quad 10x + 6x = 32 \times 5$$

$$\Rightarrow \quad x = \frac{32 \times 5}{16}$$

$x = 10$



Sides of isosceles triangle are 10, 10 and 12 cm respectively.

$AD^2 = AB^2 - BD^2$

$= 10^2 - 6^2$

$AD = \sqrt{64} = 8$ cm.

Hence, height = 8 cm.

Area of isosceles triangle $= \frac{1}{2} \times$ base $\times$ height

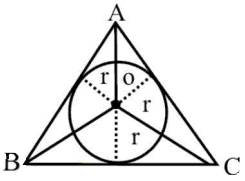
$= \frac{1}{2} \times 12 \times 8 = 6 \times 8 = 48$ cm^2

3. If the perimeter of a triangle is 28 cm. Its internal radius is 3.5 cm. Find the area of triangle.

(a) 49 cm^2 (b) 28 cm^2
(c) 35 cm^2 (d) 42 cm^2

RRB JE - 22/05/2019 (Shift-III)

Ans : (a)



Area of ΔABC = Area of ΔOBC + Area of ΔOAC + Area of ΔOAB

$$= \frac{1}{2} \times r \times BC + \frac{1}{2} \times r \times AC + \frac{1}{2} \times r \times AB$$

$$= \frac{1}{2} \times r \times (BC + AC + AB)$$

$$= \frac{1}{2} \times 3.5 \times 28 = 49 \text{ cm}^2$$

4. The ratio of their corresponding sides of two similar triangles is 1:3 and the area of the larger triangle is 72 cm^2. Find the area of smaller triangle.

(a) 18 cm^2 (b) 8 cm^2
(c) 14 cm^2 (d) 9 cm^2

RRB JE - 29/05/2019 (Shift-I)

Ans : (b) According to rule of similar triangle -

$$\frac{(\text{Area})_1}{(\text{Area})_2} = \frac{(\text{Side})_1^2}{(\text{Side})_2^2}$$

$$\frac{(\text{Area})_1}{72} = \frac{1}{9}$$

$(\text{Area})_1 = 8$ cm^2

So area of small Δ will be 8 cm^2.

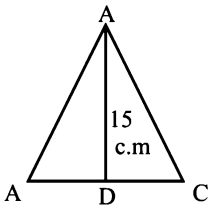
5. The altitude of an equilateral triangle is 15 cm. Find the area of this triangle.

(a) 60 cm^2 (b) $75\sqrt{3}$ cm^2

(c) 90 cm^2 (d) $60\sqrt{3}$ cm^2

RRB RPF SI – 11/01/2019 (Shift-III)

Ans : (b) Let the equilateral Triangle is ABC which altitude AD = 15 cm



$\because$ Height of equilateral triangle $(h) = \frac{\sqrt{3}}{2}a$

[where AD = h = 15 cm]

$$15 = \frac{\sqrt{3}}{2}a$$

$$a = \frac{30}{\sqrt{3}}$$

$\therefore$ Area of triangle $= \frac{\sqrt{3}}{4}a^2$ $\left[\text{where } a = \frac{30}{\sqrt{3}},\ a^2 = 300\right]$

$$= \frac{\sqrt{3}}{4} \times 300 = 75\sqrt{3}\,cm^2$$

6. The ratio of their corresponding sides of two similar triangles is 2:3 then find the ratio of their corresponding heights.

(a) 2 : 3 (b) 3 : 2

(c) 4 : 9 (d) 16 : 81

RRB RPF SI – 06/01/2019 (Shift-II)

Ans : (a) Let the height of similar triangle is h_1 and h_2.

Formula – $\frac{(Side)_1}{(Side)_2} = \frac{h_1}{h_2}$

$$\frac{2}{3} = \frac{h_1}{h_2}$$

$h_1 : h_2 = 2 : 3$

7. In a right angle triangle, the length of the hypotenuse is 2 meter less than twice the length of the shortest side. If the third side is 2 meters larger than the shortest side, what is the length of the hypotenuse of the triangle?

(a) 10 m (b) 13 m

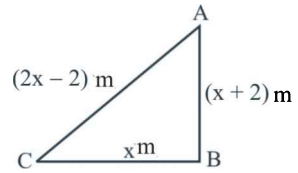
(c) 12 m (d) 7 m

RRB RPF SI – 05/01/2019 (Shift-I)

Ans :(a) Let the smallest side of right angle triangle = x m

According to Pythagoras theorem in right angle ΔABC

$$AC^2 = AB^2 + BC^2$$



$$\Rightarrow (2x-2)^2 = (x+2)^2 + x^2$$

$$\Rightarrow 4x^2 + 4 - 8x = x^2 + 4 + 4x + x^2$$

$$\Rightarrow 2x^2 - 12x = 4 - 4$$

$$\Rightarrow 2x(x-6) = 0$$

if $x - 6 = 0 \Rightarrow x = 6$

length of hypotenuse $= 2x - 2 = 2 \times 6 - 2 = 10$ m.

8. An equilateral triangle is formed by folding a rectangle whose dimensions are 4 cm and 5 cm. What will be the area of equilateral triangle so formed?

(a) $9\sqrt{3}$ cm^2 (b) $3\sqrt{3}$ cm^2

(c) $6\sqrt{3}$ cm^2 (d) $\frac{9}{4}\sqrt{3}$ cm^2

RRB RPF Constable – 18/01/2019 (Shift-I)

Ans : (a) Perimeter of rectangle = 2(length + breadth)

Perimeter of rectangle

= 2(5 + 4) = 18cm

and perimeter of equilateral triangle = 3 × side = 3a cm

According to question

Perimeter of rectangle = Perimeter of equilateral triangle

18 = 3 × side = 3a

$\therefore$ a = 6

So area of equilateral triangle $= \frac{\sqrt{3}}{4} \times 6 \times 6 = 9\sqrt{3}$ cm^2

9. An equilateral triangle is formed by folding a rectangle whose dimensions are 4 cm and 2 cm. What will be the area of equilateral triangle so formed?

(a) $4\sqrt{3}$ cm^2 (b) $6\sqrt{3}$ cm^2

(c) $\frac{9}{4}\sqrt{3}$ cm^2 (d) $2\sqrt{3}$ cm^2

RRB RPF Constable – 20/01/2019 (Shift-II)

Ans : (a) Let the length of rectangle = 4 cm and breadth = 2 cm

Perimeter of rectangle = 2(4 + 2) = 12 cm

$\because$ Perimeter of equilateral triangle = 3 × Side

$\because$ Perimeter of equilateral triangle

= Perimeter of rectangle

$\therefore$ Perimeter of equilateral triangle = 12 cm

$\therefore$ Side of equilateral triangle = 4 cm

Area of equilateral triangle $= \frac{\sqrt{3}}{4}$ $(Side)^2$

$$= \frac{\sqrt{3}}{4} \times (4)^2 = \frac{\sqrt{3}}{4} \times 16 = 4\sqrt{3}\,cm^2$$

10. Three sides of a triangle are 5 cm, 12 cm and 13 cm. A smaller triangle is formed by joining the midpoints of these three sides of this triangle. What is the area of the smaller triangle?

(a) 15 (b) 30
(c) 7.5 (d) 32.5

RRB Group-D – 16/10/2018 (Shift-I)

Ans. (c) : Biggest side of triangle = 13 cm.

Other sides are − 5 cm and 12 cm

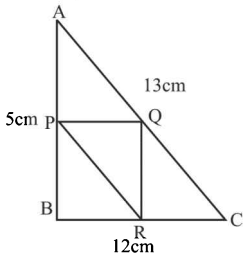
$\because (13)^2 = (12)^2 + (5)^2$

$169 = 169$

So it will be a right angle triangle

Area of ΔPQR joining the midpoint of three side of triangle

$= \frac{\Delta ABC}{4}$



Area of ΔPQR = $\frac{\frac{1}{2}\times 5\times 12}{4} = \frac{30}{4} = 7.5\ cm^2$

11. What is the cost of leveling a triangular region of land at the rate of 20 paise per square meter whose sides are 72 m, 30 m and 78 m respectively?

(a) Rs. 200 (b) Rs. 210
(c) Rs. 216 (d) Rs. 220

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (c) formula- $s = \frac{a+b+c}{2}$

(where a=72, b=30, c=78)

$= \frac{72+30+78}{2} = 90$ m

Area of triangle $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{90(90-72)(90-30)(90-78)}$

$= \sqrt{90\times 18\times 60\times 12}$

$= \sqrt{1166400} = 1080$

20 Paise $= Rs.\frac{20}{100} = Rs.\frac{1}{5}$

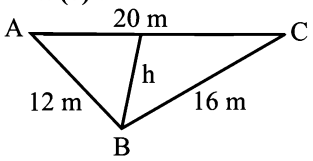
$\therefore$ Cost price $= 1080\times\frac{1}{5} =$ Rs. 216

12. The sides of a triangle are 16 m, 12 m and 20 m. Find altitude of the triangle.

(a) 9.2 m (b) 9.6 m
(c) 9.4 m (d) 9.8 m

RRB Group-D – 09/10/2018 (Shift-I)

Ans. (b) :



Area of the triangle $= \sqrt{s(s-a)(s-b)(s-c)}$

Where $s = \frac{a+b+c}{2}$

$s = \frac{16+12+20}{2} = 24$

$\therefore$ Area $= \sqrt{24(24-16)(24-12)(24-20)}$

$= \sqrt{24\times 8\times 12\times 4}$

$= 96\,m^2$

$\therefore$ The altitude $(h) = \frac{2\times \text{area}}{\text{base}}$

$= \frac{2\times 96}{20}$

$= \frac{96}{10} = 9.6\,m$

13. Find the area of a triangle if its two dimensions and circumference are 9 cm, 11 cm and 34 cm respectively.

(a) $17\sqrt{12}$sq.cm (b) $12\sqrt{15}$ sq.cm
(c) $12\sqrt{17}$ sq.cm (d) $15\sqrt{17}$ sq.cm

RRB Paramedical Exam – 20/07/2018 (Shift-I)

Ans : (c) Let the side of triangle are a, b, c

Then, a = 9 cm, b = 11 cm, a + b + c = 34 cm

9 + 11 + c = 34, c = 14 cm, $s = \frac{a+b+c}{2} = \frac{34}{2} = 17$

$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$\Delta = \sqrt{17(17-9)(17-11)(17-14)}$

$\Delta = \sqrt{17\times 8\times 6\times 3} = \sqrt{17\times 144}$

$= 12\sqrt{17}cm^2$

So area of triangle $= 12\sqrt{17}\ cm^2$

14. Find the area of a triangle whose sides are 7.8 cm, 5 cm and 11.2 cm.

(a) 18 cm^2 (b) 16.8 cm^2
(c) 17.4 cm^2 (d) 12 cm^2

RRB Group 'D' 07/12/2018 (Shift-I)

Ans : (b) Half perimeter (s) $= \frac{7.8+5+11.2}{2} = \frac{24}{2} = 12$

Area of Triangle $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{12(12-7.8)(12-5)(12-11.2)}$

$= \sqrt{12\times 4.2\times 7\times .8}$

$= \sqrt{282.24}$

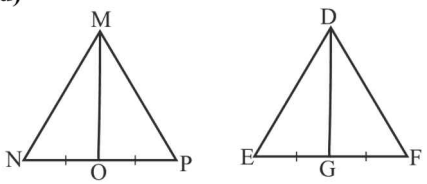
= 16.8 cm^2

15. ΔMNP is similar to ΔDEF. Area of ΔMNP is 1024 cm^2 and area of ΔDEF is 144 cm^2. If the longest side of ΔMNP is 64 cm then what is the longest side of ΔDEF?

(a) 32 cm (b) 28 cm
(c) 20 cm (d) 24 cm

RRB Group-D – 05/12/2018 (Shift-I)

Ans : (d)



$\Delta MNP \sim \Delta DEF$ By the law of similarity

$$\frac{\text{Area of } \Delta MNP}{\text{Area of } \Delta DEF} = \frac{(\text{long side of } \Delta MNP)^2}{(\text{long side of } \Delta DEF)^2}$$

$$\Rightarrow \sqrt{\frac{1024}{144}} = \left(\frac{64}{\text{long side of } \Delta DEF}\right)$$

$$\Rightarrow \frac{32}{12} = \frac{64}{\text{long side of } \Delta DEF}$$

$$\Rightarrow \text{long side of } \Delta DEF = \frac{64 \times 12}{32}$$

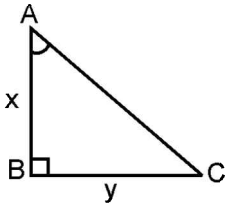
$$= 2 \times 12 = 24 \text{ cm}$$

16. Difference between the height and the base of a right angle triangle is 7 cm and the area of that triangle is 30 sq. cm. The perimeter of the triangle is:

(a) 13 cm (b) 12 cm
(c) 30 cm (d) 25 cm

RRB Group-D – 05/11/2018 (Shift-II)

Ans : (c) given



Difference between height and base of triangle = 7

$x - y = 7$ ..... (i)

Area of Triangle = 30 cm^2

So, area of right angle triangle

$$= \frac{1}{2} \times \text{base} \times \text{height}$$

$$30 = \frac{1}{2}xy \Rightarrow xy = 60 \quad \text{..... (ii)}$$

$$x + y = \sqrt{(x-y)^2 + 4xy}$$

$$= \sqrt{49 + 240}$$

$$= \sqrt{289}$$

$x + y = 17$ ..... (iii)

From the equation (i) and (iii)

$x = 12$ cm.

$y = 5$ cm.

According to Pythagoras theorem in triangle ABC

$AC^2 = AB^2 + BC^2$

$AC^2 = (12)^2 + (5)^2$

$AC^2 = 144 + 25$

$AC^2 = 169$

$AC = 13$ cm.

So, perimeter of triangle = 13 + 12 + 5 = 30 cm.

17. The numerical value of the area of an equilateral triangle is two times the numerical value of its perimeter. Find the area of the above triangle.

(a) 48 cm^2 (b) $24\sqrt{3}$ cm^2
(c) $48\sqrt{3}$ cm^2 (d) $36\sqrt{3}$ cm^2

RRB Group-D – 12/12/2018 (Shift-I)

Ans. (c) Let the side of equilateral triangle is a cm.

According to question

Area of equilateral Δ = 2 × perimeter of Δ

$$\frac{\sqrt{3}a^2}{4} = (3a) \times 2$$

$$\frac{\sqrt{3}a}{4} = 6$$

$$a = \frac{24\sqrt{3}}{3}$$

$$a = 8\sqrt{3} \text{ cm}$$

$$\because \text{ Area of equilateral } \Delta = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times 8\sqrt{3} \times 8\sqrt{3}$$

$$= 2 \times 3 \times 8\sqrt{3} = 48\sqrt{3} \text{ cm}^2$$

18. The areas of two similar triangles are 169 cm^2 and 121 cm^2. If the longest side of the larger triangle is 26 cm then the longest side of the smaller triangle is :

(a) 26 cm. (b) 18 cm.
(c) 28 cm. (d) 22 cm.

RRB NTPC 17.01.2017 Shift-2

Ans : (d)

$$\frac{\text{Largest side of large triangle}}{\text{Largest side of small triangle}} = \sqrt{\frac{169}{121}}$$

$$\Rightarrow \frac{26}{\text{Largest side of small triangle}} = \frac{13}{11}$$

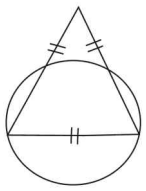
$$\Rightarrow \text{Largest side of small triangle} = \frac{11 \times 26}{13} = 22 \text{ cm.}$$

19. An equilateral triangle is constructed in such a way that the two vertices of the triangle lie on the diameter of a circle. If the area of the circle is 64π cm^2, then what will be the side of the triangle?

(a) 16 (b) 8
(c) $16\sqrt{3}$ (d) $8\sqrt{3}$

RRB NTPC 04.04.2016 Shift : 3

Ans : (a)



Area of circle = 64π

$\pi r^2 = 64\pi$

$r^2 = 64 \quad \Rightarrow r = 8$ cm.

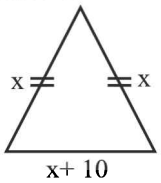
length of equilateral triangle sides = the diameter of the circle = 2 × 8 = 16 cm

20. The perimeter of a triangle is 100 cm. If its two sides are equal and the third side is 10 cm more than the equal sides. What is the length of the third side?

(a) 30 (b) 25
(c) 40 (d) 36

RRB NTPC 03.04.2016 Shift : 3

Ans :(c) ∵ Two sides of a triangle are equal
∴ Triangle will be isosceles



Perimeter of Triangle = 100 cm
x + x + x + 10 = 100
3x + 10 = 100
3x = 90 $\boxed{x = 30}$
So length of third side = x + 10 = 30 + 10 = 40 cm.

21. The area of a right angled triangle is 30 m^2. If the height is 7 m more than the base, then find the base.

(a) 5 m (b) 12 m
(c) 7 m (d) 9 m

RRB NTPC 02.04.2016 Shift : 1

Ans : (a) Let the base = x m.
height = (x + 7) m
∴ Area of right angle triangle = 30 square meter
1/2× base × height =30

$\frac{1}{2} \times x \times (x+7) = 30$

$x^2 + 7x = 60$
$x^2 + 7x - 60 = 0$
$x^2 + 12x - 5x - 60 = 0$
$x(x+12) - 5(x + 12) = 0$
$(x + 12)(x-5) = 0$
∴ x= 5
so base= 5 m.

22. The perimeter of a triangle is 200 cm. If its two sides are equal and the third side is 20 cm more than the equal sides, what will be the length of the third side?

(a) 60 (b) 50
(c) 80 (d) 70

RRB NTPC 02.04.2016 Shift : 2

Ans : (c) Let the first side = x
second side = x
Third side = x + 20
Then, x + x + x + 20 = 200
3x + 20 = 200
3x = 180
x = 60
third side = 60 + 20 = 80 cm.

23. If side of an equilateral triangle is 4 unit, then area of that triangle is :

(a) $16/\sqrt{3}$ square unit (b) $4\sqrt{3}$ square unit
(c) $2/\sqrt{3}$ square unit (d) $\sqrt{3}$ square unit

RRB NTPC 30.03.2016 Shift : 2

Ans : (b)

Area of equilateral triangle $= \frac{\sqrt{3}}{4} \times (\text{side})^2$

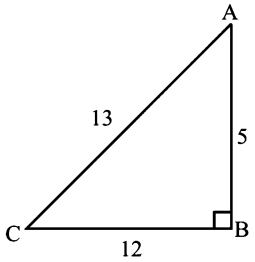
$= \frac{\sqrt{3}}{4} \times 16 = 4\sqrt{3}$ square unit

24. Find the area of the triangle whose sides are 5 cm, 12 cm and 13 cm.

(a) 46 cm^2 (b) 42 cm^2
(c) 30 cm^2 (d) 38 cm^2

RRB NTPC 18.01.2017 Shift : 1

Ans : (c)



Sides = 5, 12, 13
$5^2+12^2=13^2$
25 + 144 = 169
169 = 169
So this triangle will be right angle triangle

So area of Δ ABC $= \frac{1}{2}$ height × base

$= \frac{1}{2} \times 5 \times 12 = 30$ square cm.

25. The area of a triangle is 456 cm^2 and its height is 24 cm. Then the length of its base is:

(a) 32 (b) 36
(c) 34 (d) 38

RRB NTPC 06.04.2016 Shift : 2

Ans : (d) Area of triangle = 456 cm^2

height = 24 cm

base=?

Area of triangle $= \frac{1}{2} \times$ base $\times$ height

$456 = \frac{1}{2} \times 24 \times$ base

base $= \frac{456}{12}$, base = 38 cm.

26. The area of two similar triangles are 121 m^2 and 64 m2. If the median of the first triangle is 12.1 m then the median of the second triangle is:

(a) 6.4 m (b) 8.4 m
(c) 8.8 m (d) 9.2 m

RRB NTPC 19.01.2017 Shift : 2

Ans : (c)

$\frac{M_1}{M_2} = \sqrt{\frac{A_1}{A_2}}$ $\frac{12.1}{M_2} = \sqrt{\frac{(121)}{(64)}}$

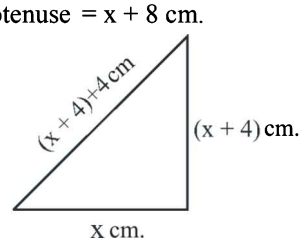
$\frac{12.1}{M_2} = \frac{11}{8}$, $\boxed{M_2 = 8.8 \text{ m.}}$

27. In a right angle triangle, the hypotenuse is 4 cm more than the perpendicular which is 4 cm more than the base. Find the length of the perpendicular.

(a) 12 cm (b) 16 cm
(c) 20 cm (d) 8 cm

RRB NTPC 27.04.2016 Shift : 3

Ans : (b) Let the base of right angle triangle = x cm.
then, perpendicular = (x + 4) cm
hypotenuse = x + 8 cm.



According to pythagoras theorem

hypotenuse^2 = perpendicular^2 + base^2

$(x + 8)^2 = (x + 4)^2 + x^2$

$\Rightarrow x^2 + 64+16x = x^2+16+8x+x^2$

$\Rightarrow x^2 + 16x - x^2-x^2-8x=16-64$

$\Rightarrow -x^2+8x= -48$

$\Rightarrow x^2 - 8x = 48$

$\Rightarrow x^2 - 8x - 48 = 0$

$\Rightarrow x^2 - 12x+4x - 48 = 0$

$\Rightarrow x(x-12)+4(x - 12) = 0$

$\Rightarrow (x+4)(x - 12) = 0$

$\Rightarrow x+4 = 0$, $x - 12 = 0$

$\Rightarrow x = -4$ (invalid)

and $x-12 = 0$ (Valid) $\boxed{x = 12}$

perpendicular = (x+4) cm. =12 + 4 = 16 cm.

28. Find the area of the triangle whose sides are 11 cm, 7 cm and 14 cm.

(a) $7\sqrt{22}$ (b) $11\sqrt{8}$
(c) $12\sqrt{10}$ (d) $13\sqrt{3}$

RRB NTPC 29.04.2016 Shift : 1

Ans : (c) a = 11 cm., b = 7 cm., c = 14 cm.

half perimeter (s) $= \frac{a+b+c}{2} = \frac{11+7+14}{2} = \frac{32}{2}$

s = 16

Area of triangle $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{16(16-11)(16-7)(16-14)}$

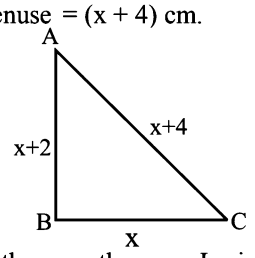
$= \sqrt{16(5)(9)(2)} = 12\sqrt{10}\ cm^2$

29. In a right angle triangle, the hypotenuse is 2 cm more than the perpendicular and the perpendicular is 2 cm more than the base. Find the length of the base.

(a) 6 cm. (b) 9 cm.
(c) 10 cm. (d) 8 cm.

RRB NTPC 29.04.2016 Shift : 2

Ans :(a) Let the length of base is x cm.
According to question
length of perpendicular = (x+2) cm.
length of hypotenuse = (x + 4) cm.



According to pythagoras theorem, In right angle ΔABC.
hypotenuse^2 = (perpendicular)^2 + (base)^2

$(x + 4)^2 = (x+2)^2 + x^2$

$x^2 + 16 + 8x = x^2 + 4 + 4x + x^2$

$\Rightarrow x^2 - 4x - 12 = 0$

$\Rightarrow x^2 - 6x + 2x - 12= 0$

$\Rightarrow x (x-6) + 2 (x-6)= 0$

$\Rightarrow (x-6) = 0$, $x + 2 = 0$

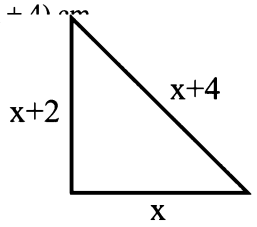
$\boxed{x = 6}$ $x \neq -2$ (It is not possible)

30. In a right angle triangle, the hypotenuse is 2 cm more than the perpendicular and the perpendicular is 2 cm more than the base. Find the length of the perpendicular.

(a) 6 cm (b) 9 cm
(c) 10 cm (d) 8 cm

RRB NTPC 29.04.2016 Shift : 3

Ans. (d) : Let the base of right angle triangle, = x cm.
∴ perpendicular = (x + 2) cm.
hypotenuse = (x + 4) cm



$\because$ hypotenuse^2 = perpendicular^2 + base^2

$(x + 4)^2 = (x + 2)^2 + x^2$

$\Rightarrow$ $x^2 + 16 + 8x = x^2 + 4 + 4x + x^2$

$\Rightarrow$ $x^2 - 4x - 12 = 0$

$\Rightarrow$ $x^2 - 6x + 2x - 12 = 0$

$\Rightarrow$ $x(x-6) + 2(x-6) = 0$

$\Rightarrow$ $(x - 6)(x + 2) = 0$

So, $x = 6$

$\therefore$ Length of perpendicular $(x + 2) = 6 + 2 = 8$ cm.

31. What is the ratio of inner radius and the circum radius of an equilateral triangle with area of $16\sqrt{3}$ sq. unit?

(a) 1/2 (b) 1/4

(c) 1/3 (d) 2/3

RRB NTPC 30.04.2016 Shift : 2

Ans : (a) $\because$ Area of equilateral $\Delta = \frac{\sqrt{3}}{4}a^2$

$\therefore$ $\frac{\sqrt{3}}{4}a^2 = 16\sqrt{3}$

$a^2 = 16\sqrt{3} \times \frac{4}{\sqrt{3}}$

$a^2 = 64 \Rightarrow a = 8$

Radius of circle in equilateral $\Delta = \frac{a}{2\sqrt{3}} = \frac{8}{2\sqrt{3}}$

Radius of circumcircle of equilateral $\Delta = \frac{a}{\sqrt{3}} = \frac{8}{\sqrt{3}}$

Intended ratio $= \frac{\frac{8}{2\sqrt{3}}}{\frac{8}{\sqrt{3}}} = \frac{1}{2}$

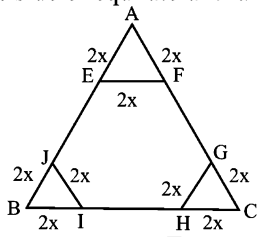
32. Three triangles are marked out of a large triangle at the three vertices such that each side of each of the small triangles is one-fourth as long as each corresponding side of the large triangle. The ratio of the area of the three small triangles taken together to that of the rest of the large triangle is:

(a) 3 : 13 (b) 1 : 5

(c) 3 : 16 (d) 4 : 15

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (a) Let the side of equilateral triangle = 8x unit



Area of equilateral triangle $= \frac{\sqrt{3}}{4} \times (8x)^2$

$= \frac{\sqrt{3}}{4} \times 64x^2 = 16\sqrt{3}x^2$

Area of three small equilateral triangle

$= 3 \times \frac{\sqrt{3}}{4} \times (2x)^2 = 3\sqrt{3}x^2$

$\therefore$ Area of Remainder equilateral triangle

$= 16\sqrt{3}x^2 - 3\sqrt{3}x^2 = 13\sqrt{3}x^2$

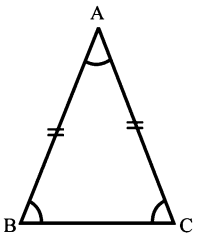
$\therefore$ Intended Ratio $= 3\sqrt{3}x^2 : 13\sqrt{3}x^2 = 3 : 13$

33. If each of the two equal angles of an isosceles triangle is twice the third angle, the measure of the third angle is:

(a) 90^0 (b) 72^0

(c) 45^0 (d) 36^0

Ans : (d)



Let the third angle $\angle A = x$

then $\angle B = \angle C = 2x$

$\because$ total of three angles of the triangle $= 180^0$

$\angle A + \angle B + \angle C = 180^0$

$x + 2x + 2x = 180^0$

$5x = 180^0$

$x = 36^0$

$\therefore$ third angle $x = 36^0$

34. Three triangles are marked out of a large triangle at the three vertices such that each side of each of the small triangles is one–third as long as each corresponding side of the large triangle. The ratio of the area of the three small triangles taken together to that of the rest of the large triangle is :

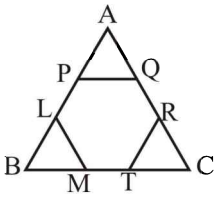
(a) 3 : 1 (b) 1 : 3

(c) 1 : 9 (d) 1 : 2

RRB ALP & Tec. (21-08-18 Shift-I)

Ans : (b) Let the side of large triangle ABC is x cm.

Area will be equal of Δ APQ, ΔBLM and ΔCRT.



Side of each small triangle $= \frac{1}{3}$ side of large triangle

$= \frac{x}{3}$ cm.

According to the question-

$$\text{Ratio} = 3 \times \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2 : \frac{\sqrt{3}}{4}(x)^2 = \frac{x^2}{3} : x^2 = 1:3$$

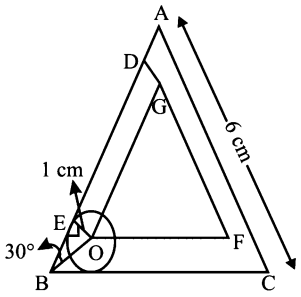
35. ABC is an equilateral triangle of side 6cm. If a circle of radius 1 cm is formed inside and along the sides of the triangle, the locus of the centre of the circle is an equilateral triangle of side is :

(a) 4 cm (b) $\left(6-2\sqrt{3}\right)$ cm

(c) 5 cm (d) $\left(3+\sqrt{3}\right)$ cm

RRB ALP & Tec. (14-08-18 Shift-III)

Ans : (b)



From ΔOEB

$\tan 30° = \frac{OE}{EB}$

$\frac{1}{\sqrt{3}} = \frac{1}{EB}$

$EB = \sqrt{3}$ cm

So, side of equilateral triangle formed from the locus center in circle

= AB – (EB + DA) = AB – 2 EB = $\left(6-2\sqrt{3}\right)$ cm

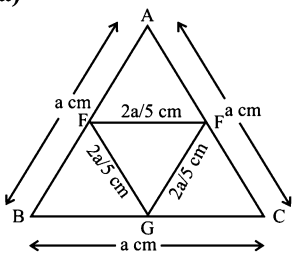
36. Three small triangles are so formed from the three corners of a large triangle in such a way that each side of the small triangle is equal to $\frac{2}{5}$ times of the corresponding side of the large triangle. What is the ratio between the total areas of the three small triangle and the remaining area of the large triangle?

(a) 12 : 13 (b) 1 : 5

(c) 12 : 25 (d) 4 : 25

RRB Group-D – 31/10/2018 (Shift-I)

Ans : (a)



Area of equilateral triangle ABC = $\frac{\sqrt{3}}{4}a^2$ cm

Area of Small equilateral ΔAEF, ΔBEG and ΔCFG

$= \frac{\sqrt{3}}{4}\left\{\left(\frac{2a}{5}\right)^2+\left(\frac{2a}{5}\right)^2+\left(\frac{2a}{5}\right)^2\right\}$

$= \frac{3\sqrt{3}}{4}\times\frac{4a^2}{25} = \frac{3\sqrt{3}}{25}a^2$ cm

$\frac{\text{Area of } \Delta ABC}{\text{Area of small three triangles}} = \frac{\sqrt{3}/4a^2}{3\sqrt{3}/25a^2}$

$\frac{\text{Area of } \Delta ABC}{\text{Area of small three triangles}} - 1 = \frac{25}{12} - 1$

$\frac{\text{Area of } \Delta ABC - \text{Area of } \Delta \text{ small three triangles}}{\text{Area of small three triangles}}$

$= \frac{25-12}{12}$

$= \frac{\text{Area of remainced triangle}}{\text{Area of small three triangles}} = \frac{13}{12}$

$= \frac{\text{Area of small three triangles}}{\text{Area of remained triangle}} = \frac{12}{13}$

Type - 2

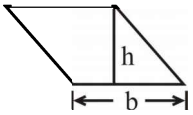
37. The base of a parallelogram is twice its height. If the area is 392 sq.m. What is its height?

(a) 14 m (b) 28 m

(c) 12 m (d) 24 m

RRB RPF Constable – 22/01/2019 (Shift-III)

Ans : (a) ∵ b = 2h



Area of Parallelogram = b×h

392 = 2h×h

$196 = h^2$

h= 14 m

38. The area of a field of trapezoidal shape is 1440 sqm. The perpendicular distance between parallel sides is 24m. The ratio of the parallel sides is 5:3. Find the length of the largest parallel side.

(a) 30 m (b) 60 m

(c) 45 m (d) 75 m

RRB JE - 30/05/2019 (Shift-III)

Ans : (d) Let the parallel sides of trapezium is 5x meter and 3x meter

area of trapezoid = $\frac{1}{2}$(total of parallel sides) × distance between them

$1440 = \frac{1}{2}(5x+3x)\times 24$

$\frac{1440\times 2}{24} = 8x$

8x = 120

x = 15

length of largest parallel sides of trapezoid = 5x m

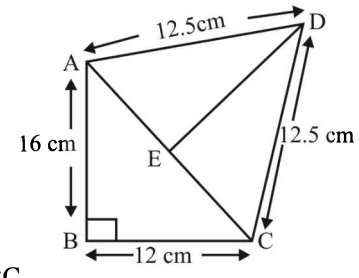
= 5 × 15 = 75 m

39. A quadrilateral has two sides 16 cm and 12 cm. and the angle between them is 90°. Its other two sides are $\frac{25}{2}$ cm each. The other three angles of the quadrilateral are not the right angles. Find the area of the quadrilateral.

(a) 202 cm^2 (b) 151 cm^2
(c) 193 cm^2 (d) 171 cm^2

RRB RPF SI – 10/01/2019 (Shift-III)

Ans. (d)



From ΔABC

$AC = \sqrt{16^2 + 12^2}$

$= \sqrt{256 + 144}$

$= \sqrt{400} = 20cm$

ΔADC in

$ED = \sqrt{\left(\frac{25}{2}\right)^2 - 10^2}$

$= \sqrt{\frac{625-400}{4}} = \sqrt{\frac{225}{4}} = \frac{15}{2}$

Area of □ ABCD = Area of Δ ABC + Area of Δ ACD

$= \frac{1}{2}\left(16 \times 12 + \frac{15}{2} \times 20\right)$

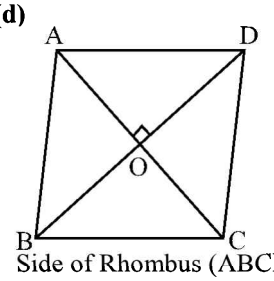
$= 16 \times 6 + 15 \times 5 = 96 + 75 = 171\ cm^2$

40. If one side of a rhombus and one diagonal are 16 cm each, then the area in cm^2 of the rhombus is:

(a) $64\sqrt{3}$ (b) $256\sqrt{3}$
(c) 128 (d) $128\sqrt{3}$

RRB RPF Constable – 20/01/2019 (Shift-II)

Ans : (d)



Side of Rhombus (ABCD) = 16 cm

and a diagonal BD = 16 cm = d_1

in ΔAOB —

$AO = \sqrt{(AB)^2 - (BO)^2}$

$= \sqrt{(16)^2 - (8)^2} = \sqrt{256 - 64}$

$AO = \sqrt{192}$

$AO = 8\sqrt{3}$ cm

Second diagonal $(d_2) = 8\sqrt{3} + 8\sqrt{3} = 16\sqrt{3}$ cm

Area of Rhombus $= \frac{1}{2}(d_1 \times d_2)$

$= \frac{1}{2}\left(16 \times 16\sqrt{3}\right)$

$= 128\sqrt{3}\ cm^2$

41. Area of a rhombus is 216 cm^2 and the length of its one diagonal is 24 cm. What is the length of each side of the rhombus?

(a) 14 cm (b) 13 cm
(c) 15 cm (d) 12 cm

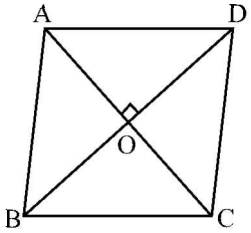
RRB Group-D – 20/09/2018 (Shift-II)

Ans : (c) Given-

Length of diagonal of Rhombus is (d_1) = 24 cm, d_2 = ?

According to question,

Area of Rhombus $= \frac{1}{2} \times d_1 \times d_2$



AO = 12 cm

DO = 9 cm

$216 = \frac{1}{2} \times 24 \times d_2$

d_2 = 18 cm

in ΔAOD

According to pythagoras theorem,

$AD^2 = 12^2 + 9^2$ {Diagonal of rhombus

$AD^2 = 144 + 81$ intersect at the angle of 90°

$AD^2 = 225$ each other}

AD = 15 cm.

So length of rhombus each side will 15 cm.

42. What is the area of a piece of metal that is in the form of a parallelogram with a base of 20 m and a height of 5.4 m?

(a) 108 sq.m. (b) 801 sq.m.
(c) 180 sq.m. (d) 810 sq.m.

RRB Group-D – 20/09/2018 (Shift-II)

Ans : (a) Area of parallelogram = (base × height)

$= 20 \times 5.4\ m^2$

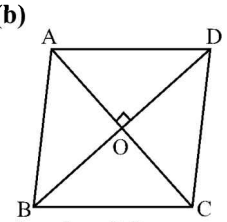
Area = 108 m^2

43. The length of each side of a rhombus is 5 m and the length of one of its diagonal is 2.8 m. Find the area of this rhombus.

(a) 6.72 m^2 (b) 13.44 m^2
(c) 14 m^2 (d) 7 m^2

RRB Group-D – 20/09/2018 (Shift-II)

Ans : (b)



Diagonal $= d_1 = 2.8$m
Each side = 5m.
ΔAOD in,

$AO = \frac{2.8}{2} = 1.4$m,

AD = 5m.
Diagonal of rhombus intersect at the angle of 90° to each other
According to Pythagoras theorem

$OD^2 = (5)^2 - (1.4)^2$
$OD^2 = 23.04$
OD = 4.8 m

So diagonal $(d_2) = 4.8 \times 2 = 9.6$ m.

So area of rhombus $= \frac{1}{2} \times d_1 \times d_2$

$= \frac{1}{2} \times 9.6 \times 2.8$

Area = 13.44 m^2

44. The side of a rhombus is 37 cm and its area is 840 cm^2. Find the sum of the length of its diagonals.

(a) 84 cm (b) 94 cm
(c) 42 cm (d) 47 cm

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (b) Area of rhombus $= \frac{1}{2} d_1 d_2$

$\Rightarrow \frac{1}{2} d_1 d_2 = 840$

$d_1 d_2 = 1680$ --------(i)

$d_1^2 + d_2^2 = (\text{side})^2 \times 4$

$d_1^2 + d_2^2 = 37 \times 37 \times 4$

$d_1^2 + d_2^2 = 5476$ ----(ii)

$\because \quad (d_1 + d_2)^2 = d_1^2 + d_2^2 + 2d_1 d_2$

from the equation (i) and (ii) -

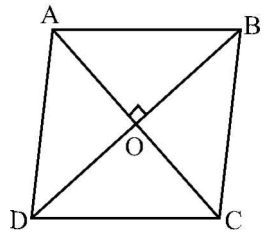
$\Rightarrow (d_1 + d_2)^2 = 5476 + 2 \times 1680$
$\Rightarrow (d_1 + d_2)^2 = 5476 + 3360$
$\Rightarrow (d_1 + d_2)^2 = 8836$
$\Rightarrow (d_1 + d_2)^2 = (94)^2$
$\Rightarrow \boxed{d_1 + d_2 = 94 \text{ cm}}$

45. The perimeter of a rhombus is 56 cm and the length of one of its diagonals is 26 cm. Find the length of the other diagonal of the rhombus.

(a) $6\sqrt{7}$ cm (b) $6\sqrt{2}$ cm
(c) $6\sqrt{3}$ cm (d) $6\sqrt{5}$ cm

RRB Group-D – 10/10/2018 (Shift-II)

Ans : (c)



OD = 13 cm
Perimeter of rhombus = 56 cm

4a = 56
Side (a) = 14 cm
According to Pythagoras theorem

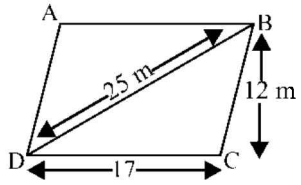
$(AO)^2 = (AD)^2 - (DO)^2$
$(AO)^2 = (14)^2 - (13)^2$
$(AO)^2 = 196 - 169$
$(AO)^2 = 27$
$AO = \sqrt{27}$
Diagonal (AC) = 2 × AO
Diagonal (AC) = $2 \times \sqrt{27}$
(AC) = $6\sqrt{3}$ cm

46. The sides of a parallelogram are 12 m and 17 m. If the length of one diagonal is 25m, then find the area of the parallelogram:

(a) 190 m^2 (b) 150 m^2
(c) 160 m^2 (d) 180 m^2

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (d)



Half perimeter of ΔBCD (s) $= \frac{12+25+17}{2} = \frac{54}{2} = 27$

Area of ΔBCD $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{27(27-12)(27-25)(27-17)}$

$= \sqrt{27 \times 15 \times 2 \times 10}$

$= \sqrt{3 \times 9 \times 3 \times 5 \times 2 \times 2 \times 5}$

$= 9 \times 5 \times 2 = 90$

Area of parallelogram ABCD $= 2 \times \Delta$BCD

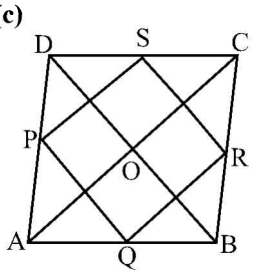
$= 90 \times 2 = 180$ m^2

47. The diagonals of a rhombus are 12 cm and 16 cm. The area of shape formed by joining the midpoints of all the sides of the rhombus is:

(a) 192 cm^2 (b) 64 cm^2
(c) 48 cm^2 (d) 96 cm^2

RRB Group-D – 25/09/2018 (Shift-I)

Ans : (c)



Area of rhombus $= \frac{1}{2} \times d_1 \times d_2$

Area $= \frac{1}{2} \times 12 \times 16$

Area $= 96\ cm^2$

A rhombus is formed by joining the midpoints of all the sides of the large rhombus and the area of this small rhombus is area $= \frac{\text{Area of larger rhombus}}{2}$

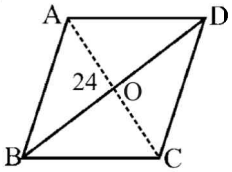
Area $= \frac{96}{2} = 48\ cm^2$

48. The area of a rhombus is 216 m^2 and the length of one of its diagonal is 25m. What is the length of each side of the rhombus?

(a) 12 m (b) 18 m
(c) 15 m (d) 30 m

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (c) :



AC = 18 m
BD = 24 m

Area of rhombus $= \frac{1}{2} \times AC \times BD$

$216 = \frac{1}{2} \times AC \times BD$

$\frac{1}{2} \times AC \times 24 = 216$

$AC = 18\,cm$

Length of each side of rhombus $= \sqrt{(OC)^2 + (OB)^2}$

$= \sqrt{9^2 + 12^2}$

$= \sqrt{81 + 144}$

$= \sqrt{225} = 15$ Meter

49. Find the distance between two parallel sides of a non- parallelogram. If the area of the non-parallelogram is 150 sqm and the length of the two parallel sides are 10 m and 15 m respectively.

(a) 12 m (b) 14 m
(c) 10 m (d) 15 m

RRB Group-D – 22/10/2018 (Shift-III)

Ans : (a) Area of a parallelogram or trapezoid =

$= \frac{1}{2}$ (sum of parallel side × distance)

$= \frac{1}{2}(10 + 15) \times$ Distance

$150 = \frac{1}{2} \times 25 \times$ Distance

Distance $= \frac{300}{25} = 12$ meter

50. The area of a rhombus is 324 cm^2 and the length of one of its diagonal is 36 cm. What is the length of each side of this rhombus?

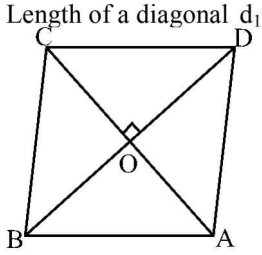
(a) 10cm (b) $18\sqrt{5}$ cm
(c) $9\sqrt{5}$ cm (d) $8\sqrt{6}$ cm

RRB Group-D – 05/12/2018 (Shift-III)

Ans : (c)

Area of rhombus = $324 cm^2$

Length of a diagonal $d_1 = 36cm$



Area of rhombus $= \frac{1}{2} \times d_1 \times d_2$

$324 = \frac{1}{2} \times 36 \times d_2$

$d_2 = 18$ cm.

According to figure $(AB)^2 = (OA)^2 + (OB)^2$

$= 9^2 + 18^2$

$= 81 + 324$

$(AB)^2 = 405$

$AB = 9\sqrt{5}$ cm.

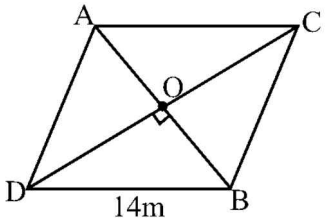
So length of each side of rhombus is $9\sqrt{5}$ cm .

51. The side of the rhombus and one diagonals are 14 cm each. Find the area of the rhombus.

(a) $49\sqrt{3}$ (b) $196\sqrt{3}$
(c) $98\sqrt{3}$ (d) 98

RRB Group-D – 03/12/2018 (Shift-III)

Ans. (c) : The diagonals of a rhombus intersect each other at 90º angle.



$OB = OA = \frac{14}{2}$ cm = 7cm

$\therefore$ According to Pythagoras theorem-

$DB^2 = OB^2 + OD^2$

$196 = 49 + OD^2$

$OD^2 = 147$

$OD = \sqrt{3 \times 7 \times 7}$

$= 7\sqrt{3}$

$DC = 2 \times 7\sqrt{3} = 14\sqrt{3}$

Area of rhombus $= \frac{1}{2} \times AB \times CD$

$= \frac{1}{2} \times 14 \times 14\sqrt{3} = 98\sqrt{3}\,cm^2$

52. The height of an equilateral triangle is equal to the diagonal of a square. Find the ratio of the areas of the triangle and the square.

(a) $\sqrt{3} : 2$ (b) $2 : \sqrt{3}$

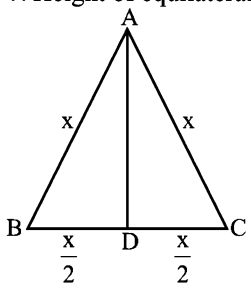
(c) 3 : 4 (d) $\sqrt{3} : \sqrt{2}$

RRB Group-D – 03/12/2018 (Shift-III)

Ans. (b) : Let the side of square= a

$\therefore$ Diagonal of square $= a\sqrt{2}$

$\therefore$ Height of equilateral triangle (h) = $a\sqrt{2}$



In equilateral ΔABC,

Let the side of Δ = x

$AD = a\sqrt{2}$

$AD^2 = AC^2 - DC^2$

$2a^2 = x^2 - \frac{x^2}{4}$

$2a^2 = \frac{3x^2}{4}$

$x^2 = \frac{8}{3}a^2$

Area of equilateral $\Delta = \frac{\sqrt{3}}{4} \times x^2$

$= \frac{\sqrt{3}}{4} \times \frac{8}{3}a^2$

$= \frac{2}{\sqrt{3}}a^2$

Area of square $= a^2$

Area of equilateral Δ : area of square $= \frac{2}{\sqrt{3}}a^2 : a^2$

$= 2 : \sqrt{3}$

53. A rhombus was formed from a rectangle with sides 12 cm and 8 cm, whose perimeter was equal to that of the rectangle and one of its angle was 120°. Find the area of the rhombus.

(a) $25\sqrt{3}\,cm^2$ (b) $50\sqrt{3}\,cm^2$

(c) $40\sqrt{3}\,cm^2$ (d) $\frac{100\sqrt{3}}{3}cm^2$

RRB Group-D – 01/10/2018 (Shift-I)

Ans. (b) :

Perimeter of rectangle = 2 (Length + breadth)

= 2 (12 + 8)

= 40 cm.

Given that

Perimeter of rhombus = Perimeter of rectangles ..... (I)

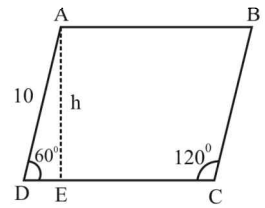
Let the side of rhombus is a cm

From equation (I),

4a = 40

a = 10

In Δ ADE,



$\sin 60^0 = \frac{h}{10}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{h}{10}$

Height $(h) = 5\sqrt{3}\,cm$

Area of rhombus = Base × Height

$= 10 \times 5\sqrt{3}$

$= 50\sqrt{3}\,cm^2$

54. If the diagonals of a rhombus are in the ratio 5:4, then find the ratio of the area of the rhombus and the product of its diagonals.

(a) 4:1 (b) 2:1

(c) 1:2 (d) 1:4

RRB Paramedical Exam – 21/07/2018 (Shift-II)

Ans : (c) Let the diagonal of rhombus = 5x, 4x

Area of rhombus $= \frac{1}{2}$ product of diagonal

$\frac{1}{2} \times 5x \times 4x = 10x^2$

$\therefore \frac{\text{Area of rhombus}}{\text{Product of rhombus diagonal}} = \frac{10x^2}{20x^2} = 1 : 2$

55. The height and the area of a parallelogram are 7 cm and 161 cm². Find the length of its base.

(a) 25 cm (b) 23 cm

(c) 21 cm (d) 19 cm

RRB NTPC 05.04.2016 Shift : 2

Ans : (b) $\because$ Area of parallelogram = Base × Height

161 = Base × 7

$\therefore$ Base= 23 cm

56. The area of the trapezium is 18 sq. cm. Its height and base are 3 cm and 5 cm respectively. Find the length of the side parallel to base.

(a) 4 cm (b) 6 cm
(c) 8 cm (d) 7 cm

RRB NTPC 06.04.2016 Shift : 2

Ans : (d) Let the length of side parallel base = x cm.

height = 3 cm

length of base = 5 cm

area= 18 cm.

Area of trapezium $= \frac{1}{2} \times$ height × sum of parallel sides

$18 = \frac{1}{2} \times 3 \times (5 + x)$

36 = 15 + 3x

3x = 36 – 15

3x = 21

x = 7

57. The length of one side of a rhombus is 17cm and one of the diagonals is 16 cm. Find the length of the other diagonal.

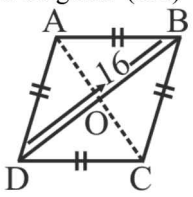
(a) 30 cm (b) 20 cm
(c) 32 cm (d) 16 cm

RRB ALP & Tec. (31-08-18 Shift-II)

Ans. (a) : In ABCD Rhombus

AB = BC = CD = DA = 17 cm

BD = 16 cm diagonal (AC) = ?



so $OB = \frac{1}{2} BD = \frac{1}{2} \times 16 = 8$ cm

In right angle ΔAOB,

$(AB)^2 = (AO)^2 + (OB)^2$

$(17)^2 = (AO)^2 + 8^2$

$(AO)^2 = 289 - 64$

$AO = \sqrt{225} = 15$ cm

Then diagonal AC = 2 × AO

AC = 2 × 15 = 30 cm.

58. The length of one side of a rhombus is 61 cm and its area is 1320 cm^2. Find the sum of the lengths of its diagonals.

(a) 120 cm (b) 122 cm
(c) 142 cm (d) 71 cm

RRB ALP & Tec. (30-08-18 Shift-I)

Ans : (c) If side of rhombus is 'a' and diagonal d_1 and d_2

So,

a = 61 cm and A = 1320 cm^2

so, $1320 = \frac{d_1 \times d_2}{2}$

or $d_1 \times d_2 = 2640$

and, $61^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$

$3721 = \frac{d_1^2 + d_2^2}{4}$

or $d_1^2 + d_2^2 = 14884$

$(d_1 + d_2)^2 = 14884 + 2 \times 2640$

$(d_1 + d_2)^2 = 20164$

$(d_1 + d_2)^2 = (142)^2$

$\boxed{d_1 + d_2 = 142 \text{ cm}}$

59. A rectangle having 10 cm and 6 cm as its dimensions was reconstructed to make a rhombus having the same perimeter as that of the rectangle and 60° as one of its angles. The area of the rhombus, in cm^2, is:

(a) $24\sqrt{3}$ (b) $8\sqrt{3}$
(c) $16\sqrt{3}$ (d) $32\sqrt{3}$

RRB ALP & Tec. (29-08-18 Shift-I)

Ans : (d) In following question–

Perimeter of rectangle = 2 × (length × breadth)

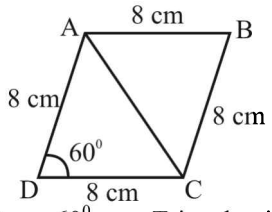
= 2 × (10 + 6) = 32 cm

here perimeter of rectangle

= Perimeter of rhombus= 32 cm

Perimeter of rhombus = 4 × side = 32 cm

or length of each side of rhombus = 8 cm.



$\because$ here is $\angle D = 60^0$ so Triangle 'ADC' will be equilateral triangle.

Area of ΔADC $= \left(\frac{\sqrt{3}}{4} a^2\right)$

$= \frac{\sqrt{3}}{4} \times 8^2 \Rightarrow \frac{64\sqrt{3}}{4} \Rightarrow 16\sqrt{3} \text{cm}^2$

Now area of rhombus will be

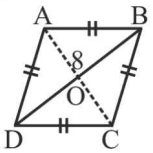
$\Rightarrow 2 \times 16\sqrt{3} = 32\sqrt{3} \text{cm}^2$

60. The area of a rhombus is 24 m^2 and the length of one of its diagonals is 8m. The length of each side of the rhombus will be:

(a) 5 m (b) 10 m
(c) 4 m (d) 6m

RRB ALP & Tec. (21-08-18 Shift-III)

Ans : (a) In rhombus ABCD
diagonal AC = x meter
BD = 8 meter



$$\text{area} = \frac{1}{2} \times \text{product of diagonal}$$

$$24 = \frac{1}{2} \times 8 \times x$$

x = 6 m.

$AO = \frac{1}{2}AC$ $\quad$ $BO = \frac{1}{2}BD$

$AO = \frac{1}{2} \times 6 = 3$ m $\quad$ $BO = \frac{1}{2} \times 8 = 4$ m

in right angle ΔAOB

$(AB)^2 = (OA)^2 + (OB)^2$

$(AB)^2 = 3^2 + 4^2$

$AB = \sqrt{25}$

AB = 5 m

side of rhombus AB = 5 m.

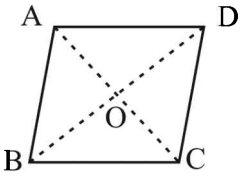
61. Find the length of one side of a rhombus whose area is 24 cm^2 and the sum of the length of its diagonals is 14 cm.

(a) 3 cm (b) 6 cm
(c) 4 cm (d) 5 cm

RRB ALP & Tec. (17-08-18 Shift-II)

Ans : (d) Let the length of diagonal AC = x cm.

$\therefore$ length of diagonal BD = (14 – x)



$$\text{area of rhombus} = \frac{1}{2} \times AC \times BD$$

$$24 = \frac{1}{2} \times x \times (14 - x)$$

$$48 = 14x - x^2$$

$$\Rightarrow x^2 - 14x + 48 = 0$$

$$x^2 - 8x - 6x + 48 = 0$$

$$(x-8)(x-6) = 0 \Rightarrow x = 8, 6$$

In right angle ΔBOC — $BC^2 = BO^2 + CO^2$

$$BC^2 = \left(\frac{BD}{2}\right)^2 + \left(\frac{AC}{2}\right)^2$$

$$BC^2 = \left(\frac{8}{2}\right)^2 + \left(\frac{6}{2}\right)^2 = 16 + 9 = 25$$

BC = 5 cm.

so length of side of rhombus = 5 cm

62. The length of each side of a rhombus is 25 m and the length of one of its diagonals is 14 m. Find the area of the rhombus

(a) 175 m^2 (b) 336 m^2
(c) 168 m^2 (d) 350 m^2

RRB ALP & Tec. (14-08-18 Shift-III)

Ans : (b) length of side of rhombus

$$= \frac{1}{2}\sqrt{d_1^2 + d_2^2}$$

where d_1, d_2 is diagonal

$$\therefore \quad 25 = \frac{1}{2}\sqrt{14^2 + d_2^2}$$

$$\therefore \quad 50 \times 50 = 14^2 + d_2^2$$

$$d_2^2 = 2500 - 196$$

$$d_2^2 = 2304$$

$$\therefore \quad d_2 = 48 \text{ m}$$

So area of rhombus $= \frac{1}{2} d_1 \times d_2 = \frac{1}{2} \times 14 \times 48$

= 336 Square meter

63. The length of one side of a rhombus is 41 cm and its area is 720 cm^2. What is the sum of the lengths of its diagonals ?

(a) 82 cm (b) 90 cm
(c) 98 cm (d) 80 cm

RRB ALP & Tec. (13-08-18 Shift-I)

Ans : (c) Area of rhombus $= \frac{\text{Product of diagonal}}{2}$

product of diagonal $(d_1 \times d_2) = 2 \times 720 = 1440$ cm^2

$$\because \text{side } (a) = \sqrt{\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2}$$

$$\Rightarrow (41)^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$$

$$\Rightarrow 41 \times 41 \times 4 = d_1^2 + d_2^2$$

$$d_1^2 + d_2^2 = 6724$$

$$(d_1 + d_2)^2 = d_1^2 + d_2^2 + 2d_1 \times d_2$$

$$(d_1 + d_2)^2 = 6724 + 2 \times 1440 \Rightarrow 9604 \text{ cm}^2$$

or $\boxed{d_1 + d_2 = 98 \text{ cm}}$

64. The length of one side of a rhombus is 13 cm and one of the diagonals is 10 cm. What is the length of the other diagonal ?

(a) 24 cm (b) 23 cm
(c) 25 cm (d) 22 cm

RRB ALP & Tec. (13-08-18 Shift-I)

Ans : (a) Side of rhombus (a) = 13 cm
First diagonal (d_1)=10 cm
Second diagonal (d_2)=?

$$\text{Side of rhombus}(a) = \sqrt{\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2}$$

$$\Rightarrow 13 = \sqrt{\left(\frac{10}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2}$$

$$\Rightarrow 169 = 25 + \frac{d_2^2}{4}$$

$$\Rightarrow 144 \times 4 = d_2^2$$

$$\Rightarrow \boxed{d_2 = 24 \text{ cm}}$$

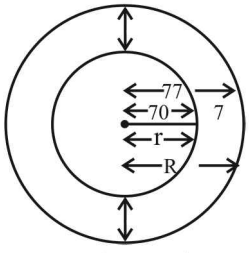
Type - 3

65. The area of a circle is 616 sq.m. Find its diameter. (π = 22/7)
(a) 7 m (b) 14 m
(c) 28 m (d) 56 m
RRB RPF SI – 13/01/2019 (Shift-III)

Ans : (c) $\because$ Area of circle = 616 sq.m.
$\pi r^2 = 616$
$r^2 = \frac{616 \times 7}{22}$, $r^2 = 196$
r = 14m
diameter = 2×14
diameter = 28m.

66. Find the area of the circular path, which is formed around a circle of circumference 440 m and whose width is 7 m.
(a) 3856 sq.m. (b) 3234 sq.m.
(c) 3900 sq.m. (d) 3204 sq.m.
RRB JE - 27/05/2019 (Shift-II)

Ans : (b) Circumference of a circle = $2\pi r$
$2\pi r = 440$
$2 \times \frac{22}{7} \times r = 440$
$r = \frac{440 \times 7}{2 \times 22}$
r = 70 m
Let r is the radius of the circle and the radius of the circle with circular Path is R.



area of circular track $= \pi\left(R^2 - r^2\right)$
$= \pi\left[(77)^2 - (70)^2\right]$
$= \pi[5929 - 4900]$
$= \pi \times 1029$
$= \frac{22}{7} \times 1029$
= 3234 Square meter

67. What is the circumference of a semicircle which diameter is 28 cm?
(a) 36 cm. (b) 144 cm.
(c) 72 cm. (d) 44 cm.
RRB RPF Constable – 24/01/2019 (Shift-I)

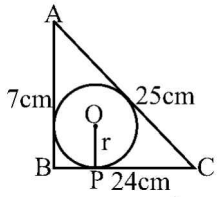
Ans : (c) diameter = 28 cm,
radius = $\frac{\text{diameter}}{2} = \frac{28}{2} = 14$ cm
circumference of a semi circle = $\pi r + 2r = r(2+\pi)$
$= 14\left(2 + \frac{22}{7}\right) = 14 \times \frac{36}{7} = 72$ cm.

68. The radius of a circle is increased by 5%. Find the percentage increase in its area.
(a) 10.25% (b) 21.5%
(c) 10.5% (d) 25%
RRB JE - 28/05/2019 (Shift-I)

Ans : (a) Let the radius of circle = x
increase % in area of the circle
$= \left(2x + \frac{x^2}{100}\right)\%$
$= \left(2 \times 5 + \frac{(5)^2}{100}\right)\%$
$= (10 + .25)\%$
$= 10.25\%$

69. What will be the radius of inner circle of triangle whose sides are 7 cm, 24 cm and 25 cm?
(a) 4 cm (b) 2.5 cm
(c) 3.5 cm (d) 3 cm
RRB RPF SI – 12/01/2019 (Shift-I)

Ans : (d) The sides of the given triangle form a right angle triangle.
Then,



area of Triangle $(\Delta) = \frac{1}{2} \times 24 \times 7 = 84$ cm^2
semi perimeter (s) $= \frac{1}{2}[24 + 25 + 7] = 28$
Hence radius of inner circle (r) $= \frac{\Delta}{s} = \frac{84}{28} = 3$cm

70. The cost of installing a paved floor in a circular room is 1540 Rs. at the rate of 10 Rs. per square meter. What is the cost of fencing at the rate of 6 Rs. per meter?

(a) Rs. 260 (b) Rs. 264
(c) Rs. 250 (d) Rs. 265

RRB Group-D – 24/10/2018 (Shift-I)

Ans : (b)
Area of the circle = πr^2
$\because$ 10 Rs. per square meter cost = Rs. 1540
$\therefore$ Area of circle $= \frac{1540}{10} = 154$

$\pi r^2 = 154$

$r^2 = \frac{154 \times 7}{22}$

$r^2 = 49$

$r = 7$ m

Perimeter of circle $= 2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ m

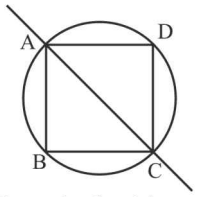
Cost of fencing at the rate of Rs. 6 per meter
$= 6 \times 44 =$ Rs. 264

71. ABCD is a quadrilateral inscribed in a circle (radius r). The bisectors of ∠DAB and ∠BCD intersect the circle at X and Y respectively. What is the length of the straight line X Y?

(a) 2r (b) (r +2)
(c) $\frac{\pi r^2}{2}$ (d) πr^2

RRB Group-D – 03/10/2018 (Shift-I)

Ans : (a)



In a cyclic quadrilateral, the bisector of the opposite angle will pass through the center.
XY = AC
and AC is the diameter of the circle
$\boxed{XY = 2r}$

72. The circumference of a circle is 15 m more then its diameter. Find the radius of the circle.

(a) 7 cm (b) 3.5 cm
(c) 4 cm (d) 8 cm

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (b) According to question-

$2\pi r - 2r = 15$

$2r(\pi - 1) = 15$

$2r\left(\frac{22}{7} - 1\right) = 15$

$2r\left(\frac{15}{7}\right) = 15$

$2r = 7$

$\boxed{r = 3.5 \text{ m.}}$

73. The inner and outer circumference of the circular ring is 22 cm and 44 cm respectively. Find the thickness of the ring.

(a) 5.5 cm (b) 1.5 cm
(c) 3.5 cm (d) 2.5 cm

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (c) Let internal radius of ring $= r_1$
and outer radius $= r_2$
as per questions –

$2\pi r_1 = 22, \quad 2\pi r_2 = 44$

$2 \times \frac{22}{7} \times r_1 = 22, \quad 2 \times \frac{22}{7} r_2 = 44$

$r_1 = \frac{7}{2}$cm, $\quad r_2 = 7$cm

thickness of ring $= r_2 - r_1$

$= 7 - \frac{7}{2} = \frac{7}{2}$cm $= 3.5$ cm

74. An outer circular path is built around a circular garden. If the outer and the inner perimeter of the path are 220 m and 44 m respectively. Find the area of the path.

(a) 3960 m^2 (b) 3696 m^2
(c) 3069 m^2 (d) 3096 m^2

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (b) Let the outer radius $= r_1$
internal radius $= r_2$
given,
circumference of circle $\Rightarrow 2\pi r_1 = 220, \quad 2\pi r_2 = 44$

$r_1 = 35$m, $\quad r_2 = 7$m

Area of the garden with the path $= 2\pi r_1^2$

$= \frac{22}{7} \times 35 \times 35 = 3850$ m^2

Area of garden $= 2\pi r_2^2 = \frac{22}{7} \times 7 \times 7 = 154$ m^2

area of path $= 3850 - 154 = 3696$ m^2

75. The area of an equilateral triangle inscribed in a circle is $9\sqrt{3}$ cm^2. What is the area of the circle?

(a) 16π (b) $12\sqrt{3}\pi$
(c) $15\sqrt{3}\pi$ (d) 12π

RRB Group-D – 05/12/2018 (Shift-I)

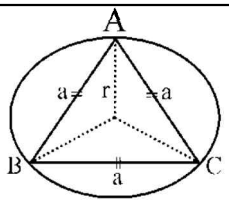
Ans : (d)

Area of equilateral $\Delta = \frac{\sqrt{3}}{4}a^2$ where a = side of Δ

$\frac{\sqrt{3}}{4} \times a^2 = 9\sqrt{3}$

$a^2 = 36$

$a = 6$



Note- radius of circumcircle of equilateral triangle with the side a $= \frac{a}{\sqrt{3}} = \frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= 2\sqrt{3}$

area of circle $= \pi r^2$

$= \pi \times 2\sqrt{3} \times 2\sqrt{3}$

Hence area of circle $= 12\pi$

76. A race ground is in the form of a ring. Its inner and outer circumference are 88 m and 154 m respectively. What is the width of the race ground?

(a) 12m (b) 15m
(c) 16.5 m (d) 10.5 m

RRB Group-D – 03/12/2018 (Shift-II)

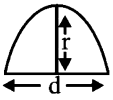
Ans : (d) Let the outer radius of the field = R
Internal radius of the field = r
Outer circumference of the field = $2\pi R$
$2\pi R = 154$
$\pi R = 77$
$R = 49/2$
Internal circumference of the field = $2\pi r$
$2\pi r = 88$
$\pi r = 44$
$r = 14$
Width of the field $= R - r = \frac{49}{2} - 14$
$= \frac{49-28}{2} = \frac{21}{2} = 10.5\text{m}$

77. If the diameter of a semicircle is 14 m. What will be the perimeter of the semicircle?

(a) 44 m (b) 22 m
(c) 36 m (d) 58 m

RRB Group-D – 03/12/2018 (Shift-III)

Ans. (c) :



$\because$ Diameter of semicircle = 14 m
$2r = 14$
$r = 7$ m
Perimeter of semicircle $= \pi r + 2r$
$= \frac{22}{7} \times 7 + 14$
$= 22 + 14 = 36$ m

78. If the diameter of the wheel of a car is 56 cm, then how many times will the wheel of the car rotate during the journey of 88 km?

(a) 500 (b) 50,000
(c) 5,000 (d) 5,00,000

RRB Group-D – 15/10/2018 (Shift-III)

Ans. (b) : Diameter of wheel = 56 cm.
$d = 56$ cm
$\therefore r = \frac{d}{2}$
$r = \frac{56}{2} = 28$ cm
Let the wheel will rotate n time in the distance of 88 km
$n \times 2\pi r = 88$ km [1 km = 100000 cm]
$n = \frac{88 \times 100000}{2 \times \frac{22}{7} \times 28}$
$n = \frac{88 \times 100000}{2 \times 88}$
$n = 50000$

79. A circular land of radius 7 m has 3.5 m wide path around it. Find the area of the path. $\left(\pi = \frac{22}{7}\right)$

(a) 202 sq. m. (b) 154 sq. m.
(c) 192.5 sq. m. (d) 346.5 sq. m.

RRB NTPC 17.01.2017 Shift-2

Ans : (c) Area of path = area with land of path – area of land.
$= \pi(7+3.5)^2 - \pi(7)^2$
$= \frac{22}{7}[(10.5-7)(10.5+7)]$
$= \frac{22}{7} \times 17.5 \times 3.5 = 192.5 \text{ m}^2$

80. If the circumference of a circle is 18π cm, then the area of the circle is:

(a) 18π sq. cm. (b) $18\pi^2$ sq. cm.
(c) 81π sq. cm. (d) 9π sq. cm.

RRB NTPC 05.04.2016 Shift-1

Ans : (c) Given-
Circumference of a circle = 18π
$2\pi r = 18\pi$
$r = \frac{18}{2}$
$r = 9$ cm
$\therefore$ area of circle $= \pi r^2$
$= \pi \times (9)^2 = 81\pi$ square cm

81. If the diameter of a circle is 7 cm then find its area.

(a) 49 cm^2 (b) 38.5 cm^2
(c) 154 cm^2 (d) 98 cm^2

RRB NTPC 04.04.2016 Shift : 2

Ans : (b) Radius of the circle (r) = $= \frac{\text{diameter}}{2}$

$= \frac{7}{2}$cm

area of circle $= \pi r^2$

$= \frac{22}{7} \times \left(\frac{7}{2}\right)^2 = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} = \frac{77}{2} = 38.5 \text{ cm}^2$

82. If the circumference of a circle is 22 cm, find the area of the semicircle.

(a) 38.5 sq.cm. (b) 19.25 sq.cm.
(c) 44 sq.cm. (d) 77 sq.cm.

RRB NTPC 03.04.2016 Shift : 2

Ans :(b) Circumference of the circle = 22 cm

$2\pi r = 22$

$r = \frac{22}{2\pi} = \frac{22 \times 7}{2 \times 22}$ $r = \frac{7}{2}$ cm

area of semicircle $= \frac{\pi r^2}{2} = \frac{22 \times 7 \times 7}{2 \times 7 \times 2 \times 2}$

$= \frac{11 \times 7}{4} = \frac{77}{4} = 19.25$ square cm.

83. When a square is made by bending a wire then the area of the square is 484 cm^2. If the same wire is bent as a circle then its area will be:

(a) 264 sq. cm. (b) 616 sq. cm.
(c) 488 sq. cm. (d) 492 sq. cm.

RRB NTPC 02.04.2016 Shift : 3

Ans : (b) Area of square = 484 square cm.

$(\text{side})^2 = 484 = (22)^2$

side = 22 cm

Perimeter of square = 4 × side = 4 × 22

= 88cm

Perimeter of circle = Perimeter of square

= 88cm

$\therefore 2\pi r = 88 \Rightarrow 2 \times \frac{22}{7} \times r = 88$

$\Rightarrow r = \frac{4 \times 7}{2} = 14$ cm

area of circle = πr^2

$= \frac{22}{7} \times (14)^2 = \frac{22}{7} \times 14 \times 14$

$= 22 \times 2 \times 14 = 616 \text{ cm}^2$

84. The largest chord of a circle measures 10 cm and the shortest chord measures 4 cm. Find the radius of the circle.

(a) 20 cm. (b) 5 cm.
(c) 8 cm. (d) 2 cm.

RRB NTPC 29.03.2016 Shift : 2

Ans : (b) diameter of circle = length of the largest chord of the circle = 10 cm

$\therefore$ radius of circle = $\frac{10}{2}$ = 5cm

85. Find the area of a circular region whose circumference is 22 cm.

(a) 22 sq. cm. (b) 11 sq. cm.
(c) 44 sq. cm. (d) 38.5 sq. cm.

RRB NTPC 29.03.2016 Shift : 2

Ans : (d) Circumference of a circle = $2\pi r$

From the question–

$2\pi r = 22$

$r = \frac{22}{2 \times \pi}$

$r = \frac{22 \times 7}{2 \times 22} \Rightarrow \boxed{r = \frac{7}{2}}$

$\therefore$ area of circle = πr^2

$= \frac{22}{7} \times \left(\frac{7}{2}\right)^2$

$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2}$

$= \frac{77}{2} = 38.5$ Square cm.

86. Find the increase in the circumference of the circle of radius 14 cm, if the radius is increased by 7cm. $\left(\pi = \frac{22}{7}\right)$

(a) 44 cm (b) 22 cm
(c) 66 cm (d) 88 cm

RRB NTPC 18.01.2017 Shift : 1

Ans : (a) Circumference of a circle of radius of 14 cm = $2\pi r$

$= 2 \times \frac{22}{7} \times 14 = 88$ cm

When increase 7 cm. in radius

Then radius = 14 + 7 = 21

Perimeter $= 2 \times \frac{22}{7} \times 21 = 132$ cm.

intended increase = 132 – 88= 44 cm

87. A piece of wire is folded and shaped into a square with a side of 44cm. Again the square is shaped into a circle. What is the radius of this circle?

(a) 108 cm. (b) 56 cm.
(c) 14 cm. (d) 28 cm.

RRB NTPC 18.01.2017 Shift : 3

Ans : (d) From the question,

side of square = 44 cm

Perimeter = 4 × side = 4 × 44 $\Rightarrow$ 176 cm

Perimeter of square = Circumference of a circle = 176 cm

$\because 2\pi r = 176\text{cm}$ $\left(\because \pi = \frac{22}{7}\right)$

Radius (r) = $\frac{176 \times 7}{22 \times 2} = 28$ cm

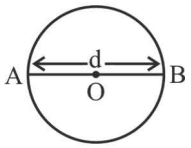
88. If the circumference of a circle is πd then what is the area of the circle?

(a) $\pi d^2/4$ (b) $2\pi d$
(c) $\pi d^2/2$ (d) πd^2

RRB NTPC 12.04.2016 Shift : 1

Ans : (a) Circumference of a circle $= \pi d$

$$2\pi r = \pi d \qquad r = \frac{d}{2}$$



$\therefore$ Area $= \pi\left(\frac{d}{2}\right)^2 = \pi\frac{d^2}{4}$

89. A cycle wheel completes 5000 revolution in 44 km. What is the radius of the wheel?

(a) 140 cm (b) 270 cm
(c) 70 cm (d) 120 cm

RRB NTPC 09.04.2016 Shift : 3

Ans : (a) Distance travelled = 44 km = 44000 m
= 4400000 cm.
revolution = 5000
distance travelled in a revolution by wheel = $2\pi r$
Hence $2\pi r \times 5000 = 4400000$

$$2 \times \frac{22}{7} \times r \times 5000 = 4400000$$

$$r = \frac{4400000 \times 7}{2 \times 22 \times 5000}, \qquad r = 140\,\text{cm}.$$

90. What is the cost of the levelling a circular ground of 28 meters diameter at the rate of 125 Rs. per square meter? $(\pi = 22/7)$

(a) Rs. 76,000 (b) Rs. 76,400
(c) Rs. 76,800 (d) Rs. 77,000

RRB NTPC 06.04.2016 Shift : 1

Ans : (d) Radius of circular field $(r) = \frac{28}{2} = 14\,\text{m}.$

$$\text{area} = \pi r^2 = \frac{22}{7} \times 14 \times 14 = 616\,\text{m}^2$$

cost on 1 m^2 = Rs.125
cost on 616 m^2 = 125×616 = Rs. 77,000

91. The diameter of a wheel is 84 cm. How many rounds will it take to cover a distance of 792 meters? (π = 22/7)

(a) 298 (b) 300
(c) 312 (d) 256

RRB NTPC 19.01.2017 Shift : 2

Ans (b) diameter = 84 cm.
r = 42 cm.

circumference of circle $= 2\pi r = 2 \times \frac{22}{7} \times 42 = 264\,\text{cm}$

distance travelled by in 1 round = 264 cm.
Number of revolution in distance of 729 m.

$$\Rightarrow \frac{79200}{264} = 300$$

92. If the radius (r) of a circle is increased by x units, how many units will increase in its circumference?

(a) π (b) 2π
(c) 2πr (d) 2πx

RRB NTPC 27.04.2016 Shift : 2

Ans : (d)
radius of circle = r
increases radius of circle = r + x
increase in circumference of circle
$= 2\pi(r + x) - 2\pi r$
increase in circumference $= 2\pi(r + x - r) = 2\pi x$

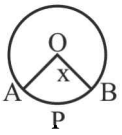
93. A square with a side of 120π cm is made from a piece of a wire. If a circle is made by bending the same wire, what will be the radius of the circle?

(a) 30 cm (b) 120 cm
(c) 240 cm (d) 60 cm

RRB NTPC 28.04.2016 Shift : 3

Ans :(c) Side of square = 120 π cm
Perimeter of square = 4 × 120π = 480π cm
∵ A circle made by folding wire
Hence it will be equal as perimeter of square and circumference of circle.
$\Rightarrow 480\pi = 2\pi r$ {r = radius of circle}
$\Rightarrow r = \frac{480}{2}$ r = 240 cm

94. In figure 'O' is the centre of a circle. The area of sector OAPB is $\frac{5}{18}$ of the area of the circle. Find x.



(a) 120 degrees (b) 100 degrees
(c) 125 degrees (d) 115 degrees

RRB ALP & Tec. (31-08-18 Shift-III)

Ans : (b) According to question $\frac{\pi r^2 x}{360^\circ} = \pi r^2 \times \frac{5}{18}$

$$\frac{x}{20^\circ} = 5$$

$$x = 100^\circ$$

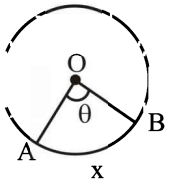
95. The length of the arc is $\frac{2}{9}$ times of the circumference of the circle. What is the measure of the angle (in degrees) subtended by the arc at the centre of the circle :

(a) 50 (b) 80
(c) 60 (d) 30

RRB ALP & Tec. (21-08-18 Shift-I)

Ans : (b) Let the arc in circle is = A×B

or Arc $(A \times B) = \frac{2\pi r\theta}{360^0}$



According to question $\frac{2\pi r\theta}{360^0} = 2\pi r \times \frac{2}{9}$

($\because$ $2\pi r$ = Circumference of Circle)

$\frac{\theta}{360^0} = \frac{2}{9}$ $\theta = 80^0$

At the centre of circle the angle made by Arc

$(\theta) = \angle AOB = 80^0$

96. A copper wire when bent in the form of a square encloses an area of 121 cm^2. If the same wire is bent into the form of a circle, find the area of the circle: (Use $\pi = \frac{22}{7}$)

(a) 154 cm^2 (b) 153 cm^2

(c) 155 cm^2 (d) 150 cm^2

RRB ALP & Tec. (09-08-18 Shift-III)

Ans : (a) area of square = a^2

According to the question,

a^2 = 121 cm^2

$\therefore$ a = 11 cm

$\because$ circumference of square = 4a = length of wire

$\therefore$ length of wire = 11 × 4 = 44 cm

circumference of circle = $2\pi r$

circumference of circle = perimeter of square

$\therefore 2\pi r = 44$

$\therefore$ r = 7 cm

$\therefore$ Area of circle = $\pi r^2 = \frac{22}{7} \times (7)^2$ = 154 cm^2

Type - 4

97. The sum of the lengths of the edges of a cube is equal to four times the perimeter of a square. If one-fourth of the numerical value of the volume of the cube is equal to the numerical value of the area of the square, then the length of the side of the square is:

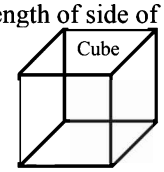
(a) 27 unit (b) 10.5 unit

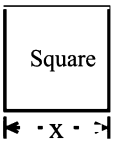
(c) $\frac{9}{4}$ unit (d) $\frac{27}{16}$ unit

RRB RPF Constable – 17/01/2019 (Shift-III)

Ans : (d) Let the length of the edge of cube = y unit

Let the length of side of square = x unit





The first condition –

Sum of length of the edge of cube

= 4 (Perimeter of square)

$12y = 4(4x)$ $\left[\begin{array}{l}\because \text{ edges of cube} = 12 \\ \text{Perimeter of square} = 4x\end{array}\right]$

$12y = 16x$

$3y = 4x$

$\boxed{y = \frac{4}{3}x}$ ..... (1)

second condition –

$\frac{\text{numerical value of volume of cube}}{4}$ = area of square

$\frac{y^3}{4} = x^2$

$y^3 = 4x^2$ ..... (2)

Putting the value of y from equation (1) in equation (2)

$\left(\frac{4}{3}x\right)^3 = 4x^2$

$\frac{64}{27}x^3 = 4x^2$

$\frac{16}{27}x = 1$

$\boxed{x = \frac{27}{16}}$

Hence side of square $= \frac{27}{16}$ unit

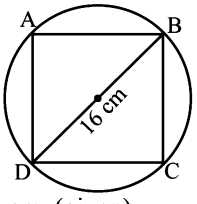
98. Find the area of a square made by touching its circumference inside a circle of radius 8cm.

(a) 144 cm^2 (b) 128 cm^2

(c) 64 cm^2 (d) 136 cm^2

RRB JE - 23/05/2019 (Shift-I)

Ans : (b)



radius of circle = 8 cm, (given)

diameter of circle = diagonal of square (According to question)

$8 \times 2 = \text{side}\left(\sqrt{2}\right)$

$\text{side} = 8\sqrt{2}$

Area of square = (side)^2

Area of square $= \left(8\sqrt{2}\right)^2 = 128$ cm^2

99. Find the embodied area between a circle and a square with side 'a' inscribed in it.

(a) $(a^2/2)(\pi - 2)$ sq. unit
(b) $2a^2(\pi - 2)$ sq. unit
(c) $a^2(2 - \pi)$ sq. unit
(d) $a^2(2\pi - 1)$ sq. unit

RRB JE - 26/05/2019 (Shift-I)

Ans : (a) Area of circle $= \pi R^2$

$$= \pi \times \left(\frac{a\sqrt{2}}{2}\right)^2$$

$$= \pi \times \frac{a^2 \times 2}{4} = \frac{2\pi a^2}{4} = \frac{\pi a^2}{2}$$

Area of square $= (\text{side})^2 = (a^2) = a^2$

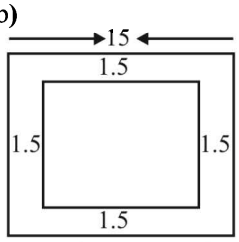
$\therefore$ Contiguous area $= \frac{\pi a^2}{2} - a^2 = \frac{a^2}{2}(\pi - 2)$

100. A room is 15 feet long and 12 feet wide. A mat is to be laid on the floor of the room leaving a space of 1.5 feet from the walls. What is the cost of the mat at the rate of Rs. 3.50 per square feet?

(a) Rs. 630 (b) Rs. 378
(c) Rs. 472.50 (d) Rs. 496

RRB JE - 30/05/2019 (Shift-II)

Ans : (b)



According to the figure left at 1.5 feet

length $= 15 - (2 \times 1.5) = 12$ feet

breadth $= 12 - (2 \times 1.5) = 9$ feet

$\therefore$ Area of room $= 12 \times 9$

$= 108$ Square feet

$\because$ Cost of 1 square feet $=$ Rs. 3.50

$\therefore$ Cost of 108 square feet $= 3.50 \times 108$

$=$ Rs. 378

101. The area of a square field is 24200 sqm. How long time will it take for a person walking at a speed of 4.4 km/hr to cross it diagonally?

(a) 3 minute (b) 4 minute
(c) 2.5 minute (d) 2 minute

RRB JE - 01/06/2019 (Shift-I)

Ans : (a) As per question,

Area of square $= 24200\ m^2$

$a^2 = 24200\ m^2$

$a = 110\sqrt{2}$ m

distance travelled diagonally by person

$= a\sqrt{2} = 110\sqrt{2} \times \sqrt{2} = 220$ m

$\because$ Speed $= 4.4$ km/hour $= \frac{4.4 \times 1000}{60}$ m/minute

$= \frac{220}{3}$ m/minute

So time $= \frac{\text{distance}}{\text{Speed}} = \frac{220}{220/3} = 3$ minute

102. The difference between the areas of the two squares is $32cm^2$. If the difference of their sides is 4cm. Find the value of the sides of both squares.

(a) 6 cm., 2 cm. (b) 12 cm., 8 cm.
(c) 4 cm., 4 cm. (d) 4 cm., 2 cm.

RRB JE - 01/06/2019 (Shift-III)

Ans. (a) Let sides of squares is x, y

area of first square $= x^2$

area of second square $= y^2$

$x^2 - y^2 = 32$ ..... (i)

$x - y = 4$ ..... (ii)

from equation (i)–

$(x-y)(x+y) = 32$

$4(x+y) = 32$

$x + y = 8$ ..... (iii)

from equation (ii) and (iii)–

$x + y = 8$

$x - y = 4$

$2x = 12$

$x = 6$

putting the value of x in equation (ii)

$x - y = 4$

$6 - y = 4$

$y = 2$

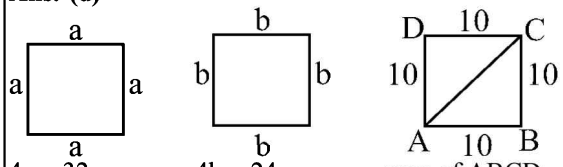
So, sides of squares is 6 cm and 2cm.

103. The perimeter of a square is 32m and the other is 24m. Find the diagonal of the square whose area is equal to the area of these two squares.

(a) $2\sqrt{10}$m (b) 20m
(c) $10\sqrt{4}$m (d) $10\sqrt{2}$m

RRB RPF SI – 16/01/2019 (Shift-I)

Ans. (d)



$4a = 32$ m
$a = 8$ m
$a^2 = 64$,

$4b = 24$
$b = 6$ m
$b^2 = 36$,

area of ABCD
Square $= 64 + 36$
$AB \times AB = 100\ m^2$
$AB = \sqrt{100}$
$AB = 10$ m
$\{\because AB = AC\}$

length of diagonal (AC) $= \sqrt{2}a$

$= \sqrt{2} \times 10 = 10\sqrt{2}$ m

104. The sum of the lengths of the cores of the cube is the $\frac{3}{5}$th of the perimeter of the square. If the numerical value of the volume of the cube is equal to the numerical value of the area of the square, then find the perimeter of the square.

(a) 500 unit (b) 360 unit
(c) 480 unit (d) 300 unit

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (a) : If length of each core of cube = a unit

$\therefore$ Sum of length core's in cube = 12 a unit

If each side of square = b, then perimeter = 4b unit

In first condition,

$$12a = \frac{3}{5} \times 4b$$

$$\text{or } a = \frac{b}{5}$$

then According to second condition, $a^3 = b^2$

$$\text{or } \left(\frac{b}{5}\right)^3 = b^2$$

$$\text{or } \frac{b^3}{125} = b^2$$

or, $\boxed{b = 125}$

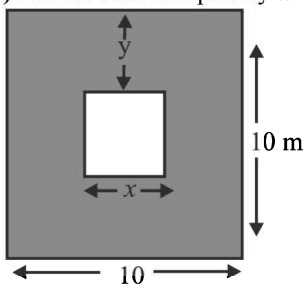
Now perimeter of square $= 4b = 4 \times 125 = 500$ unit

105. The middle area of a square room of 10 m sides is covered with square tank carpet and the remaining floor is covered with oil cloth. The carpet and the oil cloth are priced at Rs. 15 and Rs. 6.5 respectively, and their total price is Rs. 1338.50. What will be the width of the oil cloth border?

(a) 2 m (b) 5 m
(c) 1 m (d) $\frac{1}{2}$m

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (d) Let the side of squarely tank = x m.



$\therefore$ Area of square shape tank (which cover with carpet) $= x^2$

Area of oil cloth's = $10 \times 10 - x^2$
$= 100 - x^2$

$\therefore$ $15x^2 + 6.50\,(100 - x^2) = 1338.50$
$15x^2 + 650 - 6.5\,x^2 = 1338.50$
$8.5\,x^2 = 688.50$

$$x^2 = \frac{688.5}{8.5}$$

$x^2 = 81$
$x = 9$

$\therefore$ Breadth of oil cloth's border $= \frac{10 - x}{2}$

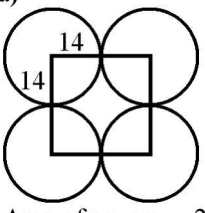
$$= \frac{10 - 9}{2} = \frac{1}{2} \text{ m}$$

106. Four equal circles are formed on the four vertices of a square in such a way that each circle touches two other circles. What will be the area outside the perimeter of the circles towards the middle of the square. If the length of each side of the square is 28 cm?

(a) 168 cm^2 (b) 40 cm^2
(c) 42 cm^2 (d) 32 cm^2

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (a)



Area of square = 28^2 = 784 cm^2

So area of Four radius section

$$= 4 \times \frac{\theta}{360} \pi r^2$$

$$= 4 \times \frac{90}{360} \times \frac{22}{7} \times 14 \times 14$$

= 616

perimeter of middle of square = area of square – area of radius section

= 784 - 616 = 168 cm

107. The area of the square field is 31684 m^2 on which the wire has to be tied at 1, 2, 3 and 4 m above the ground. If the length required for each wire is 5% more than the circumference of the field, then what length of the wire is required?

(a) 2090 m (b) 2099 m
(c) 2909 m (d) 2990.4 m

RRB Group-D – 22/09/2018 (Shift-III)

Ans. (d) : Let the side of square = a m

area of square (a^2) = 31684 m^2

a = 178 m

Circumference of square = 4a

= 4 × 178 = 712 m

as per question-

$$\text{Length of 1 wire} = 712 \times \frac{105}{100} = 747.60$$

Total length of four wire = 747.60 × 4

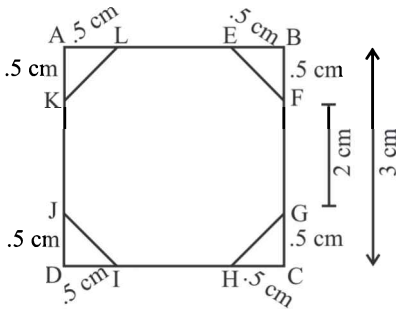
= 2990.40 m

108. A 0.5 cm line segment was cut from each corner of a square with an edge of 3 cm, the cut section has a vertex. What is the area of the octagon so formed?

(a) $10\sqrt{2}$cm and 8 cm^2
(b) 8 cm and 8 cm^2
(c) $8\sqrt{2}$cm and 8 cm^2
(d) $\left(8+2\sqrt{2}\right)$cm and 8.5 cm^2

RRB Group-D – 10/10/2018 (Shift-I)

Ans : (d)



$$EF = GH = IJ = KL = \sqrt{\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2}$$

$$=\frac{1}{\sqrt{2}}$$

Perimeter of octagon= LE+EF+FG+GH+HI+IJ+JK+KL

$$=2+\frac{1}{\sqrt{2}}+2+\frac{1}{\sqrt{2}}+2+\frac{1}{\sqrt{2}}+2+\frac{1}{\sqrt{2}}$$

$$=8+\frac{4}{\sqrt{2}}$$

$$=\left(8+2\sqrt{2}\right)\text{cm}$$

Area of octagon = area of square − 4 × are of triangular Part

$$=(3)^2-4\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}$$

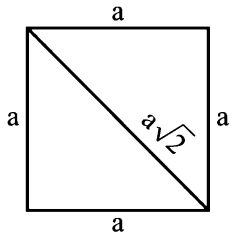
$$= 8.5\ cm^2$$

109. The numerical value of the area of a square is equal to half the numerical value of each of its diagonals. What is the numerical value of diagonal?

(a) 1 (b) $\sqrt{2}$
(c) 2 (d) $\frac{\sqrt{2}}{2}$

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (a)



As per question,
Area of square = half of the diagonal of square

$$a^2 = a\sqrt{2}\times\frac{1}{2}$$

$$a=\frac{1}{\sqrt{2}}$$

diagonal of square $= a\sqrt{2}$

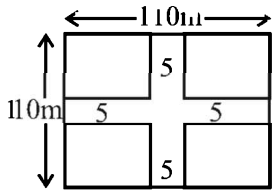
$$=\frac{1}{\sqrt{2}}\times\sqrt{2}=1$$

110. The side of a square field is 110 m. Two routes 5 m wide pass through the middle of the square field parallel to the sides, intersects each other. Find the area of the routes.

(a) 1000 m^2 (b) 1100 m^2
(c) 1075 m^2 (d) 975 m^2

RRB Group-D – 27/11/2018 (Shift-III)

Ans. (c)



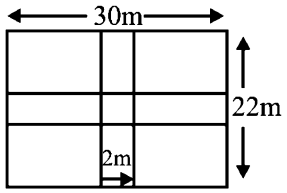
Area of square shape = 110 × 110 = 12100 m^2
Remaining area of after road = 105 × 105 = 11025 m^2
So area of road = 12100 – 11025 = 1075 m^2

111. A rectangular park measuring 30 m × 22 m has two footpath 2m wide. One from north to south and the other from east to west and they both intersect each other in the middle of the park. If the cost of construction of the road is Rs. 15 per square meter, then calculate the total cost of construction to the road.

(a) Rs. 1545 (b) Rs. 1560
(c) Rs. 1490 (d) Rs. 1500

RRB Group-D – 12/11/2018 (Shift-I)

Ans. (d)



area of path = 30 × 2 + 22 × 2 – 2 × 2 = 100 square meter
total cost of 1 square meter = Rs. 15
100 square meter = 15 × 100 = Rs. 1500

112. The sum of the length of the cores of a cube is equal to twice the perimeter of a square. If the numerical value of the volume of the cube is equal to the numerical value of the area of the square then what is the area of the square?

(a) 10.5 unit (b) 27 unit
(c) 13.5 unit (d) 12.5 unit

RRB Group-D – 12/11/2018 (Shift-III)

Ans : (c) Let the length of core of cube = a unit
and length of side of square = b unit
∵ Number of core of cube = 12

As per question,

$12a = (4b) \times 2$

$12a = 8b$

$a = \frac{2}{3}b$ unit

Volume of cube = area of square

$a^3 = b^2$

$\left(\frac{2}{3} \times b\right)^3 = b^2$

$\therefore b = \frac{27}{8}$ unit

So perimeter of square = 4× side = 4 × $b = 4 \times \frac{27}{8} = 13.5$ unit

113. What will be the edge of the cube formed by melting the cubes with edges of 4 cm, 8 cm and 10 cm?

(a) 13 (b) 11
(c) 12 (d) 14

RRB NTPC 04.04.2016 Shift : 3

Ans : (c) $\because$ Volume of cube $= a^3$

Volume of three cubes $= (6)^3 + (8)^3 + (10)^3$

$= 216 + 512 + 1000 = 1728\ cm^3$

side of new cube $= (1728)^{1/3} = (12^3)^{1/3} = 12$ cm

114. A wire is in the shape of a rectangle. Its length and width are 42.7 m and 21.8 m respectively. If this wire is bent again into a square shape, then what will be the measure of the side of the square?

(a) 16.125 (b) 32.25
(c) 11.35 (d) 22.70

RRB NTPC 03.04.2016 Shift : 1

Ans : (b) Perimeter of square = Perimeter of rectangle

side × 4 = 2 (length + breadth)

side $= \frac{2}{4}(42.7 + 21.8) = \frac{1}{2}(64.5)$

Side of square = 32.25 m

115. If the length of the diagonal of a square is $13\sqrt{2}$ unit then what will be its area?

(a) 104 sq. unit (b) 169 sq. unit
(c) 338 sq. unit (d) 676 sq. unit

RRB NTPC 29.03.2016 Shift : 3

Ans : (b) Length of diagonal $= 13\sqrt{2}$ unit

$a\sqrt{2} = 13\sqrt{2}$ $a = 13$

$\therefore$ Area of square $= a^2 = (13)^2 = 169$ Square unit.

116. The length of the diagonal of a square is 26 cm, find the side (in cm) of the square.

(a) $13\sqrt{2}$ (b) $\pm 13\sqrt{2}$
(c) 13 (d) $26\sqrt{2}$

RRB NTPC 11.04.2016 Shift : 3

Ans : (a) Length of diagonal of square = $a\sqrt{2} = 26$ cm

$a = \frac{26}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$ side of square (a) $= 13\sqrt{2}$ cm.

117. The length of the diagonal of a square is 24 cm, find the side in cm of the square

(a) $12\sqrt{2}$ (b) $\pm 12\sqrt{2}$
(c) 12 (d) $24\sqrt{2}$

RRB NTPC 07.04.2016 Shift : 1

Ans : (a) Length of diagonal of square =side $\times\sqrt{2} = 24$

side $= \frac{24}{\sqrt{2}}$

side $= 12\sqrt{2}$

118. The order of a rotational symmetry of a square is:

(a) 2 (b) 6
(c) 4 (d) 8

RRB NTPC 07.04.2016 Shift : 3

Ans : (c) The four sides of the square are equal and each angle is 90°. Hence is rational symmetry will be 4 in order.

119. Find the area of a square whose diagonal is half of 12 cm.

(a) 18 sq.cm. (b) 64 sq.cm.
(c) 36 sq.cm. (d) 72 sq.cm.

RRB NTPC 19.01.2017 Shift : 2

Ans : (a) Diagonal of square $= a\sqrt{2}$ (where a = side)

as per question –

$\frac{12}{2} = a\sqrt{2}$

$a = \frac{6}{\sqrt{2}}$

$a = \frac{6\sqrt{2}}{2}$ $\boxed{a = 3\sqrt{2}}$

area of square $= a^2$

$= (3\sqrt{2})^2 = 18$ square cm

120. Find the area of a square which is inscribed in a circle of radius of 13 cm.

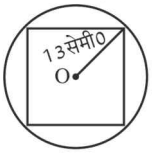
(a) 169 sq.cm. (b) 338 sq.cm.
(c) 507 sq.cm. (d) 676 sq.cm.

RRB NTPC 26.04.2016 Shift : 1

Ans : (b) diagonal of square = 2 × radius

= 2 × 13 = 26 cm

$\therefore$ side of square $= \frac{\text{diagonal of square}}{\sqrt{2}}$



$= \frac{26}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = 13\sqrt{2}$ cm.

Area of square $= (13\sqrt{2})^2 = 169 \times 2 = 338\ cm^2$

121. The area of a square field is 313600 m^2. How long will it take for a woman to cross the field diagonally at the rate of $4\sqrt{2}$ m/s.?

(a) 3 minute (b) 2 minute 20 sec.
(c) 2 minute 40 sec. (d) 3 minute 10 sec.

RRB NTPC 28.04.2016 Shift : 3

Ans : (b) Given–
Area of square field = 313600 square meter
$(\text{side})^2 = 313600$
$\text{side} = \sqrt{313600} = 560$ meter
$\therefore$ diagonal $= \text{side} \times \sqrt{2} = 560\sqrt{2}$ meter (distance)
Speed $= 4\sqrt{2}$ meter/second
Hence time taken to cross the diagonal field = distance/speed
$= \frac{560\sqrt{2}}{4\sqrt{2}} = 140$ second = 2 minute 20 second

122. The sum of the lengths of the core of a cube is equal to half the perimeter of a square. If the numerical value of the volume of the cube is equal to one-sixth of the numerical value of the area of the square, then the length of one side of the square is :

(a) 18 units (b) 36 units
(c) 31.5 units (d) 27 units

RRB ALP & Tec. (31-08-18 Shift-II)

Ans. (b) : Let the core of cube = y
and side of square = x
as per question-

$$12y = \frac{4x}{2}$$

$$y = \frac{1}{6}x \quad \text{.....(i)}$$

as per second condition-

$$y^3 = \frac{1}{6}x^2$$

$$\left(\frac{1}{6}x\right)^3 = \frac{1}{6}x^2 \text{ from equation (i)}$$

$$\frac{1}{216}x^3 = \frac{1}{6}x^2$$

$$x = \frac{216}{6}$$

$$x = 36$$

side of square (x) = 36 unit

123. The sum of the lengths of the core of a cube is equal to half the perimeter of a square. If the numerical value of the volume of the cube is equal to the numerical value of the area of the square, then the length of one side of the square is:

(a) 108 units (b) 36 units
(c) 216 units (d) 288 units

RRB ALP & Tec. (29-08-18 Shift-I)

Ans : (c) If length of each core of cube = a_1
and length of each side of square is = a_2 then –

$$\text{Sum of core's length of cube} = \frac{\text{Perimeter of square}}{2}$$

$$12a_1 = \frac{4a_2}{2}$$

$$\text{or } 12a_1 = 2a_2$$

$$\boxed{a_1 = \frac{a_2}{6}}$$

$$\text{Now volume of cube} = (a_1)^3 = \left(\frac{a_2}{6}\right)^3 = \frac{a_2^3}{216}$$

$$\text{Hence as per question, } \frac{a_2^3}{216} = a_2^2$$

$$\text{or, } \boxed{a_2 = 216 \text{ unit}}$$

Hence length of intended side = 216 unit

124. The area of the square field is 196 sq.m. Its each side is:

(a) 16 m (b) 17 m
(c) 14 m (d) 13 m

RRB ALP & Tec. (10-08-18 Shift-II)

Ans : (c)
area of the square field = $(\text{side})^2$
$196 = (\text{side})^2$
Side $= \sqrt{196} = \sqrt{14 \times 14}$
Side = 14 m

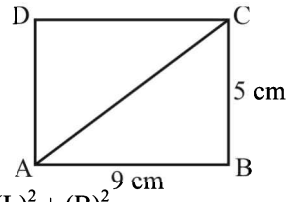
Type - 5

125. Find the length of the diagonal (in cm) of a rectangle whose length and breadth are 9 cm and 5 cm respectively.

(a) $\sqrt{106}$ (b) $\pm\sqrt{106}$
(c) $2\sqrt{14}$ (d) $\pm 2\sqrt{14}$

RRB RPF SI – 10/01/2019 (Shift-II)

Ans : (a)



$(\text{diagonal})^2 = (L)^2 + (B)^2$
$(AC)^2 = 9^2 + 5^2$
$(AC)^2 = 81 + 25$
$(AC)^2 = 106$ $AC = \sqrt{106}$ cm

126. The difference between the length and breadth of a rectangle is 6 m. If its perimeter is 64 m, then its area is:

(a) 256 sq. m. (b) 247 sq. m.
(c) 264 sq. m. (d) 238 sq. m.

RRB RPF Constable – 19/01/2019 (Shift-III)

Ans : (b) Let the length of rectangle = x m

Breadth = (x –6) m

∵ Perimeter = 64 m

2(x + x – 6) = 64

2x – 6 = 32

2x = 38

x = 19

∴ area of rectangle = x × (x -6)

= 19 × (19 – 6)

= 19 ×13 = 247 Square meter

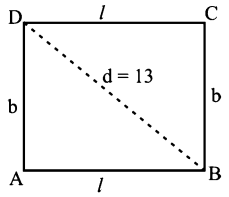
127. Suresh took 15 seconds to cross a rectangular ground diagonally at a speed of 52 m/min. and Rajesh took the same time to cross the same ground by its sides at a speed of 68 m/min. Find the area of the ground.

(a) 30 square m (b) 40 square m

(c) 50 square m (d) 60 square m

RRB JE - 26/05/2019 (Shift-II)

Ans : (d)



Distance $= 52 \times \frac{15}{60}$

$\boxed{d = 13}$

as per question,

$\ell + b = 68 \times \frac{15}{60}$

$\boxed{\ell + b = 17}$.....(i)

diagonal of rectangle (d) $= \sqrt{\ell^2 + b^2}$

$13 = \sqrt{\ell^2 + b^2}$

take both side of square

$169 = \ell^2 + b^2$

square on equation (i)

$(l+b)^2 = (17)^2$

$l^2 + b^2 + 2lb = 289$ $\quad \because [l^2 + b^2 = 169]$

169 + 2lb = 289

2lb = 289 – 169

2lb = 120

area of field $\boxed{\ell b = 60 \text{ square meter}}$

128. The perimeter of a rectangle is two times the perimeter of a square whose side is 18 units. If the breadth of the rectangle is 45 m then find its area.

(a) 1360 square unit

(b) 1050 square unit

(c) 1215 square unit

(d) 1152 square unit

RRB JE - 23/05/2019 (Shift-I)

Ans : (c) given— side of square = 18 unit

Breadth of rectangle = 45 m

Perimeter of rectangle = 2 × perimeter of square

2 (Length + breadth) = 2 × 18 × 4

2 (Length + 45) = 2 × 72

L = 72 – 45 = 27

Area = Length × breadth = 27 × 45

Area = 1215 square unit

129. What will be the cost of gardening at the rate of Rs. 10 per square meter in a 1 m wide space around a rectangular plot whose perimeter is 340 m?

(a) can not be determined

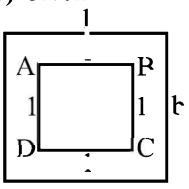
(b) 3440

(c) 3400

(d) 1700

RRB JE - 23/05/2019 (Shift-III)

Ans : (b) Given-



Perimeter of rectangular plot = 340 m

2 (l + b) = 340

(l + b) = 170 m

New perimeter = 2 (l + 1 + b +1)

= 2 (l + b + 2)

= 2 (170 + 2)

= 2 (172)

= 344 m

cost of gardening = 344 × 10 = ₹ 3440

130. The diagonal of the floor of a rectangular room is 7.5 feet. The smaller side of the room is 4.5 feet. Find the area (in sq. feet) of the room.

(a) 27 foot^2 (b) 13.5 foot^2

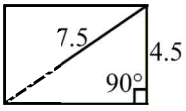
(c) 37 foot^2 (d) 5.25 foot^2

RRB RPF SI – 05/01/2019 (Shift-II)

Ans : (a) Diagonal floor of rectangular room= 7.5 foot

Small side of rectangular room (breadth)

= 4.5 foot



∴ Large of rectangular room (length)

$= \sqrt{(7.5)^2 - (4.5)^2}$

$= \sqrt{36} = 6$ foot

∴ Area of the room = $\ell \times b = 6 \times 4.5 = 27$ foot^2

131. The length of a rectangular plot is three times of its width. If the area of rectangular plot is 867 m^2, then what is the length of rectangular plot?

(a) 34 m (b) 17 m
(c) 51 m (d) 68 m

RRB JE - 25/05/2019 (Shift-II)

Ans : (c) Let the breadth of rectangular Plot (b) = x meter

Length of rectangular Plot (l) = 3x meter
area of rectangular plot = l.b
$867 = 3x \times x$
$867 = 3x^2$
$289 = x^2$
$\boxed{x = 17}$
length of rectangular plot (l) = 3x = 3 × 17
= 51 m

132. The area of a rectangular carpet is 120 m^2 and its perimeter is 46 m. Find the length of its diagonal.

(a) 15 m (b) 16 m
(c) 20 m (d) 17 m

RRB JE - 29/05/2019 (Shift-III)

Ans : (d) Area of rectangle = $\ell \times b$

$\ell \times b = 120$ ..... (i)

Perimeter of rectangle = 2 (ℓ + b)

$(\ell + b) = 23$ ..... (ii)

diagonal of rectangle $= \sqrt{l^2 + b^2}$
from equation (i) and (ii)–
$(\ell + b)^2 = \ell^2 + b^2 + 2\ell \times b$
$(23)^2 = \ell^2 + b^2 + 2 \times 120$
$529 - 240 = \ell^2 + b^2$
$289 = \ell^2 + b^2$
$\ell^2 + b^2 = 289$
diagonal = 17 m

133. A circular wire of radius 7 cm is folded to form a rectangle whose sides are in the ratio of 4:7. Find the area of the rectangle so formed.

(a) 56 cm^2 (b) 60 cm^2
(c) 112 cm^2 (d) 84 cm^2

RRB JE - 31/05/2019 (Shift-I)

Ans : (c) $2\pi r = 2\ (l + b)$

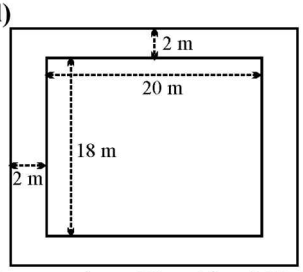
$2 \times \frac{22}{7} \times 7 = 2\ (7x + 4x)$
$22 = 11x$
$x = 2$
Area of rectangle = 7x ×4x
Area of rectangle = $28x^2 = 28 \times 4 = 112$ cm^2

134. Find the area of the jogging track which has a width of 2m and is built around a rectangle with dimension 20m × 18m.

(a) 200 m^2 (b) 140 m^2
(c) 136 m^2 (d) 168 m^2

RRB JE - 31/05/2019 (Shift-III)

Ans. (d)



Area of rectangle = 20 × 18 = 360 Square meter
Area of rectangle with jogging track =24 × 22 = 528 m^2
Area of jogging track = 528 – 360 = 168 m^2

135. Find the length (in cm) of the diagonal of a rectangle whose length and breadth are 5 cm and 6 cm respectively.

(a) $\sqrt{61}$ (b) $\pm\sqrt{61}$
(c) $\sqrt{11}$ (d) $\pm\sqrt{11}$

RRB RPF Constable – 24/01/2019 (Shift-III)

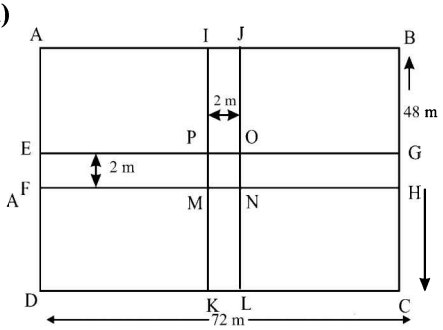
Ans : (a) diagonal of rectangle $= \sqrt{l^2 + b^2}$
$= \sqrt{5^2 + 6^2} = \sqrt{25 + 36} = \sqrt{61}$

136. In a rectangular park of 72 m × 48 m, 2m wide paths Nehru Road and Zoo Park Road intersect each other at right angles in the centre of the park. Each path is parallel to one of the dimensions of the park. What will be the cost of constructing the road at the rate of Rs. 150 per square meter?

(a) Rs. 35, 300 (b) Rs. 35, 350
(c) Rs. 35, 395 (d) Rs. 35, 400

RRB RPF SI – 12/01/2019 (Shift-I)

Ans. (d)



Area of path
= area of EFHG + Area of IJKL - Area of MNOP
= 72×2+2×48-2×2
= 2 (72+48)-4 = 2×120-4 = 236
Hence cost of path= 236×150
= ₹ 35,400

137. Shrinesh wants to upgrade his office floor, which is made of 2800 marble tiles. Each tile is 3 cm long and 5 cm wide. Calculate the total cost of polishing the floor at the rate of Rs. 25 per square meter.

(a) Rs. 100 (b) Rs. 115
(c) Rs. 125 (d) Rs. 105

RRB Group-D – 16/10/2018 (Shift-I)

Ans. (d) : Length of each tile = 3 cm.
and breadth = 5 cm.
So tile is a rectangle.
area of tiles = $3\times5cm^2 = 15cm^2$
$15cm^2 = \frac{15}{100\times100}m^2$
area of floor = $\frac{15}{100\times100}\times2800m^2$
$= 4.2m^2$
cost of polishing floor at the rate of Rs. 25 per square meter. = 4.2 × 25 = Rs.105

138. The sides of a rectangular plot of a land are given 35 m and 25 m in the nearest integer. What is the maximum value of the perimeter of a rectangular plot?
(a) 60 m (b) 122 m
(c) 120 m (d) 61 m
RRB Group-D – 29/10/2018 (Shift-III)

Ans : (b) Taking minimum integer of side of rectangular plot-
length = 35.5 m, b = 25.5 m
maximum value of perimeter of plot
= 2 (length + breadth)
$= 2(35.5+25.5)$
$= 2\times61$
= 122 m

139. The area of a rectangle is 12.46 m^2. If the length of the rectangle is 3.5 m then find the breadth (in m) of the rectangle.
(a) 3.56 (b) 3.58
(c) 3.62 (d) 3.54
RRB Group-D – 10/10/2018 (Shift-I)

Ans : (a) Given–
area of rectangle = 12.46 m^2
length (l) = 3.5 m
breadth $= \frac{12.46}{3.5m} = 3.56m$
$\boxed{b = 3.56m}$

140. The dimensions of a rectangular plot is given to 28 m and 22 m to the nearest integer. What is the minimum possible perimeter of a rectangular plot?
(a) 49 m (b) 100 m
(c) 50 m (d) 98 m
RRB Group-D – 11/10/2018 (Shift-III)

Ans : (d) Perimeter of rectangular plot = 2 (l + b)
length and breadth of rectangular plot is minimum 28 m and 22 m
Then perimeter = 2 (28 + 22)
= 2 × 50 = 100 m
minimum Perimeter will be 98 m of rectangular plot which is predecessor of 100 m.

141. The ratio of length and breadth of a rectangular field is 4:3. If the area of the field is 1452 m^2, then find the cost of making the boundary of the field at the rate of Rs. 12 per meter.
(a) Rs. 1,858 (b) Rs. 1,848
(c) Rs. 1,845 (d) Rs. 1,868
RRB Group-D – 22/10/2018 (Shift-II)

Ans : (b) Given-
length and breadth of rectangular field is 4x m and 3x m
As per question- 4x × 3x = 1452
$\Rightarrow 12x^2 = 1452$
$\Rightarrow x^2 = 121$
$\Rightarrow x = 11$ m
So, length = 4x
= 4 × 11 = 44 m
breadth = 3x = 3 × 11 = 33 m
Perimeter of rectangular field = 2 (ℓ + b)
= 2 (44 + 33)
= 2 × (77)
= 154 m
Hence cost of making boundary of rectangular field
= 154 × 12
= Rs. 1848

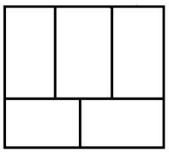
142. A farmer has a rectangular field 250 m long and 380.5 m wide. He expects to sow different types of rice which will yield 30 quintals per hectare and he will sell it in the market at Rs. 620 per quintal. Calculate his expected income.
(a) Rs. 716932.5 (b) Rs. 179632.5
(c) Rs. 176923.5 (d) Rs. 176932.5
RRB Group-D – 30/10/2018 (Shift-III)

Ans. (d) 1 hectare = 10000 square meter
income = area of field ×30 quintal × 620 quintal
$= \frac{250\times380.5\times30\times620}{10,000}$
= ₹ 176932.5

143. The side of a square flower garden is 1 m 80 cm long. It is extended by digging a 20 cm wide border from all side. Find the area of extended flower garden.
(a) 4.81 m^2 (b) 4.82 m^2
(c) 4.84 m^2 (d) 4.8 m^2
RRB Group-D – 30/10/2018 (Shift-III)

Ans. (c) It is extended by digging 20cm width from each side
Then 20 + 20 = 40 cm
Total increased part = (180 + 40) cm = 220 cm
area of square = $(side)^2$
$= (220)^2$
= 48400 cm^2
= 4.84 m^2

144. In the following figure, five congruent rectangles are given, each of which has a largest side x m. If the total area of the figure is 6000 sq.m, then find the value of x.

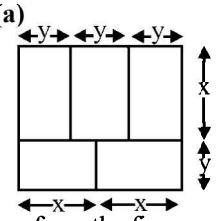


(a) $30\sqrt{2}$ (b) $40\sqrt{3}$

(c) 40 (d) $\frac{40\sqrt{2}}{3}$

RRB Group-D – 07/12/2018 (Shift-III)

Ans : (a)



It is clear from the figure

$2x = 3y$

as per question

$5xy = 6000$

$5x\frac{2x}{3} = 6000$

$x^2 = 1800 = 30\sqrt{2}$

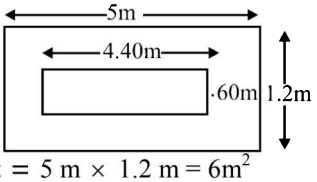
145. A carpet is 5 m long and 1.2 m wide. It has a 30 cm wide printed border on all flour sides. What will be the cost of printing the border at the rate of Rs. 225 per square meter?

(a) Rs. 854 (b) Rs. 1,027

(c) Rs. 756 (d) Rs. 902

RRB Group-D – 05/12/2018 (Shift-III)

Ans : (c)



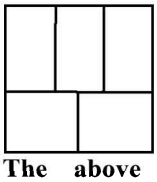
area of carpet = 5 m × 1.2 m = $6m^2$

area of remain carpet left of border = 4.40 m × 0.60 m = 2.64 m^2

area of border = 6 − 2.64 = 3.36

cost of borders print = 3.36 × 225 = Rs. 756

146.



The above rectangle has five congruent rectangles. Each of which has a largest side x m. If the total area of the figure is 7500 m^2, then find the value of x.

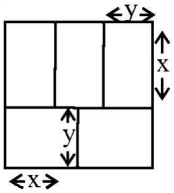
(a) $15\sqrt{10}$ (b) $10\sqrt{10}$

(c) $10\sqrt{15}$ (d) $\frac{10\sqrt{15}}{3}$

RRB Group-D – 03/12/2018 (Shift-III)

Ans. (a) : Let the small side = y m

area of large rectangle = area of five congruent rectangle



$2x \times (x + y) = 5xy$

$2x + 2y = 5y$

$2x = 3y$

$\therefore y = \frac{2}{3}x$

but

$5xy = 7500$

$5x \times \frac{2}{3}x = 7500$

$x^2 = 2250$

$x = 15\sqrt{10}$ m

147. The area of a rectangle is $2x^2 + 3x + 1$ and its length is 2x +1. Find its breadth.

(a) 2x +2 (b) x +1

(c) x – 2 (d) x – 1

Ans. (b) : area of rectangle = length × breadth

$2x^2 + 3x + 1 = (2x+1) \times b$

$\Rightarrow (2x^2 + 3x + 1) = (2x + 1) \times b$

$\Rightarrow (2x^2 + 2x + x + 1) = (2x + 1) \times b$

$\Rightarrow 2x(x + 1) + 1(x + 1) = (2x + 1) \times b$

$\Rightarrow (2x + 1) \times b = (x + 1)(2x + 1)$

$b = (x + 1)$

148. 150 cm wide carpet is to be laid on the floor of a rectangular hall. If the hall is 20 m long and 18 m wide, what will be the required cost of carpet at the rate of Rs. 12 per meter?

(a) Rs. 3,280 (b) Rs. 2,280

(c) Rs. 2,880 (d) Rs. 2,480

RRB Group-D – 05/10/2018 (Shift-III)

Ans. (c) As per question,

length of hall = 20 m

breadth of hall = 18 m

area of hall = 20 × 18 [150cm = 1.5m]

= 360 m^2

breadth of carpert = 1.5 m

cost = Rs. 12 per meter

cost of carpet = $\frac{360}{1.50} \times 12$

= Rs. 2880

149. The breath of a rectangular field is 60% of its length. If the perimeter of the field is 800 m, then find the area of the field.

(a) 18750 m^2 (b) 37500 m^2
(c) 40000 m^2 (d) 48000 m^2

RRB Paramedical Exam – 20/07/2018 (Shift-II)

Ans. (b) Let the length of rectangular field = x

breadth of rectangular field $= x \times \frac{60}{100} = \frac{3x}{5}$

perimeter of field = 800

$2\left(x + \frac{3x}{5}\right) = 800$

$2\left(\frac{5x + 3x}{5}\right) = 800$

8x = 400 × 5

x = 250 m

length of field x = 250

breadth of field $= \frac{3x}{5} = \frac{3 \times 250}{5} = 150\,m$

area of field = 250 × 150 = 37500 m^2

150. A square of the side 2 m inside a rectangle of length 5 m and breadth 2 m is shaded. What is the ratio of the area of the shaded square and the area of the unshaded region of the rectangle?

(a) 3 : 2 (b) 2 : 3
(c) 5 : 2 (d) 2 : 5

RRB NTPC 17.01.2017 Shift-2

Ans : (b) area of rectangle = 5 × 2 = 10m^2
area of shade square = 2 × 2 = 4m^2
area of unshaded square = 10 – 4 = 6m^2

Intended ratio = area of square : area of unshaded rectangle

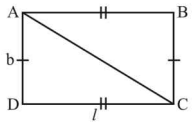
Intended ratio = 4 : 6 = 2 : 3

151. The length of the diagonal of a rectangle and its perimeter are 11 cm and 13 cm respectively. Find the area of the rectangle.

(a) 12 square cm. (b) 48 square cm.
(c) 36 square cm. (d) 24 square cm.

RRB NTPC 05.04.2016 Shift-1

Ans : (d)



Given—

Semi perimeter $(l + b) = 13$ cm. ..... (i)

diagonal of rectangle $\left(\sqrt{l^2 + b^2}\right) = 11cm$

$\Rightarrow l^2 + b^2 = 121$ cm l = length, b = breadth

$\Rightarrow (l + b)^2 - 2lb = 121$ from equation (i)

$\Rightarrow (13)^2 - 2lb = 121$

$\Rightarrow 2lb = 169 - 121 = 48$

$\Rightarrow l.b = 48/2$

$\Rightarrow l.b = 24$ Square cm.

area of rectangle = 24 Square cm.

152. The perimeter of a rectangle is 28 cm. If its one side is 4 cm then find other side of the rectangle.

(a) 24 cm (b) 7 cm
(c) 10 cm (d) 8 cm

RRB NTPC 04.04.2016 Shift : 1

Ans : (c) Let be length of second side = x cm

∵ perimeter of rectangle = 28

2 × (x + 4) = 28

x + 4 = 14 x = 10 cm.

153. The area of a rectangle is 42 sq. cm and its length is 7 cm. Find its perimeter.

(a) 14 cm. (b) 21 cm.
(c) 26 cm. (d) 24 cm.

RRB NTPC 03.04.2016 Shift : 2

Ans :(c) Area of rectangle = $\ell \times b = 42$

$7 \times b = 42$

$b = 6$ cm.

Perimeter of rectangle = $2\ (\ell + b)$

$= 2\ (7 + 6) = 2 \times 13 = 26$ cm.

154. The length of a rectangle is 6 times its breadth. If the perimeter of a rectangle is 56 cm then find the area of the rectangle.

(a) 48 square cm. (b) 96 square cm.
(c) 144 square cm. (d) 64 square cm.

RRB NTPC 03.04.2016 Shift : 3

Ans : (b) Let be breadth of rectangle = x cm

Then length of rectangle = 6x cm

∵ Perimeter of rectangle = $2\ (\ell \times b)$

$56 = 2\ (x + 6x)$ $28 = 7x$ $\boxed{x = 4}$ cm

So area of rectangle = $\ell \times b = 6x \times x$

$= 6x^2 = 6\ (4)^2 = 6 \times 16 = 96$ square cm

155. The perimeter of a rectangle and its diagonal are 34 cm and 13 cm respectively. Find its area

(a) 987 square cm. (b) 240 square cm.
(c) 120 square cm. (d) 60 square cm.

RRB NTPC 31.03.2016 Shift : 1

Ans : (d) If length and breadth of volume will be ℓ cm and b cm then.....

diagonal of rectangle = 13 cm

$\sqrt{\ell^2 + b^2} = 13$

$l^2 + b^2 = 169$ ..... (1)

∵ Perimeter of rectangle = 34 cm

$2(l+b) = 34$

$l + b = 17$..... (2)

∵ $(l+b)^2 = l^2 + b^2 + 2lb$

$(17)^2 = 169 + 2lb$

$289–169=2lb$

$2lb=120$

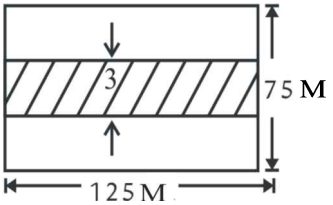
∴area of rectangle = 60cm.

156. The length of a rectangular field is 125 m and breadth is 75 m. There is a 3 m wide footpath in the middle of the field. What is the area of the field without the footpath?

(a) 9375 square m. (b) 9000 square m.
(c) 9750 square m. (d) 8625 square m.

RRB NTPC 31.03.2016 Shift : 2

Ans : (b)



area of field without foot path
= 125 × 75 – 125 × 3
= 125 × (75–3)
= 125 × 72
= 9000 square m.

157. What is the length of diagonal, if area of a rectangle is 168cm^2 and breadth is 7cm?

(a) 24 cm (b) 15 cm
(c) 17 cm (d) 25cm

RRB NTPC 30.03.2016 Shift : 1

Ans : (d) ∵ Area of rectangle = 168 Square cm
∴ length × breadth = 168
as per question
7 × length = 168
length $= \frac{168}{7} = 24$ cm
∴ length of diagonal $= \sqrt{l^2 + b^2}$
$= \sqrt{(24)^2 + (7)^2}$
$= \sqrt{576 + 49} = \sqrt{625} = 25$ cm.

158. The perimeter of a rectangle is 28 cm. If the length is $\frac{5}{2}$ times of the breadth, then find the length and breadth of the rectangle.

(a) 90 and 5 (b) 10 and 4
(c) 6 and 7 (d) 11 and 3

RRB NTPC 29.03.2016 Shift : 1

Ans : (b) As per question
length of rectangle $= \frac{5}{2} \times$ Breadth
Perimeter of rectangle = 28
2(L + B) = 28
$(\frac{5}{2}B + B) = \frac{28}{2} = 14$
$\frac{7}{2}B = 14 \Rightarrow B = 4$ cm
$\therefore L = \frac{5}{2} \times 4 = 10$ cm
So length and breadth of rectangle is 10 cm and 4 cm.

159. If the perimeter of a rectangle is 24 cm and the length is twice of the breadth, then find its area.

(a) 24 square cm. (b) 32 square cm.
(c) 28 square cm. (d) 36 square cm.

RRB NTPC 28.03.2016 Shift : 2

Ans : (b) Perimeter of rectangle = 24 cm
Let be length = 2x breadth = x
Perimeter of rectangle = 2(l + b)
2(2x + x) = 24
$2 \times 3x = 24 \Rightarrow x = \frac{24}{6} = 4$
L= 2x = 2 × 4 = 8cm
B = x = 4cm
Area of rectangle = $\ell \times$ b
= 8 × 4 = 32 cm^2

160. Ankita stands at the corner of a rectangular field of 40 m length and 30 m breadth. If Ankita runs only along then diagonal and returns to the starting point, then what is the total distance covered by Ankita?

(a) 100 m (b) 80 m
(c) 140 m (d) 120 m

RRB NTPC 28.03.2016 Shift : 3

Ans : (a) Length of rectangle's diagonal
$= \sqrt{(40)^2 + (30)^2}$
$= \sqrt{1600 + 900} = \sqrt{2500} = 50$ m
∴ Total distance covered by Ankita
= 2 × 50 = 100 m

161. The length of a rectangular board is 4 times its breadth, if the area of the board is 256 sq. m, then find its length?

(a) 8 m (b) 16 m
(c) 24 m (d) 32 m

RRB NTPC 18.01.2017 Shift : 1

Ans : (d) Let be breadth = b
ℓ = 4b, Area = 256
∴4b × b = 256
b^2 = 64, b = 8
∴ L = 4b = 4 × 8 = 32m

162. The order of the rotational symmetry of a rectangle is:

(a) 1 (b) 4
(c) 2 (d) 0

RRB NTPC 11.04.2016 Shift : 1

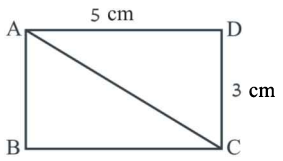
Ans : (c) The order of the rotational symmetry of the rectangle is 2.

163. Find the length of the diagonal of a rectangle whose length and breadth are 5 cm and 3 cm respectively.

(a) $\sqrt{34}$ (b) $\pm\sqrt{34}$
(c) 4 (d) ±4

RRB NTPC 11.04.2016 Shift : 1

Ans : (a)



Length of diagonal (AC) $= \sqrt{(5)^2 + (3)^2}$

$= \sqrt{25+9} = \sqrt{34}$ cm

164. The length of a rectangle is 24 cm more than its breadth. If the perimeter of a rectangle is 112 cm then find its length.

(a) 40 cm (b) 16 cm
(c) 24 cm (d) 32 cm

RRB NTPC 10.04.2016 Shift : 3

Ans : (a) Let be breadth of rectangle = x cm

∴ length = (24 + x) cm

Now perimeter of rectangle = 2 (length + breadth)

112 = 2 (x + 24 + x)
112 = 2 (2x + 24)
112 = 4x + 48
4x = 112–48
4x = 64, x = 16

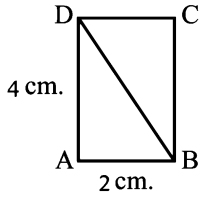
hence length of rectangle = (24 + x) = 24 + 16 = 40 cm

165. Find the length (in cm) of the diagonal of a rectangle whose length is 2 cm and breadth is 4 cm respectively.

(a) $2\sqrt{5}$ (b) $\pm 2\sqrt{5}$
(c) $2\sqrt{3}$ (d) $\pm 2\sqrt{3}$

RRB NTPC 07.04.2016 Shift : 2

Ans : (a) According to the question-



length of diagonal (BD) $= \sqrt{AD^2 + AB^2}$

$= \sqrt{(4)^2 + (2)^2}$

$= \sqrt{16+4} = \sqrt{20}$

$= 2\sqrt{5}$

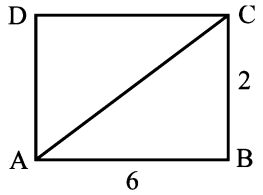
Hence the length of diagonal of rectangle $= 2\sqrt{5}$ cm

166. Find the length of the diagonal of rectangle whose length is 6 cm and breadth is 2 cm.

(a) $2\sqrt{10}$ (b) $\pm 2\sqrt{10}$
(c) $4\sqrt{2}$ (d) $\pm 4\sqrt{2}$

RRB NTPC 07.04.2016 Shift : 3

Ans : (a)



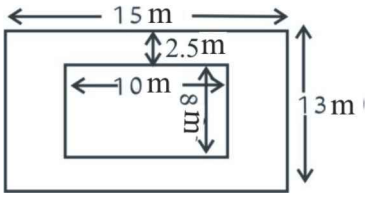
Length = 6 cm
breadth = 2 cm
$(\text{diagonal})^2 = (\text{length})^2 + (\text{breadth})^2$
$(\text{diagonal})^2 = (6)^2 + (2)^2$
$(\text{diagonal})^2 = 36+4$
diagonal $= \sqrt{40} = 2\sqrt{10}$

167. A footpath of 2.5 m breadth is built around a rectangular garden having a length of 10 m and breadth of 8 m. Find the area of the garden which includes the footpath.

(a) 130.25 square m. (b) 131.25 square m.
(c) 195.00 square m. (d) 162.50 square m.

RRB NTPC 06.04.2016 Shift : 1

Ans : (c)



Area of the garden includes the footpaths =15×13 =195 Square meter

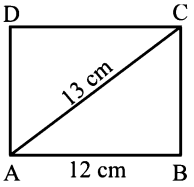
168. The one side of a rectangle is 12 m and its diagonal is 13 m, then find its area.

(a) 60 square m. (b) 55 square m.
(c) 50 square m. (d) 45 square m.

RRB NTPC 06.04.2016 Shift : 2

Ans : (a) Triangle ABC in $AC^2 = AB^2 + BC^2$

$(13)^2 = (12)^2 + BC^2$
$BC^2 = 169 - 144$
$BC = \sqrt{25}$
BC = 5 cm.



Area of rectangle = AB × BC = 12 × 5= 60 square cm.

169. Find the length of the diagonal of a rectangle whose length and breadth are 9 cm and 6 cm respectively.

(a) $3\sqrt{13}$ (b) $\pm 3\sqrt{13}$
(c) $3\sqrt{5}$ (d) $\pm 3\sqrt{5}$

RRB NTPC 22.04.2016 Shift : 2

Ans : (a) Length of rectangle = 9 cm

b = 6 cm

diagonal $= \sqrt{l^2 + b^2}$

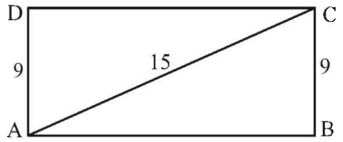
$= \sqrt{(9)^2 + (6)^2} = \sqrt{81+36} = \sqrt{117} = 3\sqrt{13}$

170. Find the area of a rectangle whose diagonal and breadth are 15 m and 9 m respectively.

(a) 42 square m. (b) 144 square m.
(c) 108 square m. (d) 225 square m.

RRB NTPC 26.04.2016 Shift : 1

Ans : (c) diagonal of rectangle AC = 15 m. and BC = 9 m.



in a ABC,

$$AB = \sqrt{AC^2 - BC^2} \Rightarrow \sqrt{15^2 - 9^2}$$
$$= \sqrt{225 - 81}$$
$$= \sqrt{144}$$

AB = 12 m

length of rectangle = 12m

area of rectangle = length × breadth

$= 9 \times 12 = 108\ m^2$

171. Find the length of the diagonal of a rectangle whose length and breadth are 6 cm and 6 cm respectively.

(a) $6\sqrt{2}$ (b) $\pm 6\sqrt{2}$
(c) 0 (d) $\sqrt{2}$

RRB NTPC 26.04.2016 Shift : 3

Ans : (a)

diagonal of rectangle $= \sqrt{(\text{length})^2 + (\text{breadth})^2}$

$= \sqrt{(6)^2 + (6)^2} = \sqrt{36 + 36} = \sqrt{6 \times 6 \times 2} = 6\sqrt{2}$

172. The area of a rectangle is 448 sq. m. If its length is 12% more than its breadth, then find its breadth.

(a) 14m (b) 16m
(c) 18m (d) 20m

RRB NTPC 27.04.2016 Shift : 2

Ans : (d)

Let the breadth = x

$\text{length} = \frac{x \times 112}{100} = \frac{112x}{100}$

area of rectangle = length × breadth

As per question $448 = \frac{x \times 112x}{100}$

$\Rightarrow \quad x^2 = \frac{448 \times 100}{112}$

$x^2 = 400$

$x = 20$

So breadth = 20 m

173. Find the maximum area of a rectangular field which is surrounded by a rope of 400 m

(a) 5000 m^2 (b) 6250m^2
(c) 4000m^2 (d) 10000m^2

RRB NTPC 27.04.2016 Shift : 3

Ans : (d) We know that the area of a rectangle will be maximum when the length of the rectangle is equal to its width

∴ For maximum area

2(L + B) = 400 (∵ L = B)

$\text{length} = \frac{400}{4} = 100\ m$

area = length × breadth

$= 100 \times 100 = 10{,}000\ m^2$

174. Find the maximum area of a rectangular field enclosed by a 40 m long rope?

(a) 160 m^2 (b) 180 m^2
(c) 200 m^2 (d) 100 m^2

RRB NTPC 30.04.2016 Shift : 2

Ans : (d) Let be length = a

breadth = b

According to question,

2 (a + b) = 40

a + b = 20

∵ ab = maximum

∴ a = 10, b = 10

so maximum area of field = ab = 10 ×10 = 100 m^2

Type - 6

175. If the surface area of a cube is 3750 cm^2, then find its volume?

(a) 14255 cm (b) 16625 cm
(c) 12225 cm (d) 15625 cm

RRB JE - 23/05/2019 (Shift-III)

Ans : (d) Surface area of the cube = $6a^2$

As per question

$6a^2 = 3750$

$\Rightarrow a^2 = 625$

a = 25

Volume of cube = $(\text{side})^3$

$= 25 \times 25 \times 25$

$= 15625\ cm^3$

176. The sum of all the sides of a cube is 48 cm. Find its volume?

(a) 27 cm^3 (b) 216 cm^3
(c) 64 cm^3 (d) 36 cm^3

RRB JE - 23/05/2019 (Shift-III)

Ans : (c) Let each side of cube = a cm

As per question

12 a = 48

a = 4

Volume of cube = $(a)^3 = (4)^2 = 64cm^3$

177. Numerical values of surface area and volume of a cube are equal. Find the volume of the cube?

(a) 512 cube unit (b) 276 cube unit
(c) 216 cube unit (d) 162 cube unit

RRB JE - 25/05/2019 (Shift-I)

Ans : (c) Surface area of the cube $= 6a^2$
volume of cube $= a^3$
As per question
$6a^3 = a^3$
$a = 6$
Hence volume of cube $= a^3 = 6^3$
$= 216$ cube unit

178. If the length of each side of a cube is increased by 3 times, then how much of its volume will be the actual volume?
(a) 9 times (b) 2 times
(c) 3 times (d) 27 times
RRB RPF SI – 11/01/2019 (Shift-I)

Ans. (d) Let side of first cube = a
Volume of first cube $= a^3$
Side of second cube = 3a
Volume of second cube $= (3a)^3 = 27\,a^3$
The volume of the first cube will be 27 times the volume of the second cube.

179. The sum of the sides of a cube is equal to one-eight of the perimeter of a square. If the numerical values of the volume of the cube and the area of the square are equals then what will be the length of the cube core?
(a) 288 units (b) 366 units
(c) 432 units (d) 576 units
RRB Group-D – 26/09/2018 (Shift-I)

Ans : (d) Let the length of cube's core = a unit
length of squares side = b unit
As per first condition
$12a = \frac{4b}{8} \Rightarrow \quad b = 24a$ _____(i)
As per second condition- $a^3 = b^2 \Rightarrow a^3 = 24 \times 24 \times a^2$
$\boxed{a = 576 \text{ unit}}$
hence length of cube's core = 576 unit

180. If the sides of a cube are doubled then its volume increased by how many times of its actual volume?
(a) 3 (b) 4
(c) 8 (d) 6
RRB Group-D – 10/10/2018 (Shift-II)

Ans : (c) Let side of cube = a unit
volume of cube $= a^3$ cubic unit
doubling the cube side = 2a
New volume $= (2a)^3 = 8a^3$ cubic unit
So its volume will increase by 8 times its actual volume.

181. The length of the edge of a cube is 80 cm. A circle is painted yellow on each surface of that cube. If the area covered by the circle is as large as possible, then what will be the total area of the surface of the cube which is not painted?
(a) 8,200.14 cm^2 (b) 8,528.57 cm^2
(c) 8,228.57 cm^2 (d) 8,127.14 cm^2
RRB Group 'D' 07/12/2018 (Shift-I)

Ans : (c) Given
$r = \frac{80}{2} = 40\text{cm}, \quad r = 80 \text{ cm}$
area of not painted layer $= 6a^2 - \{6 \times (\pi r^2)\}$
$= 6 \times (80)^2 - \{6 \times \frac{22}{7} \times (40)^2\}$
$= 38400 - \{6 \times 5028.57\}$
$= 38400 - 30171.42$
$= 8228.58 \text{ cm}^2 \approx 8228.57 \text{ cm}^2$

182. The length, breadth and height of a rightangled parallel hexagon are 3.6 m, 2.5 m and 1.8 m respectively, many cubes are made by melting this metal, measurement of each side is 0.3 m of each cube. How many such cubes are made?
(a) 60 (b) 6000
(c) 600 (d) 60,000
RRB Group-D – 11/12/2018 (Shift-III)

Ans : (c) Let the number of new cubes is n
$$n = \frac{\text{right angled parallel hexagon metal volume}}{\text{melting volume of cube}}$$
$$= \frac{3.6 \times 2.5 \times 1.8}{0.3 \times 0.3 \times 0.3}$$
$$= 12 \times 50 = 600$$

183. How many times the volume of a cube is reduced by halving its sides?
(a) $\frac{1}{4}$ (b) $\frac{1}{8}$
(c) $\frac{1}{3}$ (d) $\frac{1}{2}$
RRB Group-D – 11/10/2018 (Shift-I)

Ans : (b) Volume of cube with sides 'a' = a × a × a = a^3
Volume of cube with sides $\frac{a}{2} = \left(\frac{a}{2}\right)^3 = a^3 \times \frac{1}{8}$
so by halving the sides of the cube, Its volume will be reduced by $\frac{1}{8}$ times.

184. If one side of a cube is extended by 1 cm, the volume will increase by 169 cm^3. Find the each side of the cube?
(a) 8cm (b) 7cm
(c) 6cm (d) 9cm
RRB Group-D – 05/10/2018 (Shift-III)

Ans. (b) Let edge of cube = a cm
So, volume of cube $= a^3$
As per question,
$(a + 1)^3 - a^3 = 169$
$a^3 + 1 + 3a(a + 1) - a^3 = 169$
$3a^2 + 3a - 168 = 0$
$a^2 + a - 56 = 0$
$a^2 + 8a - 7a - 56 = 0$
$a(a + 8) - 7(a + 8) = 0$
$(a + 8)(a - 7) = 0$
$\therefore \quad a = 7$ cm $\quad a = -8$ (invalid)

185. The sum of the lengths of the edges of a cube is equal to the perimeter of a square. If the numerical value of the volume of the cube is equal to the numerical value of the area of the square, then the length of one side of the square is:

(a) 30 units (b) 9 units
(c) 27 units (d) 12 units

RRB ALP & Tec. (21-08-18 Shift-III)

Ans : (c) Let side of cube = y unit and side of square = x unit

As per question

$12y = 4x,\ y = \frac{x}{3}$ .....(i)

and $y^3 = x^2$

From equation (i)

$\left(\frac{x}{3}\right)^3 = x^2$

$x = 27$

Side of square x = 27 unit

186. If the edge of a cube is increased by 3 cm, the volume will increases by 657 cm³. Then what is the original length of each edge of the cube?

(a) 7 cm (b) 8 cm
(c) 6 cm (d) 9 cm

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (a) Let length of the edge of cube = a

so volume = a^3

As per question length increase in 3 cm

So length of new cube's edges = a + 3

So new volume = $(a+3)^3$

As per question $(a+3)^3 - a^3 = 657$

$a^3 + 27 + 9a(a+3) - a^3 = 657$

$27 + 9a^2 + 27a = 657$

$9a^2 + 27a = 657 - 27 = 630$

$9a^2 + 27a = 630$

$a^2 + 3a = \frac{630}{9} = 70$

$a^2 + 3a - 70 = 0$

$a^2 + 10a - 7a - 70 = 0$

$a(a+10) - 7(a+10) = 0$

$\Rightarrow (a+10)(a-7) = 0$

So intended length of a cube's edge = 7 cm

Type - 7

187. The sides of a cuboid are in the ratio of 1:2:3. If its surface area is 88 cm², find its volume.

(a) 48 cm³ (b) 120 cm³
(c) 64 cm³ (d) 24 cm³

RRB JE - 25/05/2019 (Shift-II)

Ans : (a) Let the length of cuboide = (l)

breadth (b) and height (h) is x, 2x and 3x

area of cuboid surface = 88 cm²

$2(l.b + b.h + h.l) = 88\ cm^2$

$2(x.2x + 2x.3x + 3x.x) = 88 cm^2$

$2(2x^2 + 6x^2 + 3x^2) = 88$

$22x^2 = 88$

$x^2 = 4$

$\boxed{x = 2}$

length of cuboid (l) = x = 2cm

breadth of cuboid (b) = 2x = 2 × 2 = 4cm

height of cuboid (h) = 3x = 3 × 2 = 6 cm

Volume of cuboid = ℓ.b.h

= 2 × 4 × 6

= $\boxed{48 cm^3}$

188. Find the volume of the cuboid whose length, breadth and height are 24 cm, 18 cm and 16 cm.?

(a) 7142 cm³ (b) 5184 cm³
(c) 7089 cm³ (d) 6912 cm³

RRB RPF Constable – 20/01/2019 (Shift-III)

Ans : (d) length = 24 cm, breadth = 18 cm, height = 16

∴ Volume of cuboid = length × height × breadth

= 24 × 16 × 18

= 6912 cm³

189. Cuboid of dimensions l × b × h is cut into the form of the frames of dimensions l × 0.5 b × 0.4 h. Find the number of frames.

(a) 4 (b) 5
(c) 20 (d) 10

RRB JE - 27/05/2019 (Shift-II)

Ans : (b)

number of getting seats = $\frac{\text{Volume of large cuboid}}{\text{Volume of small cuboide}}$

$= \frac{\ell \times b \times h}{\ell \times 0.5b \times 0.4h}$

$= \frac{1}{0.5 \times 0.4}$

$= \frac{1}{0.20} = \frac{100}{20} = 5$

190. Find the area of the cardboard required to make a box of 10 cm × 8 cm × 4 cm.

(a) 388 cm² (b) 412 cm²
(c) 488 cm² (d) 304 cm²

RRB JE - 28/05/2019 (Shift-I)

Ans : (d) Area of 10 cm × 8 cm × 4 cm =

= 2 (lb + bh + hl)

= 2 (10×8+8×4+4×10)

= 2 (80+32+40)

= 2×152 = 304 cm².

191. A wooden box measures 20 cm × 12 cm × 10 cm. The thickness of the wood is 1 cm. Find the volume of the wood used to make the box.

(a) 1120 cm³ (b) 2400 cm³
(c) 519 cm³ (d) 960 cm³

RRB JE - 29/05/2019 (Shift-II)

Ans : (d)
Wooden box measurement $= 20 \text{ cm} \times 12 \text{ cm} \times 10 \text{ cm}$
$= 2400 \text{ cm}^3$
$\because$ wood thickness is 1 cm
$\therefore$ internal measurement of wooden box
$= (20 - 1 \times 2) \times (12 - 1 \times 2) \times (10 - 1 \times 2)$
$= 18 \times 10 \times 8 = 1440 \text{ cm}^3$
$\therefore$ volume of wood used for making box
$= 2400 - 1440 = 960 \text{ cm}^3$

192. The length, breadth and height of the box of each rectangular panel are 12 cm, 8 cm and 10 cm respectively. Find the total surface area of the box.?
(a) 592 Square cm. (b) 376 Square cm.
(c) 524 Square cm. (d) 482 Square cm.
RRB JE - 30/05/2019 (Shift-III)

Ans : (a) Given
$l = 12$ cm
b = 8 cm
h = 10 cm
Total surface area of rectangular panel
$= 2(lb + bh + hl)$
$= 2(12 \times 8 + 8 \times 10 + 12 \times 10)$
$= 2(96 + 80 + 120)$
$= 2 \times 296$
= 592 square centimeter

193. The dimensions of a luggage box are 80 cm, 60 cm and 40 cm. How many square cm of fabric will be required to cover the box?
(a) 10400 Square cm. (b) 20800 Square cm.
(c) 20400 Square cm. (d) 10200 Square cm.
RRB RPF SI – 06/01/2019 (Shift-II)

Ans : (b) Need quatity of cloth = All surface of box
$= 2(80 \times 60 + 60 \times 40 + 40 \times 80)$
$= 2(4800 + 2400 + 3200) = 2 \times 10400$
= 20800 Square cm.

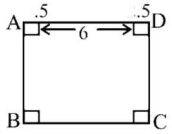
194. The surface area of a cuboid is 1372 sq. cm. Its dimensions are in the ratio of 4:2:1. Find the length?
(a) 24 cm (b) 7 cm
(c) 28 cm (d) 12 cm
RRB JE - 30/05/2019 (Shift-III)

Ans : (c) Let cuboid dimensions 4x, 2x, x cm.
Surface area of cuboid $= 2[lb + bh + hl]$
$1372 = 2[4x \times 2x + 2x \times x + 4x \times x]$
$1372 = 2[8x^2 + 2x^2 + 4x^2]$
$1372 = 2[14x^2]$
$x^2 = \frac{1372}{28}$
$x^2 = 49$
$x = 7$
length of cuboid $= 4x$
$= 4 \times 7 = 28$ cm.

195. The ratio of the length, breadth and height of a room is 3:3:4. If the length is two times, width is two-third and the height is halved then by what percentage will the cost of painting the four walls of the room be reduced?
(a) $30\frac{3}{4}$ (b) $33\frac{1}{3}$
(c) 32 (d) $28\frac{1}{4}$
RRB Group-D – 12/11/2018 (Shift-I)

Ans. (b) : Let length of room = 3x
breadth = 3x
height = 4x
area of room $= 2(L + B) \times h$
$= 2(3x + 3x) \times 4x$
$= 48x^2$
When the length is doubled, breadth is two-thirds and height is halved, then the new area obtained
$= 2\left(3x \times 2 + 3x \times \frac{2}{3}\right) \times \frac{4x}{2}$
$= 2(6x + 2x) \times 2x$
$= 16x \times 2x = 32x^2$
Decrease $= 48x^2 - 32x^2 = 16x^2$
Decrease % $= \frac{16x^2}{48x^2} \times 100 = \frac{100}{3} = 33\frac{1}{3}\%$

196. 0.25 cm^2 square is cut from each corner of a square plate of each side 7 cm. The remaining part is made with cut to form a cuboid. What will be the volume of this cuboid with its top open?
(a) 20 (b) 16
(c) 21 (d) 18
RRB Group-D – 03/10/2018 (Shift-I)

Ans : (d)


Side of square $= \sqrt{.25} \text{ cm}^2$
= 0.5 cm
So
height of cuboid = 0.5 cm
length of cuboid = 6 cm
breadth of cuboid = 6 cm
Volume of cuboid $= lbh$
$= 0.5 \times 6 \times 6$
$= 18 \text{ cm}^3$

197. The measurement of a rectangular box is in the ratio of 2:3:5. If the total surface area is 6200 cm^2, find the dimensions of the cuboid.
(a) 20cm × 40cm × 50cm
(b) 20cm × 30cm × 40cm
(c) 30cm × 40 cm × 50cm
(d) 20cm × 30cm × 50cm
RRB Group-D – 03/10/2018 (Shift-III)

Ans : (d) Let sides is 2x cm. 3x cm., and 5x cm then total surface of area or all surface of cuboid

$= 2(lb + bh + hl)$

$2(lb + bh + hl) = 6200$

$2(2x \times 3x + 5x \times 3x + 5x \times 2x) = 6200$

$2(6x^2 + 15x^2 + 10x^2) = 6200$

$31x^2 = 3100$

$x^2 = 100 \Rightarrow \boxed{x = 10 \text{ cm}}$

So dimension or sides of cuboid—

$2x = 2 \times 10 = 20\text{cm}$

$3x = 3 \times 10 = 30$ cm

$5x = 5 \times 10 = 50$ cm

198. Two cubes of 343 cm^3 volume are added at their edges. The total floor area of the cuboid so formed is?

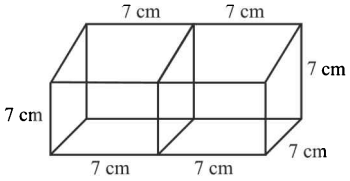
(a) 436 square cm (b) 466 square cm

(c) 346 square cm (d) 490 square cm

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (d) Area of cube = 343 cm^3

then side of cube = 7 cm



area of cuboid = 2 (lb + bh + hl)

as per question

l= 7+7=14, b= 7,h=7

∴ area of cuboid

$= 2(14\times7+7\times7+7\times14) = 490$ square cm

199. A water tank is 5 m long, 3 m wide and 1 m deep. How much litre of water can be filled in it?

(a) 810 L (b) 15, 000 L

(c) 729 L (d) 720 L

RRB Group-D – 12/10/2018 (Shift-III)

Ans : (b)

Volume of water tank $= l \times b \times h$

$= 5 \times 3 \times 1 = 15\ m^3$

$= 15 \times 1000L$ (1 cubic m = 1000 L)

$= 15000L$

200. The area of the three faces of a cube including the top is 25 m^2, 32 m^2 and 32 m^2. What is the volume of the cube?

(a) 184 m^3 (b) $\sqrt{3024}m^3$

(c) 92 m^3 (d) 160m^3

RRB Group-D – 24/09/2018 (Shift-I)

Ans : (d) Area of cuboid's first panel(lb) $= l \times b$

$= 25\ m^2$

area of cuboid's second panel (bh) $= b \times h = 32\ m^2$

area of cuboid's third panel (hl) $= h \times l = 32\ m^2$

Now,

Volume of cuboid $= \sqrt{lb \times bh \times lh}$

$= \sqrt{25\times32\times32}$

$= \sqrt{5\times5\times16\times2\times16\times2}$

$= 5 \times 4 \times 4 \times 2 = 160$

Hence volume of cuboid = 160 m^3

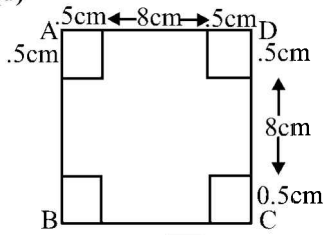
201. The pieces of 0.25 cm^2 square have been cut from a square plate of each side of 4 cm and the remaining plate is joined with cut edges from an open cuboid. What will be the volume of this open cuboid?

(a) 34 (b) 30

(c) 36 (d) 32

RRB Group-D – 26/11/2018 (Shift-III)

Ans : (d)



Side of cut square = $\sqrt{.25}$ = .5cm

∴ Volume of open top cuboid = $8 \times 8 \times 0.5 = 32\ cm^3$

202. Two cubes each volume 216 cm^2 are joined end to end. The total surface area of the resulting cuboids is?

(a) 360 cm^2 (b) 380 cm^2

(c) 330 cm^2 (d) 340 cm^2

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (a)

Let side of cube a cm

Volume of cube = $(\text{side})^3$

$216 = a^3$

$a = \sqrt[3]{216}$

a = 6 cm

related both cube–

$l = 12$, b = 6, h = 6

Surface area of cuboid $= 2(lb + bh + hl)$

$= 2(12\times6 + 6\times6 + 6\times12)$

$= 2(72 + 36 + 72)$

$= 2(180) = 360\ cm^2$

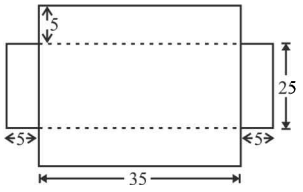
203. Four squares of 5 cm side are cut from the corners of a 45 cm long and 35 cm wide rectangular sheet and an open box is made from the remaining sheet. Find the volume of the box?

(a) 1200 $cm.^3$ (b) 875 $cm.^3$

(c) 1325 $cm.^3$ (d) 4375 $cm.^3$

RRB NTPC 02.04.2016 Shift : 1

Ans : (d)



for open box

$l = 35$ cm. b = 25 cm. h = 5 cm.

∴ volume of open box = $35 \times 25 \times 5 = 4375\ cm^3$

204. A rectangular reservoir has 4200 litres of water. Find the depth of water in reservoir if the measure of the base of this reservoir is 6m × 3.5 m.

(a) 1m (b) 2.5 m
(c) 3 m (d) 2 m

RRB NTPC 18.01.2017 Shift : 2

Ans : (d) Given,
length of reservoir 6 m
breadth of reservoir = 3.5 m
let height or depth is (h)
As per question,

Volume of rectangular reservoir = 42000 ℓ

$l \times b \times h = 42000\ l$

$\Rightarrow \quad 6 \times 3.5 \times h = \frac{42000}{1000}$ m (1m = 1000 ℓ)

$\Rightarrow \quad h = \frac{42}{21}$

$\Rightarrow \quad h = 2$ m

205. The ratio of the length, breadth and height of a room is 3:2:1. If its volume is 3072 m^3, then find its width?

(a) 18 m (b) 16 m
(c) 24 m (d) 12 m

RRB NTPC 09.04.2016 Shift : 3

Ans : (b) Ratio = 3 : 2 : 1
Let length of room = 3x
b = 2x
h = x
volume of cuboid = $l \times b \times h = 3072$
$3x \times 2x \times x = 3072$
$6x^3 = 3072 \quad \Rightarrow \quad x^3 = 512$
$x = \sqrt[3]{512}$
$x = 8$
hence, b = 2x = 2×8 = 16 m

206. The surface areas of three faces of a cuboid sharing a vertex are 20 m^2, 32 m^2 and 40 m^2. What is the volume of the cuboid?

(a) 92 m^3
(b) $\sqrt{3024}$ m^3
(c) 160 m^3
(d) 184 m^3

RRB ALP & Tec. (09-08-18 Shift-II)

Ans : (c) Let length, breadth and height of cuboid is ℓ, b and h.

$\ell b \times bh \times \ell h = 40 \times 20 \times 32$

$(\ell bh)^2 = 25600$

$\ell bh = 160\ m^3$

Hence the value of the volume of the cuboid (ℓbh) will be 160 cuboic meter.

Type - 8

207. Find the ratio of the volume of a cylinder of radius r and sphere of radius r and height 2r?

(a) 2:3 (b) 3:2
(c) 5:4 (d) 3:5

RRB JE - 22/05/2019 (Shift-I)

Ans : (a) Circle radius = r
Cylinder radius = r
height of cylinder = 2r

So $\frac{\text{Volume of circle}}{\text{Volume of cylinder}} = \frac{\frac{4}{3}\pi r^3}{\pi r^2 h}$

$= \frac{\frac{4}{3}\pi r^3}{\pi r^2 \times 2r} = \frac{2}{3}$

∴ volume of circle : volume of cylinder = 2 : 3

208. What will be the total surface area of that solid cylinder? whose radius is half of the radius of the circle having an area of 154 square meters and the height is equal to its radius?

(a) 231 m^2 (b) 123 m^2
(c) 312 m^2 (d) 132 m^2

RRB JE - 25/05/2019 (Shift-I)

Ans : (a) Area of circle = 154 m^2
$\pi r^2 = 154$
$r^2 = 154 \times \frac{7}{22} = 7 \times 7$
$r = 7$
As per question,
height of cylinder = 7 m
radius of cylinder = $\frac{7}{2}$ m
Curve surface area = $2\pi r\ (h + r)$

$= 2 \times \frac{22}{7} \times \frac{7}{2}\left(7 + \frac{7}{2}\right)$

$= 22 \times \frac{21}{2}$

$= 231\ m^2$

209. The ratio of the radius of two cylinders is 2 : 3. The ratio of their heights is 5 : 3. If the volume of one cylinder is 160 cube, find the volume of the other cylinder?

(a) 216 cubic unit (b) 400 cubic unit
(c) 240 cubic unit (d) 243 cubic unit

RRB JE - 26/05/2019 (Shift-II)

Ans : (a) Let radius of first cylinder $(r_1) = 2r$
and radius of second cylinder $(r_2) = 3r$
Similarly, height of first cylinder $(h_1) = 5h$
and height of second cylinder $(h_2) = 3h$
As per question,
volume of first cylinder = 160 cubic unit

Volume of cylinder $= \pi r^2 h$

$\pi r_1^2 h = 160$

$\pi \times 2r \times 2r \times 5h = 160$

$\pi \times 20r^2 h = 160$

$r^2 h = \frac{160}{\pi \times 20}$

$r^2 h = \frac{8}{\pi}$ _____ (i)

Volume of second cylinder $= \pi r_2^2 h_2$

$= \pi \times 3r \times 3r \times 3h$

$= \pi\, 27r^2 h$

$= \pi \times 27 \times 8 \times \frac{1}{\pi}$ {equation from (i)}

$= 27 \times 8 = 216$

So volume of second cylinder = 216 cubic unit

210. A wire made by melting a metal sphere with radius of 6 cm., The length of the wire whose radius of the transverse section is 8 cm?

(a) 4 cm (b) 3.5 cm
(c) 5 cm (d) 4.5 cm

RRB JE - 26/05/2019 (Shift-III)

Ans : (d) Let length of wire is l cm

As per question-

$\therefore$ Volume of sphere = Volume (cylinder) of wire

$\frac{4}{3}\pi \times 6 \times 6 \times 6 = \pi \times 8 \times 8 \times \ell$

$\ell = \frac{4 \times 6 \times 6 \times 6}{3 \times 8 \times 8} = 4.5$ cm.

211. Find the area of the outer curve surface of a metal pipe of radius 21 cm. and length 90 cm.?

(a) 11480 Square cm. (b) 10880 Square cm.
(c) 12880 Square cm. (d) 11880 Square cm.

RRB JE - 27/05/2019 (Shift-III)

Ans : (d) radius (r) = 21 cm

height (h) = 90 cm

area of external curved of pipe = $2\pi rh$

$= 2 \times \frac{22}{7} \times 21 \times 90$

$= 2 \times 22 \times 3 \times 90 = 11880$ Square cm.

212. Find the total surface area of semi sphere mounted on a cylinder while both have the same radius, and the height of the cylinder is twice its radius?

(a) $8\pi r^2$ (b) $4\pi r^2$
(c) $7\pi r^2$ (d) $(2\pi rh + 2\pi r^2)$

RRB JE - 29/05/2019 (Shift-II)

Ans : (c) Radius of semicircle = radius of cylinder

height of cylinder (h) = 2r

$\therefore$ Total curve surface area = curve surface area of cylinder + curve surface area of semi sphere + area of base of cylinder = $2\pi rh + 2\pi r^2 + \pi r^2$

$= 2\pi r \times 2r + 3\pi r^2$

$= 4\pi r^2 + 3\pi r^2 = 7\pi r^2$

213. Find the radius of the cylinder whose volume is 3850 cm and height is 25 cm.?

(a) 7 cm (b) 14 cm
(c) 3.5 cm (d) 10.5 cm

RRB RPF SI – 12/01/2019 (Shift-II)

Ans : (a) volume of cylinder (V) = 3850 cm^3

h = 25 cm

r = ?

$V = \pi r^2 h$

$3850 = \frac{22}{7} \times r^2 \times 25$

$r^2 = \frac{3850 \times 7}{22 \times 25} = 49$

r = 7 cm

214. the capacity of a cylindrical tank is 3080 cubic meters. The radius of its base is 7 meters. So find the depth of the tank?

(a) 10 m (b) 25 m
(c) 15 m (d) 20 m

RRB JE - 02/06/2019 (Shift-III)

Ans : (d) base radius (r) = 7 m

capacity or volume of cylinder and tank =3080 cubic m.

$\pi r^2 h = 3080$

$\frac{22}{7} \times 7^2 \times h = 3080$

$7h = \frac{3080}{22}$

$h = \frac{140}{7} = 20$ m

215. If the height 'H' of a cylinder is equal to the circumference of its base. Find the area of its curved surface in terms of H?

(a) h^3 (b) 2h/3
(c) $3h^2$ (d) h^2

RRB JE - 02/06/2019 (Shift-I)

Ans : (d) circumference $(2\pi r) = h$

curved surface of cylinder = $2\pi rh$ $[2\pi r = h]$

$= h \times h = h^2$

216. The length of a rectangular seat equal to twice its breadth. If the volume of the cylinder is V_1, V_2 when the seat is folded relative to its length and breadth, Find the ratio of V_1 and V_2?

(a) 1 : 1 (b) 1 : 3
(c) 2 : 1 (d) 2 : 3

RRB JE - 27/06/2019 (Shift-I)

Ans : (c)

On related to length –

Circumference of first cylinder $(2\pi r_1) = 2$, $h_1 = 1$

$r_1 = \frac{1}{\pi}$

On related to width –

Circumference of second cylinder $(2\pi r_2) = 1$, $h_2 = 2$

$$r_2 = \frac{1}{2\pi}$$

$$\frac{V_1}{V_2} = \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \frac{\pi \times \frac{1}{\pi^2} \times 1}{\pi \times \frac{1}{4\pi^2} \times 2} = 2:1$$

217. The total surface area of a solid cylinder is 462 cm^2. Its curve surface is one third of the entire surface area. Find its volume?

(a) 964 cm^3 (b) 810 cm^3
(c) 539 cm^3 (d) 1024 cm^3

RRB JE - 27/06/2019 (Shift-III)

Ans : (c) Given

Total Surface area of cylinder = 462 cm^2

$2\pi r (h+r) = 462$ ----(i)

Curved surface area of cylinder $= 462 \times \frac{1}{3}$

$2\pi rh = 154$ cm ----(ii)

from equation (i) and (ii)-

$$\frac{2\pi r(h+r)}{2\pi rh} = \frac{462}{154}$$

$$\frac{(h+r)}{h} = 3$$

$h + r = 3h$

$r = 2h$ ----(iii)

from equation (ii)–

$2\pi rh = 154$

$$2 \times \frac{22}{7} \times 2h \times h = 154$$

$$4 \times \frac{22}{7} \times h^2 = 154$$

$$h^2 = \frac{49}{4}$$

$$h = \frac{7}{2}$$

from equation (iii)–

$r = 2h$

$$r = 2 \times \frac{7}{2} = 7 \text{ cm}$$

Volume of cylinder ($\pi r^2 h$) =

$$\frac{22}{7} \times 7 \times 7 \times \frac{7}{2} = 11 \times 49 = 539 \text{ cm}^3$$

218. Find the volume of a cylinder whose radius is one-third of the radius of a circle with an area of 1386 cm. and the height is twice its radius?

(a) 3246 cm^3 (b) 3562 cm^3
(c) 2156 cm^3 (d) 2584 cm^3

RRB RPF Constable – 18/01/2019 (Shift-III)

Ans. (c) Area of circle = πR^2

$\therefore \pi R^2 = 1386$

$$\frac{22}{7} R^2 = 1386$$

$$R^2 = \frac{1386 \times 7}{22} = 63 \times 7$$

$$R = \sqrt{63 \times 7} = 3 \times 7$$

R = 21 cm

As per question,

Radius of cylinder = $21 \times \frac{1}{3} = 7$ cm

height of cylinder = 7 × 2 = 14 cm

Volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times 7 \times 7 \times 14 = 2156 \text{ cm}^3$$

219. A solid cylindrical metal with a diameter of 14 cm and a height of 15 cm is melted to form a solid wire whose length is 60 meters. Find the diameter of the wire.

(a) 7 mm (b) 14 mm
(c) 3.5 mm (d) 35 mm

RRB RPF Constable – 22/01/2019 (Shift-II)

Ans. (a) Diameter of cylindrical metal = 14cm

2r = 14cm

r = 7cm

The length of the wire shape

= 60m

Cylinder is melted to form of a wire, so the length of the wire.

h = 60m = 6000cm

Volume of solid cylinder = Volume of wire

$$\pi r_1^2 h_1 = \pi r_2^2 h_2$$

$$7 \times 7 \times 15 = r^2 \times 6000$$

$$r = \sqrt{\frac{7 \times 7}{400}} = \frac{7}{20} \text{cm}$$

$$r = \frac{7}{20} \times 10\text{mm}$$ $\boxed{1\text{cm} = 10\text{mm}}$

So diameter $= 2r = \frac{7}{2} \times 2 = 7$mm

220. To measure the volume of an irregular shaped stone, peter place the stone in a 5 cm semi-diameter container filled with water. If the water level of the box increases from 10 cm to 17 cm, then what is the volume of the stone.

(a) 500 cm^3 (b) 450 cm^3
(c) 550 cm^3 (d) 400 cm^3

RRB Group-D – 25/10/2018 (Shift-II)

Ans : (c) ∵ Semi diameter of box (r) = 5 सेमी.

Water level in the compartment 10 cm Greater than 17 cm.

hence height = 17-10 = 7 cm

Thus the box will be in the form of cylinder

Thus, the volume of cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times (5)^2 \times 7 = 22 \times 25 = 550 \text{ cm}^3$$

221. 42 cm diameter solid metal sphere is melted and made then a wire which has a diameter 7mm find the length of the wire.

(a) 1008 m (b) 1008 cm
(c) 2016 cm (d) 2016 m

RRB Group-D – 28/09/2018 (Shift-III)

Ans : (a) diameter of circle = 42 cm

Radius $= \frac{42}{2} = 21\text{cm} = 21 \times 10^{-2}$ m

diameter of wire (cylinder) = 7 mm

Radius (r) $= \frac{7}{2}\text{mm} = \frac{7}{2} \times 10^{-3}$

Volume of circle = Volume of cylinder (wire)

$$\frac{4}{3}\pi R^3 = \pi r^2 h$$

$$\frac{4}{3} \times (21 \times 10^{-2})^3 = \left(\frac{7}{2} \times 10^{-3}\right)^2 \times h$$

$$\frac{4}{3} \times 21 \times 21 \times 21 \times 10^{-6} = \frac{7}{2} \times \frac{7}{2} \times 10^{-6} \times h$$

$$h = \frac{4}{3} \times \frac{21 \times 21 \times 21 \times 4}{7 \times 7}$$

h = 16×3×21

h = 1008 m.

So, length of wire (h) = 1008 m

222. The diameter of the hemisphere connected to both ends of a cylinder with a diameter of 10 cm is a height of 7 cm. Find the surface area of this solid shape.

(a) 374 cm^2 (b) 1056 cm^2
(c) 594 cm^2 (d) 836 cm^2

RRB Group-D – 27/09/2018 (Shift-III)

Ans : (a) Surface are of solid shape =

Surface area of cylinder + area of circle (on both side)

$$= 2\pi r(h + r) + 2\pi r^2$$

$$= 2\pi r[(h + r) + r]$$

$$= 2 \times \frac{22}{7} \times \frac{7}{2}\left[\left(10 + \frac{7}{2}\right) + \frac{7}{2}\right]$$

$$= 22\left[\frac{27}{2} + \frac{7}{2}\right]$$

= 11 × 34

= 374 cm^2

223. The length of a hollow cylinder is 84 cm and its outer diameter is 72 cm and the inner diameter is 24 cm. If the of substance is 10 grams/cm^3, what will be the weight of the hollow cylinder?

(a) 295680 kg (b) 329680 kg
(c) 295.68 kg (d) 329.68 kg

RRB Paramedical Exam – 21/07/2018 (Shift-III)

Ans : (c) Height of a hollow cylinder = 84 cm

external diameter = 32 cm

external radius (R) = 16 cm

internal diameter = 24 cm

internal radius (r) = 12 cm

Density of substance of cylinder = 10 gr/cm^3

Density substance of cylinder $= \frac{10}{1000} = \frac{1}{100}$ kg/cm^3

Volume of hollow cylinder = $\pi h (R^2 - r^2)$

$$= \frac{22}{7} \times 84\left\{(16)^2 - (12)^2\right\}$$

= 22 × 12 × 28 × 4

= 88 × 12 × 28

= 29568 cm^3

Hence weight substance of hollow cylinder = volume × density $= \frac{29568}{100} = 295.68$ kg.

224. The outer and inner diameters of a hollow sphere are 12 cm and 8 cm respectively. It is melted to give the shape of a cylinder whose base diameter is 10 cm. what will be the height of the cylinder?

(a) 3.33 cm (b) 4.33 cm
(c) 3.17 cm (d) 4.17 cm

RRB Group-D – 05/12/2018 (Shift-I)

Ans : (c) given-

∵ External and internal diameter of hollow circle is 12cm and 8cm.

∴ External radius of hollow circle (R) = 6 cm

and internal radius (r_1) = 4 cm

diameter of cylinder (2r) = 16 cm

$$r_2 = \frac{16}{2} = 8 \text{ cm}$$

As per question-

Volume of hollow sphere = volume of cylinder

$$\frac{4}{3}\pi\left(R^3 - r_1^3\right) = \pi r_2^2 h$$

$$\frac{4}{3}\pi\left[(6)^3 - (4)^3\right] = \pi(8)^2 \times h$$

$$\frac{4}{3}\pi[216 - 64] = 64\pi h$$

$$\frac{1}{3}[152] = 16h$$

$$\Rightarrow 50.667 = 16h \quad \Rightarrow h = 3.17 \text{ cm}$$

225. The capacitance of a cylindrical tank is 20790 m^3. If its radius is 10.5 m, find its depth.

(a) 60 m (b) 120 m
(c) 30 m (d) 75 m

RRB Group-D – 03/12/2018 (Shift-III)

Ans. (a) : Capacitance of cylindrical tank = 20790 m^3

So, $\pi r^2 h = 20790$

$$\frac{22}{7} \times (10.5)^2 \times h = 20790$$

$$\frac{22}{7} \times 10.5 \times 10.5 \times h = 20790$$

$$\frac{22}{7} \times \frac{105}{10} \times \frac{105}{10} \times h = 20790$$

$$\frac{22 \times 21 \times 21}{7 \times 2 \times 2} \times h = 20790$$

$$\frac{11\times 63}{2}\times h = 20790$$
$$h = \frac{20790\times 2}{11\times 63}$$
$$h = 60 \text{ m}$$
So depth = 60 m

226. The radius of a sphere is three times the radius of the base of a cylinder. The height of a cylinder is 9 times the radius of its base. It the total surface area of the cylinder and the numerical value of the volume of the sphere are equal, so what is the height of the cylinder.

(a) 3 unit (b) 4.5 unit
(c) 5 unit (d) 2.25 unit

RRB Group-D – 10/12/2018 (Shift-I)

Ans. (c) : let radius of cylinder = r
then radius of circle $(r_2) = 3r$
height of cylinder (h) = 9r
as per question
Volume of circle = Surface area of cylinder
$$\frac{4}{3}\pi(r_2)^3 = 2\pi r(h+r)$$
$$\Rightarrow \frac{4}{3}\pi(3r)^3 = 2\pi r(9r+r)$$
$$\Rightarrow \frac{4}{3}\pi 27r^3 = 20\pi r^2$$
$$r = \frac{5}{9}\text{ unit}$$
$$h = 9r = 9\times\frac{5}{9}$$
h = 5
So height of cylinder = 5 unit

227. A rectangular Aluminium sheet with dimensions 22m × 10m is rolled into a cylinder so that the smaller side becomes the height of the cylinder. What is the volume of the cylinder formed

(a) 385 m^3 (b) 370 m^3
(c) 380 m^3 (d) 375 m^3

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (a) When the rectangular aluminum sheet is folded into a cylindrical form, the width (smaller side) will change as the height of the cylinder and the length will become the circumference of the base.
∴ height of cylinder (h) = 10 m
circumference's base of cylinder = $2\pi r$ = 22m.
$$\Rightarrow \quad 2\times\frac{22}{7}\times r = 22$$
$$r = \frac{7}{2}\text{m}$$
∵ Volume of cylinder (V) = $\pi r^2 h$
$$\therefore \quad V = \frac{22}{7}\times\frac{7}{2}\times\frac{7}{2}\times 10$$
$$V = 11\times 7\times 5 = 385 \text{ m}^3$$

228. What will be the volume of an perpendicular cylinder of radius 1 cm and height 2 cm. in cubic cm (cm^3).
$$\left(\textbf{Take } \pi = \frac{22}{7}\right)$$

(a) $\frac{22}{7}$ (b) 22
(c) 44 (d) $\frac{44}{7}$

RRB NTPC 17.01.2017 Shift-1

Ans : (d) Volume of perpendicular cylinder = $\pi r^2 h$
$$= \frac{22}{7}\times(1)^2\times 2$$
$$= \frac{22}{7}\times 2 = \frac{44}{7}\text{ cm}^3$$

229. A perpendicular cylinder whose radius is 7 cm and height is 2 cm. Their volume will be in cm^3.
$$\left(\textbf{Take } \pi = \frac{22}{7}\right)$$

(a) 308 (b) 308/21
(c) 308/7 (d) 310

RRB NTPC 17.01.2017 Shift-3

Ans : (a) Radius of cylinder (r) = 7 cm, height (h) = 2 cm then
Volume of cylinder= $\pi r^2 h$
$$= \frac{22}{7}\times 7^2\times 2$$
$$= 22\times 7\times 2 = 308 \text{ cm}^3$$

230 . What will be the volume of an perpendicular cylinder if its radius is 2.5 cm and height is 2 cm.

(a) 275 (b) 275/21
(c) 275/2 (d) 275/7

RRB NTPC 19.01.2017 Shift : 1

Ans : (d) volume of cylinder = $\pi r^2 h$
$$= \frac{22}{7}\times(2.5)^2\times 2$$
$$= \frac{275}{7}\text{cm}^3$$

231. What is the volume of a perpendicular cylinder whose radius is 2 cm and height is 2 cm.
$$\left(\pi = \frac{22}{7}\right)$$

(a) 175/7 (b) 176/21
(c) 176/7 (d) 176

RRB NTPC 19.01.2017 Shift : 3

Ans : (c) Volume of cylinder = $\pi r^2 h = \frac{22}{7}\times 2\times 2\times 2$
$$= \frac{22}{7}\times 8 = \boxed{\frac{176}{7}\text{cm}^3}$$

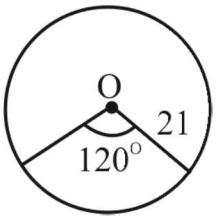
Type - 9

232. A circle sector with radius 21 cm and center is turned into a cone by turning 120°. Find the radius of the cone thus formed.

(a) 42 cm (b) 21 cm
(c) 7 cm (d) 7.5 cm

RRB JE - 29/05/2019 (Shift-I)

Ans : (c) Let radius of cone = R cm



Radius of sector = 21cm

Angle = 120^0

from, Angle $= \frac{\text{Arc}}{\text{radius}}$, $\left[\because \text{where angle will be in radian}\right]$

$120° \times \frac{\pi}{180°} = \frac{\text{Arc}}{21}$

$\frac{2\pi}{3} \times 21 = \text{arc}$

Circumference of cone base circle = length of arc.

$2\pi R = \frac{2\pi}{3} \times 21$

$\boxed{R = 7 \text{ cm}}$

233. A solid semi-sphere of metal is melted and is formed into the same radius of 'R' as a cone if the height of the cone is 'H' then.

(a) H = R (b) H = R/2
(c) H = 2R (d) H = R/3

RRB JE - 27/05/2019 (Shift-I)

Ans : (c) We know that

volume of semi circle $= \frac{2}{3}\pi r^3$

volume of come $= \frac{1}{3}\pi r^2 h$

As per question,

$\frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^2 h$ (r = R) (h=H)

$\frac{2}{3}\pi R^3 = \frac{1}{3}\pi R^2 H$

$\boxed{H = 2R}$

234. There are two cones that have a ratio of volume to 1 : 10 and a ratio of height to 2 : 5. Find the ratio of their base radius.

(a) 2:1 (b) 5:2
(c) 7:25 (d) 1:2

RRB JE - 28/05/2019 (Shift-I)

Ans : (d)

$V_1 : V_2 = 1 : 10$

$h_1 : h_2 = 2 : 5$

$\frac{V_1}{V_2} = \frac{\frac{1}{3}\pi r_1^2 h}{\frac{1}{3}\pi r_2^2 h}$

$\frac{1}{10} = \frac{r_1^2 \times 2}{r_2^2 \times 5}$

$\frac{1}{2} = \frac{r_1^2 \times 2}{r_2^2 \times 1}$

$\boxed{r_1 : r_2 = 1 : 2}$

235. The size of a solid is like a hemispherical with the same radius mounted on a cone of radius 2 cm and the height of the cone is equal to its radius. Find the volume of this solid.

(a) 2π cm^3 (b) 6π cm^3
(c) 8π cm^3 (d) 4π cm^3

RRB JE - 30/05/2019 (Shift-I)

Ans : (c)

$\because$ height of cone is equal to radius of cone

$\therefore$ total volume = Volume of semicircle + volume of cone

$= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h$ [h = r]

$= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^3 = \frac{3\pi r^3}{3}$

$= \pi \times (2)^3$ [r = 2 cm]

$= 8\pi$ cm^3

236. Find the total surface area of the cone with oblique height 2 *l* and radius r/2.

(a) $2\pi r l$ (b) $\pi r\,[l + (r/4)]$
(c) $2\pi r\,(l + r)$ (d) $\pi r\,(l + r)$

RRB RPF SI – 11/01/2019 (Shift-II)

Ans : (b) Total surface area of cone $= \pi r\,(l + r)$

where r = radius of cone

l = oblique height of cone

As per question

Total surface area $= \pi \times \frac{r}{2}\left(2l + \frac{r}{2}\right)$

$= \frac{\pi r}{2}.2\left[l + \left(\frac{r}{4}\right)\right]$

$= \pi r\left[l + \left(\frac{r}{4}\right)\right]$

237. Numerical values of the volume and surface area of a cone are equal. If 'h' and 'r' represent the height and base radius of the cone, find the value of $(1/h^2) + (1/r^2)$

(a) 2/9 (b) 1/9
(c) 1/3 (d) 3

RRB JE - 28/05/2019 (Shift-II)

Ans : (b) Volume of cone = surface area of cone

$\frac{1}{3}\pi r^2 h = \pi r l$, where, l = oblique height of cone

$\frac{1}{3} rh = l$ r – Radius, h – height

$\frac{r}{3l} = \frac{1}{h}$

Square on both side,

$\frac{1}{h^2} = \frac{r^2}{9l^2}$ .....(i)

and $\frac{1}{r^2} = \frac{h^2}{9l^2}$ .....(ii)

on equation (i + ii)

$\frac{1}{h^2} + \frac{1}{r^2} = \frac{r^2}{9l^2} + \frac{h^2}{9l^2}$

$\frac{1}{h^2} + \frac{1}{r^2} = \frac{h^2 + r^2}{9l^2}$

$\frac{1}{h^2} + \frac{1}{r^2} = \frac{l^2}{9l^2}$ $\left(\because l^2 = h^2 + r^2\right)$

$\frac{1}{h^2} + \frac{1}{r^2} = \frac{1}{9}$

238. The ratio of the area of the curved surface of two cones is 2 : 1. Their oblique height is in the ratio of 1 : 2. Find the ratio of their radius.

(a) 1 : 1 (b) 1 : 3
(c) 1 : 4 (d) 4 : 1

RRB JE - 01/06/2019 (Shift-I)

Ans : (d)

$\frac{\text{Curve surface area of first cone}}{\text{curve surface area of second cone}}$

$\frac{\pi r_1 \ell_1}{\pi r_2 \ell_2} = \frac{2}{1}$

given- ratio of oblique height $(\ell_1 : \ell_2) = 1:2$

$\frac{r_1 \times 1}{r_2 \times 2} = \frac{2}{1}$

$r_1 = 4r_2$

$\frac{r_1}{r_2} = \frac{4}{1}$

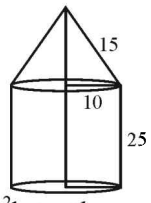
Ratio of radius $(r_1 : r_2) = 4 : 1$

239. A tent is in the shape of a cylinder and a cone has been established above it, the radius and height of the cylindrical part are 10 meters and 25 meters respectively. The radius is 10 meters of conical portion and the oblique height is 15 meters. calculate the amount of canvas required for the construction of this tent, taking an additional canvas of 20 percent for sealing, etc.

(a) 3783.26 m^2 (b) 4714.43 m^2
(c) 3772.14 m^2 (d) 2451.40 m^2

RRB JE - 26/06/2019 (Shift-III)

Ans : (d)



Area of tent = $\pi r^2 h + \pi r l$

area of tent = $\pi \times 10 \times 10 \times 15 + \pi \times 10 \times 15 = 500\pi + 150\pi$

$= 650\pi \text{ m}^2$

increased area = $650 \times \frac{120}{100} \times \frac{22}{7}$

$= \frac{780 \times 22}{7} = \frac{17160}{7} = 2451.40 \text{ m}^2$

240. Some circular cone shaped chocolates were made from a circular solid chocolates of radius 6 cm. Both the radius and height of this cone are 2 cm. The chocolate is kept in a 12 × 9 configuration in a square box. The height of the box is 2 cm. The box were filled with honey to keep the chocolate edible for a long time. The volume of honey in the box is what percentage of the total volume.

(a) 73.81 (b) 75.46
(c) 70.45 (d) 71.23

RRB RPF SI – 13/01/2019 (Shift-III)

Ans. (a) Given—

length of box = 4 × 12 = 48 cm

breadth of box = 4 × 9 = 36 cm

Number of cone = 12 × 9 = 108

Volume of honey = Volume of box − Number of cone × Volume of one cone

$= 48 \times 36 \times 2 - 108 \times \frac{1}{3} \times \frac{22}{7} \times 2^2 \times 2$

= 3456 – 905.14

= 2550.87

Honey's percentage = $\frac{2550.857}{3456} \times 100$

= 73.809%

= 73.81%

241. The height and oblique height of a perpendicular circular cone are 24 cm and 25 cm respectively. Find the curved surface area of the cone, assuming the value of π is 22/7.

(a) 572 cm^2 (b) 528 cm^2
(c) 539 cm^2 (d) 550 cm^2

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (d) According to the question, the height of the perpendicular cone is 24 cm and the oblique height is 25cm.

formula- $l^2 = h^2 + r^2$ $\begin{cases} h = 24\text{cm} \\ l = 25\text{cm} \end{cases}$

$(25)^2 = (24)^2 + r^2$

$625 = 576 + r^2$

$49 = r^2$

$\therefore$ $\boxed{r = 7 \text{ cm}}$

Curve surface area of cone $= \pi r l$

$= \frac{22}{7} \times 7 \times 25$

$= 22 \times 25$

$= 550 \text{ cm}^2$

242. The diameter and slant height of a conical tent are 16 meters and 5.6 meters respectively. If the width is to be maintained at 4 meters, how much cloth will be required for the construction of the tent.

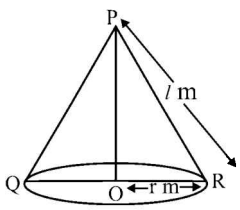
(a) 35m (b) 32m

(c) 32.5 m (d) 35.2 m

RRB Group-D – 24/09/2018 (Shift-I)

Ans : (d) Given- diameter of tent(d) = 16 m

breadth = 4 m



$\therefore$ Radius (OR = OQ) = 8 m $\left(\because r = \frac{d}{2}\right)$

Slant height of cone (l) = 5.6 m

curve surface of cone $= \pi r l$

$= \frac{22}{7} \times 8 \times 5.6 = 0.8 \times 8 \times 22 = 140.8 \text{ m}^2$

So require quantity of cloths for making tent

$= \frac{\text{Curve page of cone}}{\text{breadth}}$

$= \frac{140.8}{4} = 35.2 \text{ m}$

243. Curve surface of cone X is 5 times that of Curve surface of cone Y. Slant height of cone Y is 5 times slant height of cone X. what will be the ratio of the area of the base of the cone X and the cone Y.

(a) 5 : 1 (b) 125 : 1

(c) 625 : 1 (d) 25 : 1

RRB Group-D – 05/12/2018 (Shift-III)

Ans : (c) Curve surface area of cone $= \pi r l$

Curve surface of X cone = Curve surface of y cone $\times 5$

$\pi r_1 l_1 = \pi r_2 l_2 \times 5$

$r_1 l_1 = r_2 l_2 \times 5$.....(1)

$5 \times$ oblique height of cone x = oblique height of cone y

$\boxed{5 l_1 = l_2}$

on keeping l_2 value in equation (1)

$r_1 l_1 = r_2 \times 5 l_1 \times 5$

$\frac{r_1}{r_2} = \frac{25}{1}$

Ratio of base area of cone X and cone Y =

$\frac{\pi r_1^2}{\pi r_2^2} = \frac{(25)^2}{(1)^2} = 625 : 1$

244. The oblique height and diameter of a cone are 15 cm and 28 cm respectively. The volume of the cone is. ($\pi = 22/7$)

(a) $\frac{616}{3}\sqrt{29}\text{cm}^3$ (b) $\frac{308}{3}\sqrt{43}\text{cm}^3$

(c) $\frac{616}{3}\sqrt{43}\text{cm}^3$ (d) $\frac{308}{3}\sqrt{29}\text{cm}^3$

RRB Group-D – 04/12/2018 (Shift-III)

Ans. (a) As per question,

Slant height (ℓ) = 15cm

diameter of cone = 28cm

2r = 28cm

r = 14cm

$h = \sqrt{\ell^2 - r^2}$

$h = \sqrt{(15)^2 - (14)^2}$

$h = \sqrt{225 - 196}$

$h = \sqrt{29}$

then volume of cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 14 \times 14 \times \sqrt{29}$

$= \frac{22 \times 2 \times 7 \times 14 \times \sqrt{29}}{3 \times 7}$

$= \frac{44 \times 14}{3}\sqrt{29}$

$= \frac{616}{3}\sqrt{29}\text{cm}^3$

245. A solid round metal ball of 36 cm diameter is melted and small solid cones 12 cm in diameter and 12 cm high are made from it. Describe the number of cones made using this molten metal.

(a) 52 cone (b) 60 cone

(c) 48 cone (d) 54 cone

RRB Group-D – 11/12/2018 (Shift-III)

Ans : (d) diameter of solid sphere ball = 36 cm

radius of solid sphere ball (R) = 18 cm

diameter of cone = 12 cm

radius of cone (r) = 6 cm

height of cone (h) = 12cm

Suppose the number of cones formed by melting the solid sphere ball is n, then

$$n = \frac{\text{the volume of the solid sphere ball}}{\text{volume of cone}}$$

$$n = \frac{\frac{4}{3}\pi R^3}{\frac{1}{3}\pi r^2 h}$$

$$n = \frac{4\times18\times18\times18}{6\times6\times12}$$

n= 54 cone

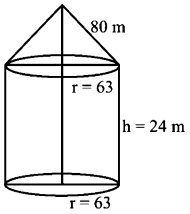
So the number of cone's of melted metal = 54 cone

246. A tent as such that its lower part is like a cylinder of 24 m height having 126 m diameter. Its apex is cone shaped with a base of the same diameter of 126 m and is 80 m slant high. Its canvas is 8 m wide. Calculate the length of canvas required to construct the tent.

(a) 3296 m (b) 3020 m
(c) 3168 m (d) 3190 m

RRB Group-D – 10/12/2018 (Shift-I)

Ans. (c) We know that $r = \frac{D}{2} = \frac{126}{2} = 63\text{m}.$



curve surface of tent = $2\pi rh + \pi rl$

$= \pi r(2h + l)$

$= \frac{22}{7}\times63(2\times24+80)$

= 198 (48 + 80)

= 198 (128)

= 25344 m^2

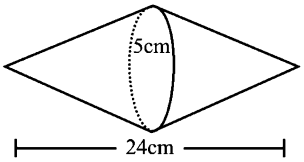
area of canvas = (l × b) = 25344

$l \times 8 = 25344$

$\therefore l = \frac{25344}{8}$

length of require canvas = 3168 m

247. The following structure is formed by connecting two perpendicular cones with their bases. The radius of each base is 5 cm. And the total length of this structure is 24 cm the total volume of this structure will be in cubic centimeters. ($\pi = 3.14$)



(a) 240 (b) 628
(c) 314 (d) 376.8

RRB Group-D – 08/10/2018 (Shift-I)

Ans. (b) : Volume of cone = $\frac{1}{3}\pi r^2 h$

radius = 5 cm

height = 12 cm

Volume of given figure = 2 × volume of one perpendicular cone

Volume of figure = $\left(\frac{1}{3}\times3.14\times5\times5\times12\right)\times2$

[where $\pi = 3.14$]

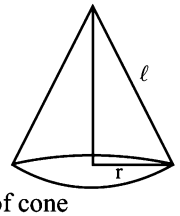
$= \frac{1884}{3} = 628\,\text{cm}^3$

248. What is the entire surface area of a cone, If 'r' is the radius and 'l' is the oblique height?

(a) $\pi r\,(l + r)$ (b) $2\pi r(l + r)$
(c) πrl (d) $2\pi rl^2$

RRB NTPC 29.04.2016 Shift : 3

Ans: (a)



Total surface area of cone

= area of base + area of curve surface

$= \pi r^2 + \pi r\ell \quad = \pi r(r + \ell)$

Type - 10

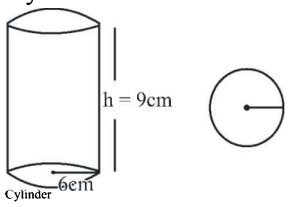
249. What will the radius of a sphere which is formed, by melting a cylinder. Whose radius is 6 cm. and height is 9 cm.

(a) $3\sqrt[3]{9}$ (b) $4\sqrt[3]{3}$
(c) $3\sqrt[3]{3}$ (d) 27

RRB RPF Constable – 24/01/2019 (Shift-III)

Ans. (a) According to question,

Volume of solid cylinder = Volume of molten sphere



$\pi r^2 h = \frac{4}{3}\pi R^3$

$(6)^2 \times 9 = \frac{4}{3}R^3$

$\frac{36\times9\times3}{4} = R^3$

$R = \sqrt[3]{9\times9\times3}$

$R = \sqrt[3]{3\times3\times3\times3\times3}$

radius of sphere (R) = $3\sqrt[3]{9}$

250. Two perpendicular cones of height 16.4 cm and 17.2 cm respectively whose base radius is 8.4 cm Both these cones are melted and formed into a sphere. Find the diameter of the sphere.

(a) 8.6 cm (b) 8.2 cm
(c) 8.4 cm (d) 16.8 cm

RRB JE - 22/05/2019 (Shift-III)

Ans : (d) Given radius of both cone $r_1 = r_2 = 8.4$ cm
height h_1 of first cone = 16.4 cm
height h_2 of second cone = 17.2 cm
Suppose radius of sphere = R
Volume of both cone = Volume of sphere

$$\frac{1}{3}\pi r_1^2 h_1 + \frac{1}{3}\pi r_2^2 h_2 = \frac{4}{3}\pi R^3$$

$$\frac{1}{3}\pi\left[r_1^2 h_1 + r_2^2 h_2\right] = \frac{4}{3}\pi R^3$$

$$\left[(8.4)^2 \times 16.4 + (8.4)^2 \times 17.2\right] = 4R^3$$

$$(8.4)^2\left[16.4 + 17.2\right] = 4R^3$$

$$\frac{8.4 \times 8.4 \times 33.6}{4} = R^3$$

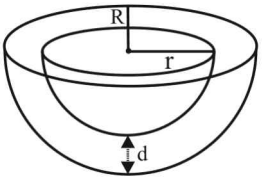
$R^3 = 8.4 \times 8.4 \times 8.4$
R = 8.4
diameter of sphere (d) = 2R
= 2 × 8.4
= 16.8 cm

251. Find the surface area of a semicircular bowl of thickness 'd' and internal radius 'r'.

(a) $\pi(4r^2 + 6rd + 3d^2)$ (b) $4\pi r^2 + 4\pi rd + 3d^2$
(c) $\pi(4r^2 + 3rd + d^2)$ (d) $4\pi r^2 + 6\pi rd + 3d^2$

RRB JE - 25/05/2019 (Shift-II)

Ans : (a)



Internal radius of semi-bowl sphere = r
thickness (d) = R – r
external radius of semi-bowl = R = d + r
surface area of semi-bowl sphere
$= 2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2)$
$= 2\pi\{(d+r)^2\} + 2\pi r^2 + \pi\{(d+r)^2 - r^2\}$
$= 2\pi\{d^2+r^2+2dr\} + 2\pi r^2 + \pi\{d^2+r^2+2rd - r^2\}$
$= 2\pi d^2 + 2\pi r^2 + 4\pi rd + 2\pi r^2 + \pi d^2 + 2\pi rd$
$= 3\pi d^2 + 4\pi r^2 + 6\pi rd$
$= \pi(4r^2 + 6rd + 3d^2)$

252. The height of a cone is 32 cm and the radius of its base is 8 cm. It is melted and converted into a sphere. Find the radius of that sphere.

(a) 6.5 cm (b) 2.5 cm
(c) 4 cm (d) 8 cm

RRB JE - 26/05/2019 (Shift-III)

Ans : (d) According to the question the volume of sphere = volume of cone

$$\Rightarrow \quad \frac{4}{3}\pi R^3 = \frac{1}{3} \times \pi \times 8 \times 8 \times 32$$

$$R^3 = \frac{8 \times 8 \times 32}{4}$$

$R^3 = 512$
R = 8 cm
∴ radius of sphere= 8 cm

253. A semi sphere with maximum volume is cut out from a cube of side 6 cm. Find the remaining volume.

(a) 36 – (3π/3) (b) [27 –(2 π /3) × 6]
(c) 216 – 12 π (d) 216 – 18 π

RRB JE - 29/05/2019 (Shift-III)

Ans : (d) cube volume of 6 cm side = $(6)^3 = 216$ cm^3

Volume of semi sphere $= \frac{2}{3}\pi r^3 = \frac{2}{3}\pi(3)^3 = \frac{2}{3} \times 27 \times \pi$

intended difference $= 216 - \frac{2}{3} \times 27 \times \pi$
= 216 – 18 π

254. The volume of a sphere divided by its surface area gives the answer 27 cm. Find the radius of the sphere.

(a) 243 cm (b) 81 cm
(c) 27 cm (d) 9 cm

RRB JE - 02/06/2019 (Shift-II)

Ans. (b) Volume of sphere = $\frac{4}{3}\pi r^3$
Surface area = $4\pi r^2$

$$\frac{\frac{4}{3}\pi r^3}{4\pi r^2} = 27 \text{ cm.}$$

r = (27 × 3) cm.
= 81 cm.

255. Find the volume of the sphere whose surface area is 1386 m^2.

(a) 3850 m^3 (b) 4851 m^3
(c) 4651 m^3 (d) 5711 m^3

RRB JE - 01/06/2019 (Shift-I)

Ans : (b) Given, surface area of sphere = 1386 m^2

$$4\pi r^2 = 1386$$

$$4 \times \frac{22}{7} \times r^2 = 1386$$

$4 \times r^2 = 63 \times 7$
$4 \times r^2 = 7 \times 3 \times 3 \times 7$

$$r = \sqrt{\frac{7 \times 7 \times 3 \times 3}{4}}$$

$$r = \frac{7 \times 3}{2}$$

$$r = \frac{21}{2} \text{ m}$$

volume of sphere $= \frac{4}{3}\pi r^3$

$= \frac{4}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2}$

$= 21 \times 21 \times 11$

$= 4851\ m^3$

256. A solid cylinder made of glass is 1 m high and the diameter of its base is 1.5 m wide. It is melted and turned into a solid sphere. The diameter of the sphere is.

(a) 1.5 m (b) 1 m
(c) 0.5 m (d) 2.5 m

RRB Group-D – 30/10/2018 (Shift-I)

Ans : (a) Volume of cylinder $= \pi r^2 h$

$= \pi\left(\frac{1.5}{2}\right)^2 \times 1 = \frac{2.25\pi}{4}$ cubic meter

Volume of cylinder = Volume of molten sphere

$\frac{2.25\pi}{4} = \frac{4}{3}\pi R^3$

$R^3 = \frac{2.25 \times 3}{16}$

$= \frac{3 \times 3 \times 3 \times 0.5 \times 0.5}{2 \times 2 \times 2 \times 2} = \frac{3^3}{2^3 \times 2^3}$

$R = \frac{3}{4}$ m

So diameter of sphere = 2R

$= 2 \times \frac{3}{4} = \frac{3}{2}$ m = 1.5 m

257. How many balls of 1 cm radius can be made from a steal sphere with a radius of 6 cm.

(a) 64 (b) 216
(c) 27 (d) 126

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (b) Volume of steel sphere in 6 cm radius

$= \frac{4}{3}\pi r^3 = \frac{4}{3} \times \pi \times (6)^3$

$= \frac{4}{3} \times \pi \times 216$

Volume of steel sphere in 1 cm radius $= \frac{4}{3} \times \pi r^3$

$= \frac{4}{3}\pi \times (1)^3$

$= \frac{4}{3}\pi \times 1$

Intended number of balls $= \frac{\text{Volume of large sphere}}{\text{Volume of small sphere}}$

$= \frac{\frac{4}{3}\pi \times 216}{\frac{4}{3}\pi \times 1} = 216$

= 216 (balls can be made)

258. What is the surface area of the sphere of radius 1 cm. $\left(\pi = \frac{22}{7}\right)$

(a) 89/7 (b) 89/21
(c) 88/21 (d) 88/7

RRB NTPC 17.01.2017 Shift-1

Ans : (d) Surface area of sphere $= 4\pi r^2$

$= 4 \times \frac{22}{7} \times (1)^2$

Surface area of sphere $= \frac{88}{7} cm^2$

259. What will be the area of the surface of a solid sphere whose radius is 3.5 cm. (given that $\pi = \frac{22}{7}$)

(a) 154/7 cm^2 (b) 154 cm^2
(c) 54 cm^2 (d) 54/7 cm^2

RRB NTPC 17.01.2017 Shift-3

Ans : (b) Surface area of solid sphere $= 4\pi r^2$

$= 4 \times \frac{22}{7} \times 3.5 \times 3.5$

$= 4 \times 22 \times 0.5 \times 3.5 = 154\ cm.^2$

260. The diameter of a cylinder is 6 cm and it is filled with water to a height of 10 cm. If a spherical ball is put in it, then the water level increase by 0.5 cm. Find the diameter of the ball.

(a) 3 (b) 1.5
(c) 2.4 (d) 2.8

RRB NTPC 03.04.2016 Shift : 3

Ans :(a) Increase of water volume by 0.5 cm in the cylinder = Volume of ball

$\pi \times (3)^2 \times 0.5 = \frac{4}{3} \times \pi \times r^3$

$r^3 = \frac{27 \times 0.5}{4} = \frac{27}{8}$ r = 3/2 cm

∴ diameter of ball = 2r = 2 × 3/2 = 3 cm

261. What would be the surface area of a solid sphere its radius is 1.5 cm.? ($\pi = \frac{22}{7}$)

(a) 190/21 (b) 190/7
(c) 198/7 (d) 198/21

RRB NTPC 19.01.2017 Shift : 1

Ans : (c) Surface area of soild sphere $= 4\pi r^2$

r = 1.5cm., $\pi = \frac{22}{7}$

$= 4 \times \frac{22}{7} \times (1.5)^2 = \frac{198}{7}$ square cm.

262. Find the area of a solid sphere whose radius is 2 cm. (Given $\pi = \frac{22}{7}$)

(a) 352/7 (b) 350/21
(c) 352/21 (d) 350/7

RRB NTPC 19.01.2017 Shift : 3

Ans : (a) Area of sphere $= 4\pi r^2 = 4\times\frac{22}{7}\times(2)^2$

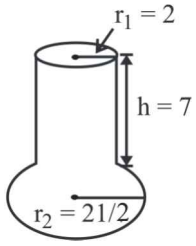
$= \frac{16\times 22}{7} = \boxed{\frac{352}{7}\text{cm}^2}$

263. A spherical glass vessel has a cylindrical upper part 7 cm long and 4 cm in diameter. The diameter of the spherical part is 21cm. Find the quantity of water it can hold in cm^3.

(a) 4932 cm^3 (b) 4939 cm^3
(c) 4930 cm^3 (d) 4929 cm^3

RRB ALP & Tec. (17-08-18 Shift-III)

Ans : (b)



Intended quantity of water

= Volume of cylinder + Volume of sphere

$= \pi r_1^2 h + \frac{4}{3}\pi r_2^3 = \pi\left[r_1^2 h + \frac{4}{3}r_2^3\right]$

$= \frac{22}{7}\left[2^2\times 7 + \frac{4}{3}\times\left(\frac{21}{2}\right)^3\right]$

= 4939 cubic cm.

264. The area of the base of a cone is 64π cm^2 while its slant height is 17 cm. This cone is remolded to obtain a solid sphere. The radius of this sphere will be

(a) $2\sqrt[3]{30}$ cm (b) $2\sqrt[3]{40}$ cm
(c) $8\sqrt[3]{30}$ cm (d) 6.5 cm

RRB ALP & Tec. (14-08-18 Shift-II)

Ans : (a) Given—

base area of cone $=64\pi$ square cm.

$\Rightarrow$ $\pi r^2 = 64\pi$

$\Rightarrow$ r = 8 cm

$\because$ Slant height of cone $(l) = 17$ cm

$\therefore$ height of cone $(h) = \sqrt{\ell^2 - r^2}$

$= \sqrt{(17)^2 - (8)^2}$

$= \sqrt{289 - 64} = \sqrt{225}$

= 15 cm.

According to question $\rightarrow$ Volume of cone = Volume of sphere

$\Rightarrow$ $\frac{1}{3}\pi r^2 h = \frac{4}{3}\pi R^3$

$\Rightarrow$ $64\times 15 = 4R^3$

$\Rightarrow$ $240 = R^3$

$\Rightarrow$ $R = \sqrt[3]{240}$

$= \sqrt[3]{8\times 30} = 2\sqrt[3]{30}$ cm.

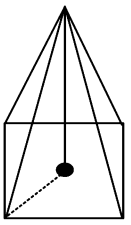
Type - 11

265. The base of a right pyramid is a square with a 16 unit diagonal. Its one slant edge is 17 units. Find its vertical height.

(a) 30 (b) 12
(c) 25 (d) 15

RRB JE - 24/05/2019 (Shift-III)

Ans : (d)



$h = \sqrt{(17)^2 - (8)^2}$

$h = \sqrt{289 - 64}$

$h = \sqrt{225}$

h = 15 unit

266. The base of a triangular prism is a triangle with 8, 15, 17 unit sides and its height is 20 units. Find its total surface area.

(a) 920 (b) 1020
(c) 960 (d) 940

RRB JE - 02/06/2019 (Shift-I)
RRB RPF Constable – 17/01/2019 (Shift-I)

Ans : (a) Given,

Sides of triangular prism is 8, 15, 17 (unit)

height (h) = 20 unit

Total surface area = perimeter of base × height + 2 × area of base

$= (8 + 15 + 17) \times 20 + 2 \times \frac{1}{2} \times 8 \times 15$

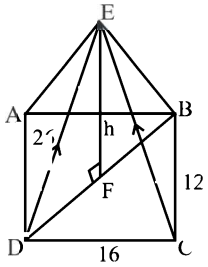
= 40 × 20 + 120 = 920 square unit

267. The base of the pyramid is a rectangle whose length and width are 16 cm and 12 cm respectively. If all the lateral edges passing through the top of the right rectangular pyramid are 26 cm in length, find the volume of the pyramid in cubic centimeters.

(a) 1536 (b) 1024
(c) 718 (d) 2072

RRB Group-D – 03/10/2018 (Shift-I)

Ans : (a)



Diagonal of rectangle

$= \sqrt{(16)^2 + (12)^2} \Rightarrow \sqrt{400} = 20\,\text{cm}$

We draw the perpendicular from the Vertex E at the base of the pyramid which is at point F. And the height of the pyramid is h and the hypotenuse is 26 cm.

$\text{height}(h) = \sqrt{(26)^2 - (10)^2} = \sqrt{576} = 24\text{ cm}$

Volume of pyramid $= \frac{1}{3} \times$ area of rectangle $\times$ height

$= \frac{1}{3} \times 16 \times 12 \times 24$

$= 1536\text{ cm}^3$

268. Ratio of the sides of the base of a symmetrical triangular prism 8 : 15 : 17 is. If the height of the prism is 21 cm and the sum of the area of the lateral surfaces is 840 cm^2, Find the volume of the prism in cubic centimeters.

(a) 1200 (b) 1260
(c) 1280 (d) 1240

RRB Group-D – 12/10/2018 (Shift-II)
RRB Group-D – 01/10/2018 (Shift-II)

Ans : (b) Suppose bases of prism are 8x, 15x and 17x.
total area of lateral surfaces = 840 cm^2
lateral surface area = perimeter of base (right triangular) × h

$840 = (8x + 15x + 17x) \times 21$
$840 = 40x \times 21$
$x = \frac{840}{40 \times 21} = 1$

So sides of triangular is 8cm, 15 cm, 17 cm
Triangle (base of prism) is right angle triangled

$\therefore (17)^2 = (8)^2 + (15)^2$
$289 = 289$

Volume of prism = area of base × height

$= \frac{1}{2} \times 8 \times 15 \times 21 = 60 \times 21 = 1260\text{ cm}^3$

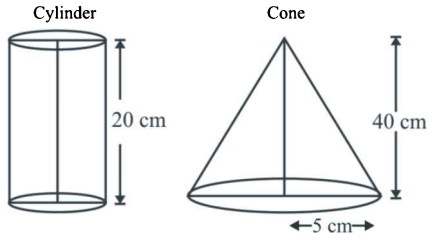
Type - 12

269. Find the percentage of material wasted in converting a cylinder of base diameter 10 cm and height 20 cm into same base but double the height of a cone.

(a) 25% (b) 20%
(c) 35% (d) 33.3%

RRB JE - 23/05/2019 (Shift-I)

Ans : (d)



Volume of cylinder $= \pi r^2 h = \pi \times 25 \times 20$
$= 500\pi\text{ cm}^3$

Volume of formed cone $= \frac{1}{3}\pi R^2 H$

$\left\{\begin{matrix} H = 2h \\ r = R \end{matrix}\right\}$ given

$= \frac{1}{3}\pi \times 25 \times 40$

$= \frac{1000\pi}{3}$

quantity of waste materials $= 500\pi - \frac{1000\pi}{3} = \frac{500\pi}{3}$

Percentage of waste materials % $= \frac{\frac{500\pi}{3}}{500\pi} \times 100$

$= 33.3\%$

270. The ratio of the radii of the bases of a cylinder and cone is $\sqrt{3} : \sqrt{2}$ and the ratio of their heights is $\sqrt{2} : \sqrt{6}$, then find the ratio of their volume.

(a) $\sqrt{3} : \sqrt{3}$ (b) $3\sqrt{3} : 2$
(c) $\sqrt{3} : \sqrt{2}$ (d) $3\sqrt{3} : \sqrt{2}$

RRB JE - 23/05/2019 (Shift-II)

Ans : (b)

$$\frac{\text{Volume of cylinder}}{\text{Volume of cone}} = \frac{\pi R_1^2 h_1}{\frac{1}{3}\pi R_2^2 h_2} = 3\left(\frac{R_1}{R_2}\right)^2\left(\frac{h_1}{h_2}\right) = \frac{V_1}{V_2}$$

$$\frac{V_1}{V_2} = 3\left(\frac{\sqrt{3}}{\sqrt{2}}\right)^2 \times \left(\frac{\sqrt{2}}{\sqrt{6}}\right) = 3 \times \frac{3}{2} \times \frac{1}{\sqrt{3}}$$

$$\frac{V_1}{V_2} = \frac{3\sqrt{3}}{2}$$

$\therefore V_1 : V_2 = 3\sqrt{3} : 2$

271. Find the area of the hexagon of side 6 cm.

(a) $72\sqrt{3}$ cm^2 (b) $64\sqrt{3}$ cm^2
(c) $108\sqrt{3}$ cm^2 (d) $54\sqrt{3}$ cm^2

RRB JE - 24/05/2019 (Shift-III)

Ans : (d) formula : area of regular hexagon

$= 6\frac{\sqrt{3}}{4} \times (\text{side})^2$

$= 6\frac{\sqrt{3}}{4} \times 6 \times 6$

$= 54\sqrt{3}\text{ cm}^2$

272. How much material is wasted when a solid cylinder of 6 cm. radius and 6 cm. height is made from a cone of 24 cm. height and 8 cm. radius?

(a) 57.8% (b) 37.5%
(c) 52% (d) 64%

RRB RPF SI – 16/01/2019 (Shift-II)

Ans : (a) Volume of cylinder $= \pi r^2 h$

$= \pi \times 6 \times 6 \times 6$

$= 216\pi$

Volume of cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\pi \times 8 \times 8 \times 24$

$= 512\pi$

Wasted material % $= \frac{(512\pi - 216\pi)\times 100}{512\pi}$

$= \frac{296\pi}{512\pi} \times 100 = \boxed{57.81\%}$

273. The volume of a cone, hemisphere and cylinder with equal radius and equal height are P, Q and R respectively. Find the value of P : Q : R.

(a) 1 : 3 : 2 (b) 3 : 4 : 2
(c) 2 : 1 : 3 (d) 1 : 2 : 3

RRB JE - 27/05/2019 (Shift-I)

Ans : (d)

Volume of cone $= \frac{1}{3}\pi r^2 h$

Volume of hemisphere $= \frac{2}{3}\pi r^3$

Volume of cylinder $= \pi r^2 h$

$P : Q : R = \frac{1}{3}\pi r^3 : \frac{2}{3}\pi r^3 : \pi r^3$ $(\because r = h)$

$\boxed{P : Q : R = 1 : 2 : 3}$

274. A square sheet of paper is fold into a cylinder by turning is side. Find the ratio of the side of the square and the radius of the base of the cylinder.

(a) 2π : 1 (b) 1 : 4 π
(c) 1 : π (d) 1 : 2 π

RRB JE - 29/05/2019 (Shift-III)

Ans : (a) Suppose side of square = x

radius of cylinder = r

from question–

Circumference of the base of cylinder = side of square

$2\pi r = x$

$\frac{x}{r} = \frac{2\pi}{1}$

$x : r = 2\pi : 1$

275. What is the ratio between the volume of a cube and the volume of a sphere that will fit perfectly in the cube?

(a) 3 : π (b) 6 : π
(c) 3 : 4 (d) π : 1

RRB JE - 30/05/2019 (Shift-III)

Ans : (b) diameter of sphere = side of cube

$2r = a$

$r = \frac{a}{2}$

$\frac{\text{Volume of cube}}{\text{Volume of sphere}} = \frac{a^3}{\frac{4}{3}\pi\left(\frac{a}{2}\right)^3}$

$= \frac{a^3}{\frac{4}{3}\pi\frac{a^3}{8}}$

$= \frac{1}{\frac{\pi}{6}} = 6 : \pi$

276. The height of a cube and a sphere is the same. Find the ratio of their volume.

(a) 6 : π (b) 4 : π
(c) 2 : π (d) 3 : π

RRB JE - 28/05/2019 (Shift-II)

Ans : (a) Height of sphere = height of cube

$2r = a$

$r = \frac{a}{2}$

$\Rightarrow \frac{\text{Volume of cube}}{\text{Volume of sphere}} = \frac{a^3}{\frac{4}{3}\pi r^3}$

$= \frac{a^3}{\frac{4}{3}\pi \times \left(\frac{a}{2}\right)^3} = \frac{a^3}{\frac{4}{3}\pi \times \frac{a^3}{8}}$

$= a^3 \times \frac{6}{\pi a^3} = 6 : \pi$

277. The area of a rectangle is 9/20 times the area of a square. If the length and breadth of the rectangle are in the ratio 5 : 4, find the ratio of the perimeter of the rectangle and the square.

(a) 25:48 (b) 27:20
(c) 27:40 (d) 25:45

RRB JE - 31/05/2019 (Shift-III)

Ans. (c) Let length of rectangle is 5x and breadth is 4x

area of rectangle = $5x \times 4x = 20x^2$

According to the question

Area of square $\times \frac{9}{20}$ = area of rectangle

Area of square $\times \frac{9}{20} = 20x^2$

Area of square $= \frac{400x^2}{9}$

Side of square $= \frac{20x}{3}$

$\frac{\text{Perimeter of rectangle}}{\text{Perimeter of square}} = \frac{2\times(5x+4x)}{4\times\frac{20x}{3}}$

$= \frac{18x\times 3}{80x} = \frac{27}{40}$

Required ratio = 27 : 40

278. How many carpets of 6 m, 4 m will be required to lay on the floor of a room measuring 40 m × 24 m?

(a) 40 (b) 12
(c) 18 (d) 15

RRB JE - 02/06/2019 (Shift-III)

Ans : (a) $\frac{\text{area of room's floor}}{\text{area of 1 carpet}} = \text{number of carpet}$

$\frac{40\times24}{6\times4} = \text{Number of carpets}$

Number of carpets = 40

279. The length of a room is 5.5 meters and the breadth is 3.75 meters. Find the cost of laying slab at the rate of Rupees 800 per m^2 on the floor.

(a) Rs. 15550 (b) Rs. 15600
(c) Rs. 16500 (d) Rs. 15000

RRB Paramedical Exam – 21/07/2018 (Shift-I)

Ans : (c) Given− length of room = 5.5 m
breadth of room = 3.75 m
area of room = area of floor = 5.5 × 3.75 = 20.625 m^2
cost of laying slab on the floor = 20.625 × 800
= Rs. 16500.00

280. The capacity to fill water in a drum is 0.6. when 38 liters of water is taken out, then it is 0.125 full. What will be the capacity of the drum in liters?

(a) 40 (b) 120 (c) 80 (d) 160

RRB RPF Constable – 19/01/2019 (Shift-III)

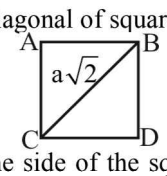
Ans : (c) Let capacity of drum is x litre
then, According to question,
$0.6x - 38 = 0.125x$
$\Rightarrow 0.6x - 0.125x = 38$
$\Rightarrow 0.475x = 38$
$x = \frac{38}{0.475} = \frac{38\times1000}{475}$ x = 80 litres

281. If only diagonal measurements are given, which of the following quadrilateral can be constricted.
1. Square 2. Rectangle 3. Rhombus

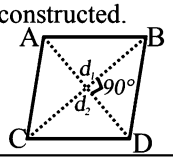
(a) 1 and 3 (b) 1 and 2
(c) only 1 (d) only 3

RRB JE - 01/06/2019 (Shift-I)

Ans : (a) Square and rhombus are geometric figures which have the same four sides.
We know, diagonal of square = $a\sqrt{2}$



With this, the side of the square can be known and the square can be constructed.
The diagonals of the rhombus intersect each other at 90°. Hence side of the rhombus will be known, so the rhombus can be constructed.



282. The ratio of length, breadth and height of room is 5 : 3 : 2. If the length is 2 times, the breadth is one third and the height is halved, then the percentage of expenditure incurred in painting the four walls of the room will be reduced by what percentage?

(a) 32 (b) 28.75
(c) 31.25 (d) 30.75

RRB RPF SI – 13/01/2019 (Shift-III)

Ans. (c) Let length = 5x
breadth = 3x
height = 2x
area of room = 2 (l + b) × h
= 2(5x + 3x) × 2x
= $32x^2$
The length is doubled, breadth is one third and height is halved, then the new area obtained
= 2(10x + x) × x = $22x^2$
Percentage reduction in expenditure

$$= \frac{(32x^2 - 22x^2)}{32x^2}\times100$$

$$= \frac{10x^2\times100}{32x^2} = \frac{500}{16} = 31.25\%$$

283. The volume of right circular cone whose base radius is equal to 5/9 of its height and equal to the volume of the sphere then what is the ratio of the radius of the cone and the sphere?

(a) $\sqrt[3]{4}:\sqrt[3]{3}$ (b) $\sqrt[3]{4}:\sqrt[3]{2}$
(c) 1:1 (d) $\sqrt[3]{60}:3$

RRB RPF SI – 12/01/2019 (Shift-II)

Ans. (d) Let height of cone =h, radius = r_1

$\therefore$ Radius $(r_1) = \frac{5}{9}h$

$\therefore \quad h = \frac{9}{5}r_1$

Let radius of sphere = r_2

$\because$ Volume of cone = Volume of sphere

$$\frac{1}{3}\pi r_1^2 h = \frac{4}{3}\pi r_2^3$$

$$r_1^2 \times \frac{9}{5}r_1 = 4r_2^3$$

$$\left(\frac{r_1}{r_2}\right)^3 = \frac{20}{9}$$

$$\frac{r_1}{r_2} = \sqrt[3]{\frac{20\times3}{9\times3}} = \frac{\sqrt[3]{60}}{3}$$

$\therefore \quad r_1 : r_2 = \sqrt[3]{60} : 3$

284. A right circular cylindrical jar with a radius of 14 cm has some water. If a solid spherical ball of radius 5.6 cm is dropped in the jar and ball is fully submerged, then calculate the approximate rise in the height of the water length in the jar.

(a) 1.2 cm (b) 1.1 cm
(c) 9.0 cm (d) 1.0 cm

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (a) Let water level will raise to the height of h in the jar.
Radius of circular cylinder = 14 c.m.
Radius of circular ball = 5.6 c.m
Volume of circular cylinder = Volume of circular ball.

$$\pi r_1^2 h = \frac{4}{3}\pi r_2^3, \quad r_1^2 h = \frac{4}{3} r_2^3$$

$$14 \times 14 \times h = \frac{4}{3} \times 5.6 \times 5.6 \times 5.6$$

$$h = \frac{4}{3} \times \frac{5.6 \times 5.6 \times 5.6}{14 \times 14}$$

$$h = 1.194 \approx 1.2 \text{ cm}$$

285. How many bricks of 20 cm × 10 cm size would be required for a 16 m long and 10 m wide floor?

(a) 8400 (b) 9000
(c) 8500 (d) 8000

RRB Group-D – 20/09/2018 (Shift-III)

Ans : (d) Number of bricks $= \frac{\text{Floor area of hall}}{\text{area of bricks}}$

$$= \frac{16 \times 100 \times 10 \times 100}{20 \times 10}$$

$$= 8000$$

286. If the perimeter of a circular plot is equal to the perimeter of a square plot, what will be the ratio of their area?

(a) 6 : 11 (b) 14 : 11
(c) 12 : 11 (d) 7 : 11

RRB Group-D – 22/09/2018 (Shift-I)
RRB Group-D – 29/10/2018 (Shift-III)

Ans : (b) Given-
circumference of a circle = perimeter of square
$2\pi r = 4a$

$$a = \frac{\pi r}{2}$$

Ratio of areas $= \frac{\pi r^2}{a^2} = \frac{\pi r^2}{\frac{\pi^2 r^2}{4}} = \frac{4 \times 7}{22} = \frac{14}{11}$

287. The size of a brick is 20 cm × 12 cm × 8 cm. How many bricks will be required to build a wall 16.8 m height, 2.5 m wide and 12 cm thick.

(a) 2652 (b) 6252
(c) 5262 (d) 2625

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (d) Number of bricks

$$= \frac{16.8 \times 100 \times 2.5 \times 100 \times 12}{20 \times 12 \times 8}$$

$$= \frac{1680 \times 250 \times 12}{20 \times 12 \times 8} = \frac{210 \times 250}{20}$$

$$= \frac{52500}{20} = 2625$$

288. A bowl of wood that is 1.8m long, 1.5 m wide and 4 m high is filled with an iron cube that is 6 cm wide. Find the number of iron cubes that can be fit in the basket.

(a) 13,500 (b) 15,000
(c) 14,500 (d) 14,000

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (b) Number of iron cubes

$$= \frac{1.8 \times 100 \times 1.5 \times 100 \times 1.2 \times 100}{6 \times 6 \times 6}$$

$$= \frac{180 \times 150 \times 120}{6 \times 6 \times 6} = 15000$$

289. The volume of a right circular cone whose base radius is one third of its height is equal to the volume of a hemisphere. Then what is the ratio of the radii of the cone and hemisphere?

(a) $2:3$ (b) $\sqrt[3]{2}:\sqrt[3]{3}$
(c) $\sqrt[3]{3}:\sqrt[3]{2}$ (d) $\sqrt[3]{2}:1$

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (b) : Let cone's radius is R and height H
given,

$$R = \frac{H}{3} \quad \text{.....(i)}$$

Let semi sphere's radius is r
As per question
Volume of cone = Volume of semi sphere

$$\frac{1}{3}\pi R^2 H = \frac{2}{3} r^3 \pi$$

cm (i) से $R = \frac{H}{3}$

$$\frac{1}{3} \times 3R^3 = \frac{2}{3} r^3$$

$$\frac{R^3}{r^3} = \frac{2}{3}$$

$$\left(\frac{R}{r}\right)^3 = \frac{2}{3}$$

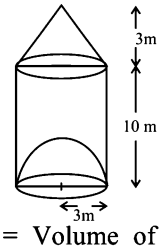
$$\frac{R}{r} = \sqrt[3]{\frac{2}{3}}$$

290. The top of the rocket is shaped like an elliptical cone with a height of 3m while its lower part is cylindrical with a radius of 3m and height 10m. To make room for the engine, the base of the rocket is cut in a semicircular shape from inside, whose radius is also 3m. Find the total volume of the rocket in cubic meters.

(a) 72π (b) 90π
(c) 100π (d) 81π

RRB Group-D – 29/10/2018 (Shift-III)

Ans : (d)



Volume of rocket = Volume of cone + Volume of cylinder – volume of semi circle

$$=\frac{1}{3}\pi r^2 h+\pi r^2 H-\frac{2}{3}\pi r^3=\frac{1}{3}\pi r^2\left(h+3H-2r\right)$$

$$=\frac{1}{3}\pi\times3\times3\left(3+3\times10-2\times3\right)$$

$$=3\pi\left(33-6\right)$$

$$=3\pi\times27=81\pi \text{ cubic meter}$$

291. A sphere was divided in the ratio 2 : 3. The larger part is inserted as a cone whose height is equal to the radius of its base, while the smaller part is inserted as a cylinder, whose height is equal to the radius of its base. What will be the ratio of the radii of the base of the cone to the height of the cylinder?

(a) 3:1 (b) $\sqrt[3]{9}:\sqrt[3]{2}$

(c) $\sqrt[3]{9}:1$ (d) $1:\sqrt[2]{3}$

RRB Paramedical Exam – 20/07/2018 (Shift-III)

Ans : (b) large part of sphere $=\frac{3}{5}$

small part of sphere $=\frac{2}{5}$

in cone $r_c = h$

Volume of cone $=\frac{1}{3}\pi r_c^2 h=\frac{3}{5}\times\frac{4}{3}\pi r^3$

$$r_c^3=\frac{12}{5}r^3 \text{.....(i)}$$

Volume r_c small part of cylinder of sphere $r_c = h$

$$\pi r_c^2 h=\frac{2}{5}\times\frac{4}{5}\pi r^3\Rightarrow h^3=\frac{8}{25}r^3 \text{.....(ii)}$$

On dividing equation (i) into equation (ii)

$$\frac{r_c^3}{h^3}=\frac{12}{5}\times\frac{15}{8}$$

$$\frac{r_c^3}{h^3}=\frac{9}{2}$$

$$\frac{r_c}{h}=\sqrt[3]{\frac{9}{2}}$$

$$\boxed{r_c:h=\sqrt[3]{9}:\sqrt[3]{2}}$$

option (b) will be correct answer

292. What will be the ratio of the area of a hexagon with the same perimeter and the area of 1 square?

(a) 2 : 3 (b) $1:\sqrt{3}$

(c) $4:3\sqrt{3}$ (d) $\sqrt{3}:2$

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (d) A circumference = equal perimeter

Let side of right hexagon X and side of square y

Then perimeter = 6x = 4y

$$\frac{x}{y}=\frac{4}{6}$$

$$\frac{x}{y}=\frac{2}{3}$$

$$\text{Ratio of square area}=\frac{\frac{3\sqrt{3}}{2}\times(x)^2}{(y)^2}$$

$$=\frac{\frac{3\sqrt{3}}{2}\times(2)^2}{(3)^2}=\frac{6\sqrt{3}}{9}=\frac{2\sqrt{3}}{3}=\frac{2\sqrt{3}}{\sqrt{3}\times\sqrt{3}}=\frac{2}{\sqrt{3}}$$

Intended ratio = $2:\sqrt{3}$ या $\sqrt{3}:2$

293. A company plans to pack spherical thin hollow balls in long circular cylinder shape cartons. where there is no lid on either side of the box. If the ball and box are made of the same material, calculate the ratio of the shell and the material used in the box.

(a) $\frac{\pi}{2}$ (b) 1

(c) π (d) 2π

RRB Group-D – 07/12/2018 (Shift-III)

Ans : (b) Let sphere's radius is r

Area of sphere = $4\pi r^2$ .....(i)

Area of made boxes for keeping sphere = Curve page of boxes

$2\pi rh = 2\pi r\times2r$

$= 4\pi r^2$ ....(ii)

from equation (i) and (ii)

$$\frac{\text{area of sphere}}{\text{area of Boxes}}=\frac{4\pi r^2}{4\pi r^2}=1$$

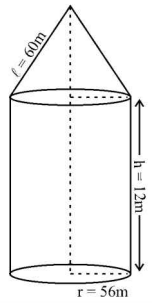
294. A tent is such that its lower part is like a cylinder of 12 meters height with a diameter of 112 meters. Its top is like a cone with a base of the same diameter of 112 meters. Its top is like a cone with a base of the same diameter of 112 meters and is 60 meters slant height. And its canvas is 4 meters wide. Calculate the required length of the canvas to make the tent.

(a) 3276 m (b) 3696 m

(c) 4686 m (d) 4096 m

RRB Group-D – 04/12/2018 (Shift-III)

Ans. (b)



Total area of tent
= Surface area of cylinder + Surface area of cone

$= 2\pi rh + \pi r\ell$

$= \frac{22}{7} \times 56(2 \times 12 + 60)$

$= 22 \times 8 \times 84 = 14784$ मी.^{2}

Let length of canvas $= l$ m

So area of canvas = area of tent

$4 \times \ell = 14784$

$\therefore \quad \ell = 3696$ m

295. A solid sphere of 5 cm radius is melted to from a solid marigold with a maximum possible number of 1 cm radius. If the balls are placed optimally in a cubic box, what percentage of the space will be used in the box?

(a) 52.38% (b) 55.76%
(c) 72.53% (d) 47.62%

RRB Group-D – 01/12/2018 (Shift-II)

Ans : (d) Side of cube = 10 cm (diameter of sphere)
radius of sphere = 5 cm
as per question
Empty place in box = volume of cube – Volume of sphere

$= 10 \times 10 \times 10 - \frac{4}{3}\pi \times 5^3$

$= 1000 - 523.80$

$= 476.19 cm^3$

Percentage of empty place in boxes $= \frac{476.19 \times 100}{1000}$

$= 47.619 \cong 47.62\%$

296. The area of the cross section of a pipe is 10.4 cm^2 and water flowed through it at a rate of 54 km/h. If the pipe is always filled with 60% flow, then in 5 minutes find the volume of water passing through the pipe.

(a) 2808 (b) 2008
(c) 2010 (d) 2101

RRB Group-D – 28/11/2018 (Shift-I)

Ans : (a) Volume of taking water by any pipe = cross section area × Velocity of water

$= 10.4 cm^2 \times 54 km/h \times 5 min \times \frac{60}{100}$

$= 10.4 \times 10^{-4} m^2 \times 54 \times \frac{5}{18} m/sec \times 5 \times 60 sec \times \frac{60}{100}$

$= 10.4 \times 10^{-4} m^2 \times 15 \times 5 \times 36$

$= 28080 \times 10^{-4} m^3$

$= 28080 \times 10^{-4} \times 1000\ \ell \quad \therefore\ 1m^3 = 1000\ \ell$

$= 28080 \times 10^{-4} \times 10^3\ \ell$

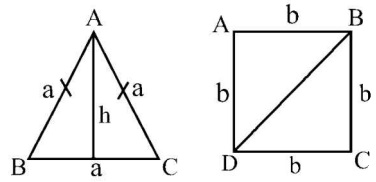
$= 2808\ \ell$

297. The height of an equilateral triangle is equal to one third of the diagonal of a square. What is the ratio between the area of the triangled and the square?

(a) $\sqrt{3} : \sqrt{2}$ (b) $2 : 9\sqrt{3}$
(c) $3 : 4\sqrt{3}$ (d) $\sqrt{3} : 6$

RRB Group-D – 26/11/2018 (Shift-III)

Ans : (b)



height of equilateral triangle = diagonal of square $\times \frac{1}{3}$

$\frac{\sqrt{3}}{2}a = \sqrt{2}b \times \frac{1}{3} \Rightarrow \frac{a}{b} = \frac{2\sqrt{2}}{3\sqrt{3}}$

$\Rightarrow \frac{\text{area of equilateral triangle}}{\text{area of square}} = \frac{\frac{\sqrt{3}}{4}a^2}{b^2} = \frac{\frac{\sqrt{3}}{4} \times 2\sqrt{2} \times 2\sqrt{2}}{3\sqrt{3} \times 3\sqrt{3}}$

$= 2 : 9\sqrt{3}$

298. If the ratio of the area of two squares is 1 : 3, what will be the ratio of the perimeter of these squares?

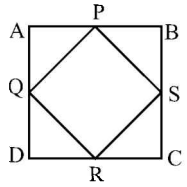
(a) $1 : \sqrt{3}$ (b) 1 : 4
(c) 1 : 3 (d) 1 : 2

RRB Group-D – 16/11/2018 (Shift-III)

Ans. (a) : Area ratio of both squares = 1 : 3
side's ratio $= 1 : \sqrt{3}$
Perimeter of square = 4a
Ratio of both perimeter of square

$= 4 \times 1 : 4 \times \sqrt{3} = 1 : \sqrt{3}$

299.

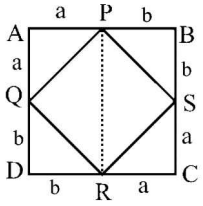


Rectangle PQRS is formed by cutting two pairs of equilateral triangles from the square ABCD in the given figure. The total area of the severed part is 288 meters^2. what will be the length of $\overline{PR}$ in meters?

(a) 24 (b) $\sqrt{288}$
(c) $\sqrt{512}$ (d) 48

RRB Group-D – 12/11/2018 (Shift-I)

Ans. (a) :



Let in ABCD square,
AP = AQ = CS = CR = a m
PB = BS = DQ = DR = b m

According to the question
Total area of part side = 288 m^2

$$\frac{1}{2}\times a\times a+\frac{1}{2}\times a\times a+\frac{1}{2}\times b\times b+\frac{1}{2}\times b\times b=a^2+b^2=288$$

So in PQRS rectangle

$$\text{Diagonal (PR)}=\sqrt{PQ^2+RQ^2}=\sqrt{2\times a^2+2\times b^2}$$

$$=\sqrt{2(a^2+b^2)}=\sqrt{2\times 288}=\sqrt{576}=24\text{ m}$$

300. A wall 4.84 meters long and 3.1 meters high is covered for 22 cm × 10cm tiles. If the value of each tile is Rs. 1.50, Find the total value of the tiles.

(a) Rs. 1,025 (b) Rs. 1,023
(c) Rs. 1,022 (d) Rs. 1,020

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (b) : Length of wall = 4.84 m = 484 cm
Height of wall = 3.1 m = 310 cm

$$\text{Number of tiles}=\frac{484\times 310}{22\times 10}=682$$

Cost of a tile = Rs. 1.50
Total cost of tiles = 682 × 1.50
= Rs. 1023

301. A cylindrical container with no lid is to be made from a rectangular seat of metal measuring 2.2 m × 2 m. The diameter and height of the container to be made is the same. The cost of manufacturing a container is Rs. 50. If the diameter of the container is to be made 14 cm, Find the total cost price to make the full container.

(a) 3,000 (b) 5,000
(c) 3,750 (d) 2,500

RRB Group-D – 02/11/2018 (Shift-II)

Ans. (a) area of rectangular sheet = $\ell \times b$
= 2.2 × 100 × 2.1 × 100
= 220 × 210
= 46200 square cm.

According to the question diameter of container is equal to their height So height is = 14 cm. and diameter 14 cm. then radius = 7cm.
area of container = surface area of container + area of base of container.

$$2\pi rh+\pi r^2=2\times\frac{22}{7}\times 7\times 14+\frac{22}{7}\times 7\times 7$$

$$=616+154=770\text{ square cm}$$

Then number of making containers to given seat

$$=\frac{46200}{770}=60$$

So manufacture cost of 60 container from according to Rs. 50 per container = 60 × 50 = 3000

302. The circumference of an equilateral hexagon is 72 cm. what is its area in cm.?

(a) $144\sqrt{3}$ (b) $216\sqrt{3}$
(c) $108\sqrt{3}$ (d) $36\sqrt{3}$

RRB Group-D – 02/11/2018 (Shift-II)

Ans. (b) Perimeter of right hexagon = 72cm
6a = 72 cm
a = 12 cm

$$\text{area of right hexagon}=6\times\frac{\sqrt{3}}{4}a^2$$

$$=6\times\frac{\sqrt{3}}{4}\times(12)^2$$

$$=6\times\frac{\sqrt{3}}{4}\times 144$$

$$=6\times 36\sqrt{3}$$

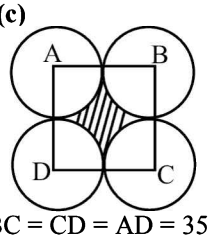
$$=216\sqrt{3}\text{ cm}^2$$

303. 4 cows are tied on four corners of a 35m square area so that he can reach each other which part of the area is left from the graze?

(a) 252.5 sq m (b) 162.5 sq m
(c) 262.5 sq m (d) 260.5 sq m

RRB Group-D – 12/11/2018 (Shift-II)

Ans : (c)



AB = BC = CD = AD = 35 m
side of square = 35 m

$$\text{radius of radius section}=\frac{35}{2}$$

$$\text{So 4 × area of radius section}=4\times\frac{\pi r^2\theta}{360^0}\qquad(\because\theta=90^0)$$

$$\text{area of four radius section}=4\times\frac{22}{7}\times\left(\frac{35}{2}\right)^2\times\frac{90^0}{360^0}$$

$$=4\times\frac{22}{7}\times\frac{35}{2}\times\frac{35}{2}\times\frac{1}{4}$$

$$=\frac{55\times 35}{2}$$

= 55 × 17.5
= 962.5 m^2

area of square = $side^2$
= $(35)^2$
= 1225 m^2

grazing left part by cow
= area of square – area of four radius section
= 1225.0 – 962.5
= 262.5 m^2

304. If the cost of painting of an 8 meter x 4.5 meter wall is Rs. 4500. Find the cost per square meter.

(a) Rs. 150 (b) Rs. 125
(c) Rs. 175 (d) Rs. 160

RRB NTPC 05.04.2016 Shift : 2

Ans : (b) area of wall = 8×4.5 = 36.0 Square m

$$\text{cost square per m.}=\frac{4500}{36}=\text{Rs. }125$$

305. Find the cost of covering a room of 60 dm × 30 dm at the rate of Rs. 45 square meter.

(a) Rs. 877.50 (b) Rs. 87750
(c) Rs. 87.75 (d) Rs. 8775

RRB NTPC 03.04.2016 Shift : 1

Ans : (a) Area of room= 65 dm × 30 dm
= 6.5 m. × 3.0 m.
= 19.5 m^2 (∴ 1m = 10 dm)
∴ Cost to cover the room = 19.5 × 45 = Rs. 877.50

306. A closed wooden rectangular box is made of one cm thick wood whose outer dimensions are as follows length 22cm, breadth 17cm and height 12cm. It is completely filled with cement what will be the amount of cement in the box?

(a) 1488 cube cm (b) 3000 cube cm
(c) 4488 cube cm (d) 2880 cube cm

RRB NTPC 02.04.2016 Shift : 2

Ans : (b) length of the wood = 22 cm
breadth = 17 cm
height = 12 cm
rectangular wood is 1 cm thick So,
length of cement = 22 – 2 = 20 cm
breadth = 17–2 = 15 cm
height = 12–2 = 10 cm
quantity of cement = 20 × 15 × 10 = 3000 cubic cm

307. How many time will it take for Ramesh to walk around a 50 meter square park if he runs at the rate of 18 km/h?

(a) 40 sec (b) 20 sec
(c) 80 sec (d) 160 sec

RRB NTPC 18.04.2016 Shift : 1

Ans : (a) given- Side of square shape park = 50 m
Perimeter of square shape park = 4 × 50
= 200 m
Speed of Ramesh = 18 Km/h
$= 18\times\frac{5}{18}$ meter/second
= 5 meter/second
Rotate one time of park by Ramesh
time = $\frac{200}{5}$ = 40 second

308. The base of a triangle is one-third of the base of a parallelogram having the same area as that of the triangle. The ratio of the corresponding heights of the triangle to the parallelogram will be:

(a) 3 : 2 (b) 4 : 1
(c) 3 : 1 (d) 6 : 1

RRB ALP & Tec. (29-08-18 Shift-I)

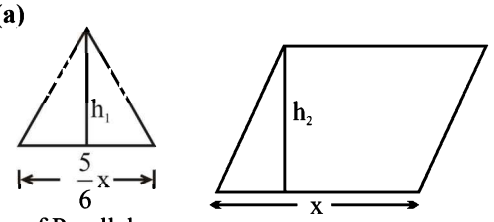
Ans : (d) Let base of parallelogram is a and height h_1 again let height of contiguous triangle is h_2.
As per question,
area of triangle = area of Parallelogram
$\frac{1}{2}\times\frac{a}{3}\times h_2 = a\times h_1$
$\Rightarrow \frac{h_1}{h_2}=\frac{1}{6}$
So height of triangle : height of parallelogram = 6 : 1

309. The base of a triangle is five-sixth of the base of a parallelogram having the same area as that of the triangle. The ratio of the corresponding heights of the triangle to the parallelogram will be:

(a) 12 : 5 (b) 5 : 12
(c) 5 : 3 (d) 6 : 5

RRB ALP & Tec. (29-08-18 Shift-III)

Ans : (a)



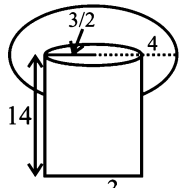
Let base of Parallelogram = x
then base of Δ triangle $=\frac{5}{6}x$
According to the question,
$\frac{1}{2}\times\frac{5}{6}x\times h_1 = x\times h_2$
$\frac{5}{12}h_1 = h_2$
$h_1 : h_2 = 12 : 5$

310. A well of diameter 3m and depth 14 m is dug. The earth, taken out of it, has been evenly spread all around it in the shape of a circular ring of breadth 4m to form an embankment. Find the height of embankment.

(a) $\frac{1}{8}$m (b) $\frac{9}{8}$m
(c) $\frac{7}{8}$m (d) $\frac{3}{8}$m

RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (b)



$h = 14,\ r = \frac{3}{2},\ R = 4+\frac{3}{2}=\frac{11}{2}$
According to the question
height of the pier **(H)** $= \frac{\pi r^2 h}{\pi(R)^2-\pi r_1^2}$
$= \frac{\pi r^2 h}{\pi(R^2-r_1^2)} = \frac{(3/2)^2\times 14}{(11/2)^2-(3/2)^2}$
$H = \frac{9/4\times 14}{\frac{121}{4}-\frac{9}{4}} = \frac{9\times 14}{112},\quad H=\frac{9}{8}$m.

311. From a solid cube of side 7cm, a conical cavity of height 7 cm and radius 3 cm is hollowed out. Find the volume of remaining solid?

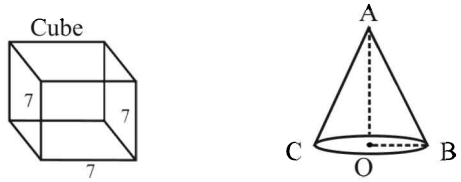
(a) 270 cm^3 (b) 277 cm^3
(c) 300 cm^3 (d) 272 cm^3

RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (b) volume of cube $= (\text{side})^3$

Volume of cone $= \frac{1}{3}\pi r^2 h$

Volume of left solid = Volume of cube – Volume of cone



OA = h = 7 cm

OB = OC = r = 3

$= (7)^3 - \frac{1}{3}\pi(3)^2 \times 7$

$= 343 - \frac{1}{3} \times \frac{22}{7} \times 3 \times 3 \times 7$

$= 343 - 66 = 277\,\text{cm}^3$

Volume of left solid $= 277\ \text{cm}^3$

312. The volume of a right circular cone, whose radius of the base is half of its altitude, and the volume of a hemisphere are equal. The ratio of the radius of the cone to the radius of the hemisphere is:

(a) $\sqrt{2}:1$ (b) $\sqrt[3]{2}:1$

(c) 1 : 1 (d) 2 : 1

RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (c) Let height of right circular cone $= h$

According to question base radius (r) $= \frac{1}{2} \times h$

$2r = h$

Volume of cone = Volume of hemisphere

$\frac{1}{3}\pi r^2 h = \frac{2}{3}\pi r_1^3$

$\left(\frac{h}{2}\right)^2 \times h = 2r_1^3$

$\left(\frac{2r}{2}\right)^2 \times 2r = 2r_1^3$

$r^3 = r_1^3 \Rightarrow \frac{r^3}{r_1^3} = 1$

$\boxed{r : r_1 = 1:1}$

313. The volume of a right circular cone, whose radius of the base is same as its altitude, and the volume of a sphere are equal. The ratio of the radius of the cone to the radius of the sphere is :

(a) 1 : 1 (b) $\sqrt[3]{4}:1$

(c) $\sqrt{2}:1$ (d) $\sqrt[3]{3}:\sqrt[3]{2}$

RRB ALP & Tec. (14-08-18 Shift-III)

Ans : (b) Let radius and height of right circular cone $= r_1$ and radius of sphere $= r_2$

$\therefore$ volume of cone = volume of sphere

$\frac{1}{3}\pi r_1^2 \times r_1 = \frac{4}{3}\pi r_2^3$

$\therefore \frac{r_1^3}{r_2^3} = \frac{4}{1}$

$\therefore \frac{r_1}{r_2} = \frac{\sqrt[3]{4}}{1} = \sqrt[3]{4}:1$

314. The volume of a right circular cone, whose radius of the base is same as one-third of its altitude, and the volume of a sphere are equal. The ratio of the radius of the cone to the radius of the sphere is :

(a) 1 : 1

(b) $\sqrt[3]{3}:\sqrt[3]{2}$

(c) $\sqrt[3]{4}:1$

(d) $\sqrt[3]{4}:\sqrt[3]{3}$

RRB ALP & Tec. (09-08-18 Shift-I)

Ans : (d) If height of right circular cone is h and radius of cone's base r_1, and sphere's radius r_2 then—

Volume of right circular cone $= \frac{1}{3}\pi r_1^2 h$

According to the question $= \frac{4}{3}\pi r_2^3$

$\because \quad r_1 = \frac{h}{3}, h = 3r_1$ .....(i)

According to the question

Volume of cone = Volume of sphere

$\frac{1}{3}\pi r_1^2 h = \frac{4}{3}\pi r_2^3$

putting $h = 3r_1$ from equation (i)

$\Rightarrow \quad \frac{1}{3}\pi r_1^2 \times 3r_1 = \frac{4}{3}\pi r_2^3$

$\Rightarrow \quad \frac{r_1^3}{r_2^3} = \frac{4}{3}$

$\Rightarrow \quad \left(\frac{r_1}{r_2}\right)^3 = \frac{4}{3}$ s

$\Rightarrow \quad r_1 : r_2 = \sqrt[3]{4}:\sqrt[3]{3}$